# Chemistry Grade 12 - Solved Exercises (BTBB)

NOTE: Chapter 20 (Carboxylic Acids and their Functional Derivatives) is PARTIAL in this edition: printed pages 282-283 of the source book are missing from the source scan, so that chapter's Multiple Choice Questions, Short Answer Questions, Long Questions 1-3 and Long Question 4 parts (i)-(ii) could not be verified and are not included. They will be added when a complete copy of the book is obtained. Nothing has been guessed

to fill the gap.

Balochistan Textbook Board (BTBB)

# Chapter 13: s- and p-Block Elements

## Exercise - Multiple Choice Questions

i) Which of the following give strong alkaline solution in water? (FLAGGED: sodium also gives a strong alkaline solution (NaOH); potassium is taken as the intended answer because it reacts most vigorously and KOH is the stronger alkali.)

- Sodium
- ✓ Potassium
- Beryllium
- Magnesium

ii) The total number of electrons in the valence shell of p-block elements varies from: (p-block groups 13-18 have 3 to 8 valence electrons.)

- 1 to 6
- 2 to 7
- ✓ 3 to 8
- 1 to 8

iii) The atomic radii in s-block elements:

- Decrease down the group
- ✓ Increase down the group
- Remains same in the group
- First decreases and then increases

iv) The general electronic configuration of p-block elements is as:

- ns1, ns2
- ns1, np6
- ns2, np6
- ✓ ns2, np1-6

v) Alkali and alkaline earth metals give flame tests when burnt in air. It is due to: (Heat excites the electrons; on de-excitation they fall back and emit light of characteristic wavelength — the colour seen is the emitted light.)

- Excitation of electrons
- ✓ De-excitation of electrons
- Smaller ionic radius of these metals
- Smaller charge density of these metals

vi) Basic oxides react with acidic oxides to form:

- Hydrides
- Halides
- Hydrogen halides
- ✓ Salts

vii) The characteristic flame colour of cesium is: (Cesium gives a blue-violet flame; blue is the listed option.)

- Green
- ✓ Blue
- Yellow
- Violet

viii) Which one of the following oxides is acidic? (GeO2, SnO2 and PbO2 are amphoteric; CO2 is acidic.)

- ✓ CO2
- GeO2
- SnO2
- PbO2

ix) Which one of the following elements only form dioxide? (Carbon forms CO and CO2; germanium and tin form monoxides and dioxides; silicon forms only SiO2.)

- Carbon
- ✓ Silicon
- Germanium
- Tin

x) The lowest melting points in carbon family (group IVA) is of: (Melting points fall down the group to tin, 232 C, then rise slightly to lead, 327 C.)

- Silicon
- Germanium
- ✓ Tin
- Lead

xi) Which one of the following reacts with water in elemental form at room temperature? (Cl2 + H2O gives HCl + HOCl.)

- Argon
- Sulphur
- ✓ Chlorine

Silicon

xii) Which one of the following is liquid at room temperature?

Fluorine

Chlorine

✓ Bromine

Iodine

xiii) Which one of the following is the strongest oxidizing agent? (Oxidizing power of halogens decreases from F2 to I2 — book Summary, p49.)

✓ Fluorine

Chlorine

Bromine

Iodine

xiv) The bond enthalpy of fluorine is: (F-F is weaker than Cl-Cl because the small F atoms crowd their lone pairs, causing repulsion.)

✓ Less than chlorine

More than chlorine

More than bromine

Less than iodine

xv) Which one of the following has giant structure? (SiO2 has a giant covalent network; the others are simple molecules.)

SO2

✓ SiO2

Cl2O7

P4O10

xvi) Which halogen of the following can oxidize all other halide ions? (Fluorine, the strongest oxidizing agent, oxidizes Cl-, Br- and I-.)

Iodine

✓ Fluorine

Bromine

Chlorine

xvii) The HCl is unable to reduce: (Resolved by Short Q24 in the same exercise: HCl CAN reduce F2 to HF, but cannot reduce Br2 or I2, because Cl- is a weaker reducing agent than Br- and I-; I2 is the listed option.)

Cl2

F2

✓ I2

HF

xviii) Which halogen acid cannot be stored in glass containers? (HF attacks silica in glass.)

✓ HF

HCl

HBr

HI

## Exercise - Short Answer Questions

### Q.1. Why are the alkali metals so reactive?

Alkali metals are very reactive because they have a single electron in their outermost s-orbital (ns1) which they lose easily to attain the stable noble-gas configuration. They have low ionization energies, large atomic radii and low electronegativity, so the valence electron is held weakly and removed readily, especially in reaction with water, oxygen and halogens.

### Q.2. Why are alkali metals stored in oil?

Alkali metals are stored in oil (kerosene) because they react violently with moisture in air and with oxygen, forming hydroxides and oxides. Oil keeps them away from air and water, preventing these reactions. (Lithium, sodium and potassium are kept under oil; lithium is stored in paraffin because it floats on ordinary oil.)

### Q.3. Why the reaction of caesium with water is very fast, violent and exothermic?

Caesium is the most reactive alkali metal: it has the largest atomic radius and the lowest ionization energy in the group, so its single valence electron is lost extremely easily. Its reaction with water is therefore very fast, violent and exothermic, producing caesium hydroxide and hydrogen gas with so much heat that the hydrogen may ignite.

### Q.4. What are the general trends in melting and boiling points going down the alkali metals?

Going down the alkali metals, melting and boiling points generally decrease. The atoms become larger, the metallic bonding weakens

because the single delocalised electron is spread over a larger volume and held less strongly, so less energy is needed to separate the atoms. (Lithium has the highest, caesium the lowest melting point in the group.)

Q.5. What is the effect of atomic radius on the chemical reactivity of the alkali metals?

The chemical reactivity of alkali metals increases with atomic radius down the group. A larger radius means the valence electron is farther from the nucleus and more shielded, so it is lost more easily (ionization energy falls). Hence caesium is the most reactive and lithium the least reactive alkali metal.

Q.6. Which are the most and the least reactive alkali metals?

The most reactive alkali metal is caesium (largest atom, lowest ionization energy) and the least reactive is lithium (smallest atom, highest ionization energy in the group). Francium is radioactive and not considered in the usual comparison.

Q.7. What are the products of a reaction between: (i) An alkali metal and water? (ii) An alkaline earth metal and water?

(i) An alkali metal and water: 2M + 2H2O -> 2MOH + H2 (e.g. 2Na + 2H2O -> 2NaOH + H2). Products: the metal hydroxide (alkaline solution) and hydrogen gas. (ii) An alkaline earth metal and water: M + 2H2O -> M(OH)2 + H2 (e.g. Ca + 2H2O -> Ca(OH)2 + H2). Products: the metal hydroxide and hydrogen gas. (Beryllium does not react with water; magnesium reacts slowly with hot water.)

Q.8. How many electrons do alkaline earth metals have to lose to attain the electronic configuration of noble gases?

Alkaline earth metals have two electrons in their outermost s-orbital (ns2) and must lose both to attain the noble-gas configuration. So they lose 2 electrons, forming M2+ ions (e.g. Mg -> Mg2+ + 2e-).

Q.9. Why are metals like sodium, potassium and aluminium not available in free state in nature?

Sodium, potassium and aluminium are not found free in nature because they are very reactive metals: they readily lose their valence electrons to oxygen, water and other substances, so they occur only in combined form — as oxides, chlorides, silicates, carbonates and other compounds in rocks, seawater and minerals. They must be extracted by electrolysis of their molten compounds.

Q.10. Why is the ionization energy of radium higher than barium?

Radium is below barium in Group II, so its ionization energy would normally be lower, but the book states radium's ionization energy is higher than barium's. This is attributed to the poor shielding of the 4f electrons (lanthanide contraction): the increased nuclear charge is poorly shielded, so the outer electrons are held more firmly, raising the ionization energy of radium above that of barium.

Q.11. Why is BeCl2 covalent and not ionic?

BeCl2 is covalent and not ionic because the Be2+ ion is very small with a high charge density, so it strongly polarizes (distorts) the electron cloud of the large chloride ions. By Fajans' rules this

polarization gives the bond a large covalent character; in the vapour phase BeCl2 exists as linear covalent molecules, and in the solid it forms a polymeric covalent chain.

Q.12. Be(OH)2 is amphoteric, how?

Be(OH)2 is amphoteric because it reacts with both acids and bases. With acids it behaves as a base: Be(OH)2 + 2HCl -> BeCl2 + 2H2O. With bases it behaves as an acid, forming a beryllate: Be(OH)2 + 2NaOH -> Na2[Be(OH)4] (sodium beryllate). This dual behaviour is due to beryllium's small size and high polarizing power.

Q.13. Can you explain why alkaline earth metals are less reactive than alkali metals?

Alkaline earth metals are less reactive than alkali metals because they have two valence electrons and a higher nuclear charge holding them: their first and especially second ionization energies are higher, their atomic radii are smaller, and removing two electrons costs more energy than removing the single electron of an alkali metal. Hence they react less vigorously with water and oxygen.

Q.14. Write an equation to show the thermal decomposition of calcium carbonate.

Thermal decomposition of calcium carbonate: CaCO3 -> CaO + CO2 (on heating). Calcium carbonate decomposes to calcium oxide (quicklime) and carbon dioxide gas.

Q.15. Write an equation to show the reaction of beryllium hydroxide

with an acid and base.

With an acid, beryllium hydroxide acts as a base: Be(OH)2 + 2HCl -> BeCl2 + 2H2O. With a base, it acts as an acid: Be(OH)2 + 2NaOH -> Na2[Be(OH)4]. These two equations together show its amphoteric character.

Q.16. Why is molten sodium chloride a good conductor of electricity?

Molten sodium chloride is a good conductor of electricity because on melting, the rigid ionic lattice breaks down and the Na+ and Cl- ions become free to move. These mobile ions carry electric charge through the melt. (Solid NaCl does not conduct because its ions are fixed in the lattice.)

Q.17. Why hydrogen and helium belong to s-block elements.

Hydrogen belongs to the s-block because its single electron enters the s-orbital (1s1). Helium belongs to the s-block because its two electrons fill the s-orbital (1s2); although it is a noble gas placed with Group 18, its electronic configuration is that of an s-block element.

Q.18. Stannous chloride is more stable than stannic chloride, why?

Stannous chloride (SnCl2, tin in +2 state) is more stable than stannic chloride (SnCl4, tin in +4 state) because of the inert pair effect: in the heavier Group 14 elements the ns2 pair becomes increasingly reluctant to take part in bonding, so the +2 oxidation state becomes more stable than +4 down the group (Ge to Sn to Pb). Hence Sn(II) is more stable than Sn(IV).

Q.19. SiO2 is a solid and CO2 is a gas, why?

SiO2 is a solid because it has a giant covalent structure: each silicon atom is bonded to four oxygen atoms in a continuous three-dimensional network, and breaking this network needs a great deal of energy. CO2 is a gas because it consists of small discrete molecules (O=C=O) held together only by weak van der Waals forces, which are easily overcome at room temperature.

Q.20. Why Fluorine is the most reactive element in Group 17?

Fluorine is the most reactive element in Group 17 because it is the smallest halogen with the highest electronegativity: its valence electrons are closest to the nucleus, it has a very high electron affinity in forming F-, and although the F-F bond is weak, the small size and large hydration energy of the fluoride ion make its reactions strongly exothermic. Reactivity decreases down the group as size increases.

Q.21. Explain why the electron affinity of fluorine is less than that of chlorine?

The electron affinity of fluorine is less than that of chlorine because fluorine is very small: the incoming electron enters the compact 2p orbital where it faces strong repulsion from the electrons already present. In the larger chlorine atom the added electron enters the more spacious 3p orbital with less repulsion, so more energy is released. Hence chlorine has the highest electron affinity in the group.

Q.22. Why the bond energy of fluorine is less than that of chlorine?

The bond energy of fluorine is less than that of chlorine because the F-F bond joins two very small atoms: their lone pairs of electrons are crowded together and repel each other strongly, weakening the bond. In Cl-Cl the atoms are larger, the lone pairs are farther apart, repulsion is less, and the bond is stronger. (F-F 158 kJ/mol, Cl-Cl 242 kJ/mol.)

Q.23. Why is HF a weak acid than other hydrohalic acids?

HF is a weaker acid than the other hydrohalic acids because the H-F bond is very strong (small atoms, short bond) and difficult to break in water, and the fluoride ion is heavily hydrated. The acidic strength of hydrogen halides increases from HF to HI as the H-X bond weakens down the group, so HI is the strongest and HF the weakest acid.

Q.24. Explain why the HCl molecule can reduce only F2 to HF but neither Br2 to HBr nor I2 to HI?

HCl can reduce F2 to HF because the chloride ion is a stronger reducing agent than the fluoride ion: 2HCl + F2 -> 2HF + Cl2 (F2 oxidizes Cl- to Cl2 and is itself reduced to F-). But HCl cannot reduce Br2 or I2, because Br- and I- are stronger reducing agents than Cl-: the reducing power of halide ions increases down the group (F- < Cl- < Br- < I-), so Cl- cannot displace Br- or I- from their elemental forms.

Q.25. Why the reactivity of halogens decreases down the group?

The reactivity of halogens decreases down the group because the

atoms become larger and the attraction of the nucleus for an incoming electron weakens: electron affinity and oxidizing power fall from fluorine to iodine, and the tendency to gain an electron to complete the octet decreases. (Oxidizing power decreases from F2 to I2 — book Summary, p49.)

Q.26. Why are iodide ions stronger reducing agents than chloride ions?

Iodide ions are stronger reducing agents than chloride ions because iodine is larger: its outer electron is farther from the nucleus, more shielded and more easily lost. The reducing power of hydrogen halides (and halide ions) increases from HF to HI (book Summary, p49), so I- gives up electrons more readily than Cl-.

Q.27. What makes one atom more electronegative than another?

An atom is more electronegative than another when it attracts the shared pair of electrons in a covalent bond more strongly toward itself. This depends on nuclear charge (more protons pull harder), atomic size (smaller atoms hold the bonding pair closer), and shielding: a high effective nuclear charge and small radius give high electronegativity (fluorine highest), while large, well-shielded atoms have low electronegativity.

## Exercise - Long Answer Questions

Q.1. Write a short note on s-block, p-block, d-block and f-block elements?

s-block elements: those in which the last electron enters the s-orbital (Groups 1 and 2, plus hydrogen and helium). They are soft,

reactive metals (except H, He) with ns1 or ns2 configurations, forming +1 or +2 ions. p-block elements: those in which the last electron enters the p-orbital (Groups 13-18). They include metals, metalloids and non-metals with ns2 np1-6 configurations, showing varied oxidation states. d-block elements: those in which the last electron enters the d-orbital (transition elements, Groups 3-12), with characteristic coloured ions, variable oxidation states and catalytic activity. f-block elements: those in which the last electron enters the f-orbital (lanthanides and actinides, placed below the main table), with very similar chemistry within each series.

Q.2. Discuss the atomic and physical properties of period 3 elements.

Atomic and physical properties of period 3 elements (Na to Ar): atomic radius decreases across the period because the nuclear charge increases while electrons are added to the same shell, pulling the electron cloud inward. Ionization energy and electronegativity generally increase across the period (with small irregularities at Mg and P due to stable configurations). Melting and boiling points rise from Na to Si (silicon has a giant covalent structure) and then fall to Ar (discrete molecules, weak forces). Electrical conductivity falls across the period: Na, Mg, Al are metals and good conductors; Si is a semiconductor; P, S, Cl, Ar are non-metals and insulators. Metallic character decreases from Na to Ar.

Q.3. Explain the reactions of period 3 elements with: (i) water (ii) oxygen (iii) chlorine.

Reactions of period 3 elements: (i) With water: Na and Mg react (Na vigorously, Mg slowly with hot water) giving hydroxides and hydrogen; Al reacts only when its oxide layer is removed; Si, P, S, Cl, Ar do not react simply with water (Cl2 disproportionates: Cl2 + H2O -> HCl + HOCl). (ii) With oxygen: all except Ar burn or react to form oxides — Na2O, MgO, Al2O3, SiO2, P4O10, SO2/SO3, Cl2O7. (iii) With chlorine: all except Ar form chlorides — NaCl, MgCl2, AlCl3, SiCl4, PCl3/PCl5, S2Cl2, Cl2O7-type oxychlorides being the exception for chlorine itself. Reactivity falls across the period as metallic character decreases.

Q.4. Describe physical properties of oxides of period 3 elements.

Oxides of period 3 elements: Na2O and MgO are ionic solids with high melting points; Al2O3 is ionic with some covalent character and very high melting point; SiO2 is a giant covalent solid with a very high melting point; P4O10 and SO2/SO3 are molecular with low melting points; Cl2O7 is a molecular liquid. The oxides become less ionic and more covalent across the period, and their melting points follow the pattern of structure: giant structures (SiO2, Al2O3) melt highest, molecular oxides lowest.

Q.5. What do you know about acid-base behaviour of the oxides of period 3 elements?

Acid-base behaviour of the oxides of period 3 elements changes across the period: Na2O and MgO are basic (dissolve in water to give alkaline solutions, react with acids to give salts). Al2O3 is amphoteric (reacts with both acids and bases). SiO2 is weakly acidic; P4O10, SO2, SO3 and Cl2O7 are acidic (dissolve in water to

give acids: P4O10 -> H3PO4, SO3 -> H2SO4, Cl2O7 -> HClO4). Thus the oxides pass from basic through amphoteric to acidic as the elements become more non-metallic.

Q.6. Describe reactions of the oxides of period 3 elements with: (i) water (ii) acids (iii) bases.

Reactions of the oxides of period 3 elements: (i) With water: basic oxides give alkalis (Na2O + H2O -> 2NaOH; MgO + H2O -> Mg(OH)2); acidic oxides give acids (SO3 + H2O -> H2SO4; P4O10 + 6H2O -> 4H3PO4; Cl2O7 + H2O -> 2HClO4); Al2O3 and SiO2 are insoluble and do not react. (ii) With acids: basic oxides give salts and water (Na2O + 2HCl -> 2NaCl + H2O; MgO + 2HCl -> MgCl2 + H2O); amphoteric Al2O3 also reacts (Al2O3 + 6HCl -> 2AlCl3 + 3H2O); acidic oxides do not react with acids. (iii) With bases: acidic oxides give salts and water (SO2 + 2NaOH -> Na2SO3 + H2O; SiO2 + 2NaOH -> Na2SiO3 + H2O); amphoteric Al2O3 also reacts (Al2O3 + 2NaOH + 3H2O -> 2Na[Al(OH)4]); basic oxides do not react with bases.

Q.7. Write note on structure, electrical conductivity, solubility, melting and boiling points of chlorides of period 3 elements.

Chlorides of period 3 elements: structure — NaCl and MgCl2 are ionic (giant lattices); AlCl3 is ionic in the solid but covalent dimeric (Al2Cl6) in the vapour; SiCl4, PCl3, PCl5 and S2Cl2 are simple covalent molecules. Electrical conductivity — molten NaCl and MgCl2 conduct (mobile ions); the covalent chlorides do not conduct. Solubility — NaCl and MgCl2 dissolve in water; AlCl3, SiCl4 and PCl3 hydrolyse in water (SiCl4 + 2H2O -> SiO2 + 4HCl); S2Cl2 is

insoluble. Melting and boiling points — ionic chlorides have high melting points (NaCl 801 C, MgCl2 714 C); covalent chlorides have low melting/boiling points (SiCl4 boils at 57 C).

Q.8. Write down properties of hydroxides of period 3 elements.
Hydroxides of period 3 elements: NaOH and KOH-type hydroxides are strong alkalis, very soluble, giving strongly alkaline solutions. Mg(OH)2 is only slightly soluble (weakly alkaline, used as an antacid). Al(OH)3 is insoluble and amphoteric (reacts with acids and with bases to give aluminates). Si(OH)4 is a very weak acid, practically insoluble. The hydroxides become less basic and less soluble across the period, passing from strongly basic through amphoteric to acidic, matching the trend of the oxides.

Q.9. What are alkali and alkaline earth metals and draw their electronic configurations?
Alkali metals (Group 1) have one valence electron, general configuration [noble gas] ns1: Li 1s2 2s1, Na 1s2 2s2 2p6 3s1, K 1s2 2s2 2p6 3s2 3p6 4s1, Rb [Kr] 5s1, Cs [Xe] 6s1. Alkaline earth metals (Group 2) have two valence electrons, general configuration [noble gas] ns2: Be 1s2 2s2, Mg 1s2 2s2 2p6 3s2, Ca 1s2 2s2 2p6 3s2 3p6 4s2, Sr [Kr] 5s2, Ba [Xe] 6s2. (Configurations are given in text form; the book asks for them 'drawn' as orbital diagrams.)

Q.10. Describe the atomic and physical properties of alkali metals.
Atomic and physical properties of alkali metals: they have large atomic radii (largest in their periods) which increase down the group; low ionization energies which decrease down the group; low

electronegativity; low melting and boiling points which decrease down the group (Li 180 C to Cs 28 C); low densities (Li, Na, K float on water); soft metals cut with a knife; good conductors of heat and electricity; silvery-white when freshly cut, tarnishing quickly in air.

Q.11. Explain the trends in reactivity with water of alkali metals.

Trends in reactivity with water of alkali metals: all react with water giving the hydroxide and hydrogen gas, and reactivity increases down the group. Lithium reacts steadily; sodium reacts vigorously, melting into a ball and moving on the surface; potassium reacts violently, the hydrogen igniting with a lilac flame; rubidium and caesium react explosively. The trend follows the falling ionization energy and increasing atomic radius, which make loss of the valence electron easier.

Q.12. Explain the reactions of alkali metals with oxygen and halogens.

Reactions of alkali metals with oxygen and halogens: with oxygen, lithium forms the normal oxide (4Li + O2 -> 2Li2O), sodium forms the peroxide (2Na + O2 -> Na2O2), and potassium, rubidium and caesium form superoxides (K + O2 -> KO2), because the larger cations stabilize larger anions. With halogens they react vigorously to form ionic halides (2Na + Cl2 -> 2NaCl), the vigour increasing down the group.

Q.13. Describe the effect of heat on nitrates, carbonates and hydrogen carbonates of alkali metals.

Effect of heat on nitrates, carbonates and hydrogen carbonates of

alkali metals: alkali metal nitrates (except LiNO3) decompose on heating to nitrites and oxygen (2KNO3 -> 2KNO2 + O2); LiNO3 decomposes to the oxide, nitrogen dioxide and oxygen (4LiNO3 -> 2Li2O + 4NO2 + O2). Alkali metal carbonates are stable to heat (do not decompose, except Li2CO3). Alkali metal hydrogen carbonates decompose on heating to carbonates, water and carbon dioxide (2NaHCO3 -> Na2CO3 + H2O + CO2).

Q.14. How are the colours produced in the flame tests? Explain.

Colours in flame tests are produced when the metal's electrons are excited by the heat of the flame to higher energy levels; when they fall back (de-excite) to lower levels, they emit energy as light of a definite wavelength, giving the characteristic colour. Each metal has its own set of energy levels, so each gives its own colour: lithium crimson, sodium golden yellow, potassium lilac, calcium brick red, barium apple green, caesium blue. The colour is identified by viewing the flame through cobalt glass where needed (to mask sodium's yellow).

Q.15. Explain atomic and physical properties of alkaline earth metals.

Atomic and physical properties of alkaline earth metals: they have smaller atomic radii than the corresponding alkali metals but the radii increase down the group; higher ionization energies than alkali metals (two electrons to remove) which decrease down the group; low electronegativity; higher melting and boiling points than alkali metals (stronger metallic bonding from two delocalised electrons per atom) which generally decrease down the group; higher

densities; harder than alkali metals; good conductors; silvery-white metals that tarnish in air.

Q.16. Discuss the trends in reactivity of alkaline earth metals with water.

Trends in reactivity of alkaline earth metals with water: reactivity increases down the group. Beryllium does not react with water; magnesium reacts slowly with hot water (Mg + 2H2O -> Mg(OH)2 + H2); calcium reacts steadily with cold water; strontium and barium react vigorously. The trend follows the falling ionization energy and increasing atomic radius, making loss of the two valence electrons easier down the group.

Q.17. Describe the reactions of alkaline earth metals with oxygen and nitrogen.

Reactions of alkaline earth metals with oxygen and nitrogen: with oxygen they form normal oxides (2Ca + O2 -> 2CaO), and barium also forms the peroxide (Ba + O2 -> BaO2). With nitrogen they form nitrides when heated (3Mg + N2 -> Mg3N2; 3Ca + N2 -> Ca3N2). Beryllium's oxide is amphoteric and its nitride formation is slower; reactivity with both gases increases down the group.

Q.18. Discuss the trends in solubility of hydroxides, sulphates, and carbonates of alkaline earth metals.

Trends in solubility of hydroxides, sulphates and carbonates of alkaline earth metals: hydroxides become more soluble down the group (Be(OH)2 insoluble, Ba(OH)2 fairly soluble) because the lattice energy falls faster than the hydration energy. Sulphates

become less soluble down the group (BeSO4 soluble, BaSO4 insoluble) — the reverse trend, because the sulphate ion is large and lattice energies dominate. Carbonates are all sparingly soluble, with solubility decreasing down the group (BeCO3 most soluble, BaCO3 least), and all decompose on heating (except BaCO3, which needs very high temperature) to the oxide and CO2.

Q.19. Explain the trends in thermal stability of the nitrates and carbonates of alkaline earth metals.

Trends in thermal stability of nitrates and carbonates of alkaline earth metals: both become more stable to heat down the group. The nitrates decompose to the oxide, nitrogen dioxide and oxygen (2Ca(NO3)2 -> 2CaO + 4NO2 + O2), and the temperature needed rises from Be to Ba because the larger cation polarizes the nitrate/carbonate ion less. The carbonates decompose to the oxide and CO2 (CaCO3 -> CaO + CO2), again needing higher temperatures down the group; BaCO3 is the most stable.

Q.20. How beryllium differs from other members of its group?

Beryllium differs from other members of its group because it is very small with a high charge density: its compounds are largely covalent (BeCl2 is covalent, not ionic); its oxide and hydroxide are amphoteric (the others are basic); it does not react with water and its salts are hydrolysed in water; it does not form a peroxide or a stable carbonate; its coordination number is 4 rather than 6; and it shows a diagonal relationship with aluminium (similar amphoteric oxides, covalent chlorides, carbides giving methane).

Q.21. Write short note on group 14 (IVA) elements. What are the atomic and physical properties of group IVA elements?

Group 14 (IVA) elements are carbon, silicon, germanium, tin and lead. Atomic and physical properties: all have four valence electrons (ns2 np2); atomic radius increases down the group; ionization energy decreases down the group; melting points fall from carbon (diamond, 3550 C) and silicon (1414 C) to tin (232 C), then rise slightly to lead (327 C); metallic character increases down the group — carbon and silicon are non-metals, germanium a metalloid, tin and lead metals. Oxidation states +4 (all) and +2 (increasingly stable down the group, inert pair effect). Catenation is strongest in carbon and decreases down the group.

Q.22. Define the term inert pair effect. Explain inert pair effect in the formation of ionic and covalent bonds.

The inert pair effect is the reluctance of the ns2 pair of electrons in the heavier p-block elements to take part in bonding. In ionic bond formation, the pair may be retained (giving the lower oxidation state, e.g. Sn2+, Pb2+); in covalent bond formation, only the np electrons may be used (e.g. SnCl2 covalent, PbCl2 ionic-covalent). Thus the +2 state becomes more stable than +4 down Group 14, and the +1 state more stable than +3 down Group 13 (Tl+ vs Tl3+).

Q.23. Write down preparation and thermal stability of chlorides of carbon, silicon and lead.

Preparation and thermal stability of chlorides of carbon, silicon and lead: carbon tetrachloride is prepared by chlorination of methane

(CH4 + 4Cl2 -> CCl4 + 4HCl) or of carbon disulphide; it is stable to heat and hydrolysis. Silicon tetrachloride is prepared by heating silicon or silica with chlorine (Si + 2Cl2 -> SiCl4) or by treating silica with carbon and chlorine; it hydrolyses readily (SiCl4 + 2H2O -> SiO2 + 4HCl). Lead chlorides: PbCl2 is prepared by precipitation (Pb(NO3)2 + 2HCl -> PbCl2 + 2HNO3) and is stable; PbCl4 is unstable and decomposes on heating (PbCl4 -> PbCl2 + Cl2), again due to the inert pair effect favouring Pb(II).

Q.24. Describe the reactions of group IVA elements with water.

Reactions of group IVA elements with water: carbon does not react with water at ordinary temperature (steam at high temperature gives water gas: C + H2O -> CO + H2). Silicon does not react with cold water but reacts with steam and with alkalis. Germanium is unaffected by water. Tin reacts slowly with steam (Sn + 2H2O -> SnO2 + 2H2). Lead is unaffected by pure water but is attacked by water containing dissolved oxygen, forming slightly soluble lead hydroxide (plumbosolvency).

Q.25. Discuss the important oxides of group IVA elements.

Important oxides of group IVA elements: carbon monoxide (CO, poisonous, reducing agent, burns with blue flame) and carbon dioxide (CO2, acidic, used in photosynthesis and fire extinguishers); silicon dioxide (SiO2, giant covalent, quartz, sand, glass-making); germanium dioxide (GeO2, amphoteric); tin oxides SnO (basic, blue-black) and SnO2 (amphoteric, white); lead oxides PbO (yellow, litharge, basic) and PbO2 (brown, strong oxidizing agent, used in lead accumulators).

Q.26. Describe the acid-base behaviour of group IVA oxides.

Acid-base behaviour of group IVA oxides: CO2 is acidic (CO2 + H2O -> H2CO3); SiO2 is weakly acidic (reacts with strong bases: SiO2 + 2NaOH -> Na2SiO3 + H2O); GeO2 is amphoteric; SnO and PbO are basic (react with acids) but SnO2 and PbO2 are amphoteric (react with both acids and bases). Thus the oxides pass from acidic (top of the group) to basic/amphoteric (bottom) as metallic character increases.

Q.27. What are halogens and draw their electronic configuration?

Halogens are the Group 17 elements — fluorine, chlorine, bromine, iodine (and astatine) — with seven valence electrons, general configuration ns2 np5: F 1s2 2s2 2p5, Cl 1s2 2s2 2p6 3s2 3p5, Br 1s2 2s2 2p6 3s2 3p6 3d10 4s2 4p5, I 1s2 2s2 2p6 3s2 3p6 3d10 4s2 4p6 4d10 5s2 5p5. They need only one electron to complete the octet, hence their high reactivity and -1 oxidation state; they occur as diatomic molecules (F2, Cl2, Br2, I2).

Q.28. Explain the atomic and physical properties of halogens.

Atomic and physical properties of halogens: atomic radius increases down the group; ionization energy and electronegativity decrease down the group (fluorine is the most electronegative element); melting and boiling points increase down the group (F2 and Cl2 gases, Br2 liquid, I2 solid) because van der Waals forces grow with molecular size; colours deepen down the group (pale yellow F2, greenish-yellow Cl2, reddish-brown Br2, violet-black I2); all are poor conductors and very reactive non-metals.

Q.29. Briefly discuss strength of halogens as oxidizing agents.

Strength of halogens as oxidizing agents: halogens are strong oxidizing agents because they readily gain an electron to complete the octet. Their oxidizing strength decreases down the group: F2 > Cl2 > Br2 > I2. Fluorine oxidizes all other halide ions (F2 + 2Cl- -> 2F- + Cl2); chlorine oxidizes Br- and I-; bromine oxidizes I- only; iodine is the weakest oxidizing agent and can even act as a reducing agent toward stronger oxidizers.

Q.30. Write note on acidity of hydrogen halides.

Acidity of hydrogen halides: HF is a weak acid, while HCl, HBr and HI are strong acids, with strength increasing from HF to HI. The trend follows the H-X bond strength, which falls down the group as the atoms get larger: the H-F bond is very strong and breaks with difficulty, whereas H-I breaks easily, releasing H+ readily. (Reducing power follows the same order: HF < HCl < HBr < HI.)

Q.31. Halide ions act as reducing agents. Explain.

Halide ions act as reducing agents because they can lose an electron to form the halogen: 2X- -> X2 + 2e-. The ease of losing the electron increases down the group as the ion becomes larger and its outer electron is held less firmly, so the reducing strength increases: F- < Cl- < Br- < I-. Thus iodide is the strongest reducing agent and fluoride the weakest; this is why HCl can reduce F2 to HF but cannot reduce Br2 or I2.

Q.32. Give the trends in reducing strength of halide ions.

Trends in reducing strength of halide ions: the reducing power increases down the group — F- < Cl- < Br- < I-. Fluoride is the weakest reducing agent (fluorine is the strongest oxidizing agent, so F- holds its electrons most firmly); chloride is stronger; bromide stronger still; iodide is the strongest reducing agent among the halides. The same order appears in the hydrogen halides: HF < HCl < HBr < HI as reducing agents.

Q.33. Write a balance chemical equation for the reaction of: i) Calcium with water ii) Sodium monoxide with water iii) Aluminium with water iv) Sulphur trioxide with water v) Chlorine(VII) oxide (Dichlorine heptaoxide) with water.

Balanced chemical equations: i) Calcium with water: Ca + 2H2O -> Ca(OH)2 + H2. ii) Sodium monoxide with water: Na2O + H2O -> 2NaOH. iii) Aluminium with water: 2Al + 6H2O -> 2Al(OH)3 + 3H2 (aluminium reacts only when the protective oxide layer is removed, e.g. as amalgam). iv) Sulphur trioxide with water: SO3 + H2O -> H2SO4. v) Chlorine(VII) oxide (dichlorine heptaoxide) with water: Cl2O7 + H2O -> 2HClO4 (perchloric acid).

# Chapter 14: d- and f-Block Elements: Transition Elements

## Exercise - Multiple Choice Questions

i) Which of the following have completely filled d-subshell? (FLAGGED: Cu, Ag and Au atoms also have d10 configurations, but Zn, Cd, Hg — Group 12 — are the textbook answer, since their d-subshell is filled in both the atom and the common ions.)

- Fe, Co, Ni
- Cu, Ag, Au
- Ni, Pd, Pt
- ✓ Zn, Cd, Hg

ii) Which one of the following does not belong to transition elements? (FLAGGED — DEFECTIVE as printed: lead, tin and arsenic are ALL non-transition elements, so three options satisfy the printed wording. The only transition element among the options is gold, which is the sensible intended answer if 'not' is a misprint for the question 'which one belongs'; that reading is used here and marked uncertain.)

- Lead
- ✓ Gold
- Tin
- Arsenic

iii) Which one of the following belongs to first transition series? (First series = Sc to Zn; iron is in it; silver, gold and mercury are in

later series.)

Silver

Gold

✓ Iron

Mercury

iv) Substance that are repelled by a magnetic field are known as:

Paramagnetic substances

✓ Diamagnetic substances

Ferromagnetic substances

Paramagnetic and Diamagnetic substances

v) Elements found in d-block of periodic table are termed as:

Main group elements

Normal elements

✓ Transition elements

Representative elements

vi) The most common oxidation states of iron are:

+1 and +2

+1 and +3

✓ +2 and +3

+3 and +4

vii) What is the correct electronic configuration for Ti2+ ion? (Ti atom is [Ar]4s2 3d2; on forming Ti2+ the 4s electrons are lost first, leaving [Ar]3d2.)

[Ar]4s2 3d0

✓ [Ar]4s0 3d2

[Ar]4s0 3d0

[Ar]4s1 3d1

viii) The number of unpaired electrons in zinc atom is: (Zn is [Ar]4s2 3d10 — all electrons paired.)

✓ 0

1

2

3

ix) Which of the following has the highest magnetic moment? (Unpaired electrons: Cu2+ 1, Ni2+ 2, Co2+ 3, Mn2+ 5; magnetic moment rises with unpaired electrons.)

Cu2+

Ni2+

Co2+

✓ Mn2+

x) In K4[Fe(CN)6], coordination number of iron is: (Six cyanide ligands surround iron.)

2

4

✓ 6

8

xi) Brass is a metal alloy that contains:

✓ Cu and Zn

Cu and Sn

Zn and Sn

Sn and Mn

xii) Give the IUPAC name for K3[Fe(CN)6]: (The complex ion [Fe(CN)6]3- contains Fe(III), since 6 CN- = -6 and the ion charge is -3.)

Potassium hexacyanoferrate (II)

✓ Potassium hexacyanoferrate (III)

Potassium hexacyanoiron (II)

Potassium hexacyanoiron (III)

xiii) The oxidation state of vanadium in V2O5 is: (2V + 5(-2) = 0, so V = +5.)

+2

+3

+4

✓ +5

xiv) The coordination number of metal atom or ion in a tetrahedral complex is:

2

✓ 4

6

8

xv) Which one of the following oxides is more acidic? (Acidic character increases with the oxidation state of the metal: Mn(VII) in

Mn2O7 is the most acidic.)

- MnO
- Mn2O3
- MnO2
- ✓ Mn2O7

xvi) Which period starts from scandium to zinc? (Sc to Zn is the first transition series, in period 4.)

- 2nd period
- 3rd period
- ✓ 4th period
- 5th period

xvii) Vanadium pentaoxide is used as a catalyst in the manufacture of: (V2O5 catalyses the oxidation of SO2 to SO3 in the Contact process.)

- HNO3
- ✓ H2SO4
- NH3
- CO(NH2)2

xviii) Which one of the following act as a reducing agent? (HNO3, KMnO4 and K2Cr2O7 are oxidizing agents; LiAlH4 is a strong reducing agent.)

- HNO3
- KMnO4
- ✓ LiAlH4
- K2Cr2O7

xix) In the reaction given below, the oxidation state of vanadium changes from: $VO_2 + 1/2\ O_2 \rightarrow V_2O_5$ (In $VO_2$, V = +4; in $V_2O_5$, V = +5.)

- +2 to +3
- +3 to +4
- ✓ +4 to +5
- +5 to +6

xx) Which is true for transition elements?

- They have low melting and boiling points
- They are insulators
- They are diamagnetic
- ✓ They show variable oxidation states

## Exercise - Short Answer Questions

Q.1. Define the following terms: (a) Central metal atom (b) Coordination number (c) Coordination sphere

(a) Central metal atom (or ion): the metal atom or ion in a coordination compound to which the ligands are attached by coordinate covalent bonds (e.g. Fe in $[Fe(CN)_6]^{4-}$). (b) Coordination number: the number of lone pairs of electrons provided by the ligands to the central metal atom or ion — i.e. the number of coordinate bonds formed (common values 2, 4 and 6). (c) Coordination sphere: the central metal atom/ion together with the ligands attached to it, written inside square brackets, e.g. $[Fe(CN)_6]^{4-}$; the ions outside the brackets are the ionization sphere.

### Q.2. Why most of the transition ions are coloured?

Most transition ions are coloured because they have partially filled d-orbitals. When ligands surround the metal ion, the five d-orbitals split into two sets of different energies; an electron can absorb visible light to jump from the lower set to the higher set (a d-d transition). The absorbed wavelength is removed from white light, and the transmitted light shows the complementary colour. Ions with empty (Sc3+, d0) or completely filled (Zn2+, d10) d-orbitals cannot undergo d-d transitions and are colourless.

### Q.3. Why majority of the transition metals are paramagnetic in nature?

The majority of transition metals are paramagnetic because their atoms/ions have unpaired electrons in the (n-1)d orbitals. Each unpaired electron behaves like a tiny magnet, so the substance is attracted into a magnetic field. The more unpaired electrons, the stronger the paramagnetism (e.g. Mn2+ with 5 unpaired electrons is strongly paramagnetic). Substances with all electrons paired are diamagnetic and are repelled by a magnetic field.

### Q.4. Why are the elements of group 12 (IIB) are excluded from the transition series elements?

Group 12 (IIB) elements (Zn, Cd, Hg) are excluded from the transition series because they have completely filled d-subshells (d10) in both the free atoms and their common ions (Zn2+, Cd2+, Hg2+). Since the d-subshell is never partially filled, they do not show the characteristic transition properties: variable oxidation states, coloured ions, paramagnetism or catalytic activity in the

same way.

Q.5. The elements of group 3 (IIIB) are excluded from the transition series elements, why?

The book treats Group 3 (IIIB) elements as excluded from the transition series because their stable ions (Sc3+, Y3+, La3+) have no d-electrons (d0 configuration): the d-subshell is empty rather than partially filled, so they do not show the typical transition properties such as coloured ions, variable oxidation states or paramagnetism, and behave like the main-group metals. (FLAGGED: this is the book's framing; many modern definitions do count scandium and yttrium as transition elements because the free atoms have partially filled d-orbitals.)

Q.6. Write down the electronic configuration of Fe3+ (Z = 26) and Zn2+ (Z = 30).

Fe (Z = 26) atom: [Ar] 4s2 3d6. On forming Fe3+, the 4s electrons are lost first and then one 3d electron: Fe3+ = [Ar] 3d5 (half-filled, extra stable). Zn (Z = 30) atom: [Ar] 4s2 3d10. On forming Zn2+, the two 4s electrons are lost: Zn2+ = [Ar] 3d10 (completely filled, stable).

Q.7. Why do transition elements show variable oxidation states?

Transition elements show variable oxidation states because the (n-1)d and ns orbitals have very similar energies, so electrons from both can take part in bonding. For example, iron shows +2 (loss of 4s electrons only) and +3 (loss of 4s and one 3d electron); manganese shows +2, +3, +4, +6 and +7. The small energy gap

allows different numbers of electrons to be lost or shared.

Q.8. Discuss catalytic behaviour of transition elements.

Transition elements are good catalysts because they can change oxidation state readily (adsorbing reactants, forming intermediate compounds, then returning to the original state), and because their surfaces provide sites where reactant molecules are adsorbed, their bonds weakened, and reaction activated. Examples: Fe in the Haber process (NH3), V2O5 in the Contact process (H2SO4), Ni in hydrogenation of oils, Pt in the Ostwald process.

Q.9. Why Ag+ ion is more stable than Ag2+ ion?

Ag+ is more stable than Ag2+ because Ag+ has the completely filled, stable d10 configuration ([Kr] 4d10). Ag2+ is d9, with an unpaired electron and an incomplete d-subshell; it is a very strong oxidizing agent that readily gains an electron to become Ag+, so it is unstable and rarely exists except in a few complexes (e.g. AgF2).

Q.10. What is binding energy? Explain the trends in binding energy.

Binding energy (cohesive energy) is the energy released when a gaseous metal atom is incorporated into the solid metal — a measure of the strength of metallic bonding. Its trend: it increases across a transition series up to the middle and then decreases. In the middle of the series the number of unpaired (n-1)d electrons available for bonding is greatest (up to 5-6), giving the strongest bonding; toward the end the d-orbitals fill and fewer unpaired electrons remain, so binding energy falls.

Q.11. Melting points of d-block elements increase up to the middle of series and thereafter decrease, why?

Melting points of d-block elements increase up to the middle of the series and thereafter decrease because the strength of metallic bonding follows the number of unpaired d-electrons available for bonding. The number of unpaired electrons rises to a maximum in the middle (chromium, manganese region) and then falls as the d-orbitals pair up, so the bonding weakens and melting points fall toward zinc (which has a filled d-subshell and the lowest melting point in the series).

Q.12. What is the oxidation state of zinc in [Zn(NH3)4]2+ ion?

In [Zn(NH3)4]2+, ammonia is a neutral ligand (charge 0). Let the oxidation state of zinc be x: x + 4(0) = +2, so x = +2. The oxidation state of zinc is +2.

Q.13. Why Cu2+ ions are paramagnetic while Zn2+ ions are diamagnetic?

Cu2+ is [Ar] 3d9 — it has one unpaired electron, so it is paramagnetic (attracted by a magnetic field). Zn2+ is [Ar] 3d10 — all d-electrons are paired, so it is diamagnetic (weakly repelled by a magnetic field). The difference is the partially filled vs completely filled d-subshell.

Q.14. Which ion shows a lower magnetic moment: Fe3+ or Cr3+?

Fe3+ is d5 (5 unpaired electrons), giving a magnetic moment of about 5.92 BM (spin-only: sqrt(5 x 7) = 5.92). Cr3+ is d3 (3 unpaired electrons), giving about 3.87 BM (sqrt(3 x 5) = 3.87).

Therefore Cr3+ shows the lower magnetic moment.

Q.15. Determine the oxidation state of cobalt in [Co(NH3)4(OH2)Cl]Cl2.

In [Co(NH3)4(OH2)Cl]Cl2, the complex ion is [Co(NH3)4(OH2)Cl]2+ (balanced by two Cl- outside). NH3 and H2O are neutral ligands; the chloride inside the sphere is -1. Let cobalt be x: x + 4(0) + 0 + (-1) = +2, so x = +3. The oxidation state of cobalt is +3.

## Exercise - Long Answer Questions

Q.1. What are transition elements? Describe the general features of transition elements?

Transition elements are the d-block elements in which the (n-1)d subshell is partially filled in the atom or in one of its common ions. General features: (1) They are hard, dense metals with high melting and boiling points. (2) They show variable oxidation states. (3) Their ions are usually coloured (d-d transitions) and paramagnetic (unpaired d-electrons). (4) They form complex (coordination) compounds readily. (5) They are good catalysts. (6) They form alloys and interstitial compounds. (7) Their compounds are often used as pigments. (8) They have high tensile strength and are good conductors of heat and electricity.

Q.2. What do you know about the transition series of d-block and f-block elements? Explain concisely.

Transition series of d-block elements: there are three main series in periods 4, 5 and 6 — the first (Sc to Zn, 3d), second (Y to Cd, 4d) and third (La to Hg, 5d) series, each of 10 elements as the

d-subshell fills. f-block transition series: the lanthanides (Ce to Lu, 4f filling) and actinides (Th to Lr, 5f filling), placed in two rows below the main table. The f-block elements within each series resemble each other closely because the f-electrons are inner and shielded, so their chemistry is very similar (all usually +3).

Q.3. What is binding energy? Discuss the trend in binding energy of transition elements?

Binding energy is the energy released when one mole of gaseous metal atoms condenses into the solid metal; it measures metallic bond strength. Trend: it increases from the start of a transition series to the middle (maximum near Groups 6-7, where the number of unpaired d-electrons is greatest) and then decreases toward the end (Zn, Cd, Hg have the lowest values, since their d-subshells are filled and only s-electrons bond). This same trend explains the melting points, hardness and enthalpies of atomization of the transition metals.

Q.4. Discuss the electronic configuration of d-block elements.

Electronic configuration of d-block elements: the general configuration is (n-1)d1-10 ns1-2. The d-subshell fills progressively across each series (Sc 3d1 4s2 to Zn 3d10 4s2). Exceptions occur where a half-filled or completely filled d-subshell is extra stable: chromium is 3d5 4s1 (not 3d4 4s2) and copper is 3d10 4s1 (not 3d9 4s2). On forming ions, the ns electrons are lost before the (n-1)d electrons (e.g. Fe2+ is 3d6, Fe3+ is 3d5).

Q.5. What is magnetic behaviour? Describe the magnetic behaviour

of substances.

Magnetic behaviour is the response of a substance to a magnetic field. Substances attracted into a magnetic field are paramagnetic (they have unpaired electrons; the more unpaired electrons, the stronger the paramagnetism, measured by the magnetic moment mu = sqrt(n(n+2)) BM). Substances weakly repelled by a magnetic field are diamagnetic (all electrons paired, e.g. Zn2+, Sc3+). Substances that retain magnetism strongly (Fe, Co, Ni) are ferromagnetic. Most transition metal ions are paramagnetic because of unpaired (n-1)d electrons.

Q.6. What are alloys? Give composition and uses of some well-known alloys.

Alloys are homogeneous mixtures of two or more metals, or a metal with a non-metal, made by melting together. Composition and uses of some well-known alloys: Brass (Cu 60-70%, Zn 30-40%) — utensils, musical instruments, cartridges. Bronze (Cu 90%, Sn 10%) — statues, medals, bells. Steel (Fe + C, with Cr, Ni, Mn) — construction, tools, cutlery. Stainless steel (Fe, Cr 18%, Ni 8%) — cutlery, surgical instruments. Duralumin (Al 95%, Cu 4%, Mg, Mn) — aircraft parts. Solder (Sn 50%, Pb 50%) — joining metals. Nichrome (Ni 60%, Cr 15%, Fe 25%) — heating elements.

Q.7. Describe some properties of an alloy that are different from the metals that compose it.

Properties of an alloy different from its component metals: (1) Hardness and strength — alloys are harder and stronger than the pure metals (steel is harder than iron; duralumin is stronger than

aluminium). (2) Melting point — usually lower than the pure metals (solder melts lower than tin or lead alone). (3) Corrosion resistance — stainless steel resists rusting, unlike iron. (4) Appearance — brass has a golden colour unlike copper or zinc. (5) Electrical resistance — nichrome has much higher resistance than its components, useful in heaters. (6) Malleability/ductility change — alloys are generally less malleable than pure metals.

Q.8. What are coordination compounds? How transition elements form these compounds?

Coordination (complex) compounds are compounds containing a central metal atom or ion attached to a group of ligands (ions or molecules donating electron pairs) by coordinate covalent bonds, existing independently as complex molecules or ions, e.g. [Cu(NH3)4]2+, [Fe(CN)6]4-. Transition elements form these compounds readily because they have small size, high charge and vacant (n-1)d orbitals of suitable energy to accept electron pairs from ligands; their variable oxidation states also allow many complexes. The complex ion is written in square brackets, with counter-ions outside.

Q.9. What are the components of complex compounds?

Components of complex compounds: (1) Central metal atom or ion — the acceptor of electron pairs (e.g. Fe3+ in [Fe(CN)6]3-). (2) Ligands — ions or molecules that donate electron pairs to the metal (e.g. NH3, H2O, CN-, Cl-, en). (3) Coordination sphere — the metal plus its ligands, written in square brackets. (4) Coordination number — the number of coordinate bonds (lone pairs donated) to the

metal. (5) Counter-ions (ionization sphere) — ions outside the brackets that balance the charge (e.g. K+ in K4[Fe(CN)6]).

Q.10. Explain different types of ligands with suitable examples.

Types of ligands with examples: (1) Monodentate (unidentate) — donate one lone pair through one donor atom: NH3 (ammine), H2O (aqua), Cl- (chloro), CN- (cyano), OH- (hydroxo). (2) Bidentate — donate two lone pairs through two donor atoms: ethylenediamine (en, H2N-CH2-CH2-NH2), oxalate (C2O4 2-). (3) Polydentate (chelating) — donate several lone pairs: EDTA (hexadentate, 6 donor atoms). (4) Ambidentate — can bind through either of two donor atoms: NO2- (nitro through N, nitrito through O), SCN- (thiocyanato through S, isothiocyanato through N). (5) Bridging ligands — bind two metal atoms simultaneously (e.g. OH-, Cl- bridges).

Q.11. Describe the rules for naming coordination compounds.

Rules for naming coordination compounds: (1) The cation is named first, then the anion. (2) Ligands are named first in alphabetical order, ignoring multiplicative prefixes (di, tri), followed by the metal. (3) Anionic ligands end in -o (chloro, cyano, hydroxo, oxalato); neutral ligands keep their names except H2O = aqua, NH3 = ammine, CO = carbonyl, NO = nitrosyl. (4) The number of ligands is shown by di, tri, tetra, penta, hexa (or bis, tris, tetrakis for complex ligand names). (5) The metal's oxidation state is given in Roman numerals in brackets. (6) If the complex ion is an anion, the metal name ends in -ate (ferrate, cobaltate, platinate).

Q.12. Give systematic names for the following compounds. i) K3[Fe(C2O4)3] ii) K2[PtCl4] iii) H[AuCl4] iv) Na2[Fe(CN)5NO] v) (NH4)2[PbCl6] vi) [Cr(NO)4] vii) [Ag(CN)2]1- viii) [Zn(NH3)4]2+ ix) [Os(en)4]Cl4 x) [Co(SO4)(NH3)4]NO3 xi) [Co(NH3)4(OH2)Cl]Cl2

Systematic names: i) K3[Fe(C2O4)3] — Potassium trioxalatoferrate(III). ii) K2[PtCl4] — Potassium tetrachloroplatinate(II). iii) H[AuCl4] — Hydrogen tetrachloroaurate(III). iv) Na2[Fe(CN)5NO] — Sodium pentacyanonitrosylferrate(II) (sodium nitroprusside). v) (NH4)2[PbCl6] — Ammonium hexachloroplumbate(IV). vi) [Cr(NO)4] — Tetranitrosylchromium(0) (FLAGGED: with NO as a neutral 3-electron donor, chromium is in the 0 state). vii) [Ag(CN)2]1- — Dicyanoargentate(I) ion. viii) [Zn(NH3)4]2+ — Tetraamminezinc(II) ion. ix) [Os(en)4]Cl4 — Tetra(ethylenediamine)osmium(IV) chloride. x) [Co(SO4)(NH3)4]NO3 — Tetraamminesulphatocobalt(III) nitrate. xi) [Co(NH3)4(OH2)Cl]Cl2 — Tetraammineaquachlorocobalt(III) chloride.

Q.13. Write chemical formulae of the following complexes: i) Hexaamminechromium(III) nitrate ii) Hexaamminenickel(II) bromide iii) Hexaaquaferrate(II) ion iv) Tetrahydroxoaluminate(III) ion v) Tetraamminedihydroxoplatimate(IV) sulphate vi) Tetraamminechloronitrocobalt(III) nitrate vii) Tetraamminechloronitroplatinum(IV) sulphate viii) Ammonium hexachlorotitanate(IV) ix) Sodium hexafluorocobaltate(III) x) Sodium hexahydroxostannate(IV) xi) Tris(ethylenediamine)cobalt(II) sulphate

Chemical formulae: i) Hexaamminechromium(III) nitrate —

[Cr(NH3)6](NO3)3. ii) Hexaamminenickel(II) bromide — [Ni(NH3)6]Br2. iii) Hexaaquaferrate(II) ion — [Fe(H2O)6]2+. iv) Tetrahydroxoaluminate(III) ion — [Al(OH)4]1-. v) Tetraamminedihydroxoplatimate(IV) sulphate — [Pt(NH3)4(OH)2]SO4. vi) Tetraamminechloronitrocobalt(III) nitrate — [Co(NH3)4Cl(NO2)]NO3. vii) Tetraamminechloronitroplatinum(IV) sulphate — [Pt(NH3)4Cl(NO2)]SO4. viii) Ammonium hexachlorotitanate(IV) — (NH4)2[TiCl6]. ix) Sodium hexafluorocobaltate(III) — Na3[CoF6]. x) Sodium hexahydroxostannate(IV) — Na2[Sn(OH)6]. xi) Tris(ethylenediamine)cobalt(II) sulphate — [Co(en)3]SO4.

Q.14. Describe the chemistry of vanadium, manganese, iron and copper.

Chemistry of vanadium: shows oxidation states +2, +3, +4, +5; V2O5 is amphoteric and a catalyst in the Contact process; vanadyl ion VO2+ is blue; vanadium steels are hard and strong. Manganese: states +2 to +7; MnO is basic, MnO2 amphoteric, Mn2O7 acidic and a strong oxidizing agent; KMnO4 is a powerful oxidizing agent in acid (MnO4- + 8H+ + 5e- -> Mn2+ + 4H2O) and is used in volumetric analysis. Iron: most common states +2 and +3; Fe2+ is a reducing agent, Fe3+ an oxidizing agent; iron forms the blood pigment haemoglobin (Fe2+ in haem) and is used in steel. Copper: states +1 (Cu2O, CuCl, diamagnetic, disproportionates) and +2 (CuO, CuSO4.5H2O blue, paramagnetic); copper is used in wires, alloys (brass, bronze) and coins.

Q.15. Explain the chemistry of chromium, cobalt and nickel.

Chromium: states +2, +3 (most stable) and +6; Cr2O3 is green and amphoteric; K2Cr2O7 is a strong oxidizing agent in acid (Cr2O7 2- + 14H+ + 6e- -> 2Cr3+ + 7H2O) and is used in volumetric analysis; chromium gives stainless steel its rust resistance. Cobalt: states +2 (pink salts, e.g. CoCl2.6H2O) and +3 (stable mainly in complexes like [Co(NH3)6]3+); cobalt is used in blue glass and pigments, and vitamin B12 contains cobalt. Nickel: states +2 (green salts, e.g. NiSO4) and +4 (in complexes); nickel is used in coins, stainless steel, and as a hydrogenation catalyst; Ni(CO)4 is used in the Mond process for purifying nickel.

Q.16. Find the charge on the central metal ion in: i) [Cu(CN)2]1- ii) [Co(NH3)6]3+ iii) Na2[CoCl4] iv) [Cr(H2O)6]Cl3

Charge on the central metal ion: i) [Cu(CN)2]1-: two CN- give -2; let Cu = x: x - 2 = -1, so Cu = +1. ii) [Co(NH3)6]3+: NH3 is neutral; Co = +3. iii) Na2[CoCl4]: the complex ion is [CoCl4]2- (balanced by 2 Na+); 4 Cl- = -4; Co - 4 = -2, so Co = +2. iv) [Cr(H2O)6]Cl3: the complex ion is [Cr(H2O)6]3+ (balanced by 3 Cl-); H2O is neutral; Cr = +3.

Q.17. Describe shapes of coordination compounds.

Shapes of coordination compounds depend on the coordination number: coordination number 2 gives a linear shape (e.g. [Ag(NH3)2]+). Coordination number 4 gives tetrahedral (e.g. [Zn(NH3)4]2+, [Ni(CO)4]) or square planar (e.g. [Pt(NH3)2Cl2], [Ni(CN)4]2-) shapes. Coordination number 5 gives trigonal bipyramidal or square pyramidal shapes. Coordination number 6 gives the octahedral shape (e.g. [Fe(CN)6]4-, [Co(NH3)6]3+), the

most common geometry. The shape is decided by the metal, its oxidation state, and the ligand field.

Q.18. Describe origin of colours of complex compounds.

Origin of colours of complex compounds: in a complex, the ligands split the metal's five d-orbitals into two sets (t2g lower, eg higher in octahedral fields). An electron absorbs a photon of visible light and jumps between these sets (d-d transition); the absorbed colour is removed from white light, and the compound shows the complementary colour. The size of the splitting depends on the metal, its oxidation state and the ligand, so different complexes have different colours. Some intense colours (e.g. purple $MnO_4^-$, orange $Cr_2O_7^{2-}$) come instead from charge-transfer transitions, since Mn(VII) and Cr(VI) have no d-electrons for d-d transitions.

# Chapter 15: Organic Compounds

## Exercise - Multiple Choice Questions

i) The conversion of coal into petroleum products is carried out by: (Coal is first converted to water gas (CO + H2), which is then converted to hydrocarbons by the Fischer-Tropsch process.)

- Fischer-wood process
- ✓ Fischer-Tropsch process
- Brikeland-Eyde's process
- Landler process

ii) Organic compounds that show weak attractive forces have: (Weak intermolecular forces are easily overcome, so melting (and boiling) points are low and vapour pressure is high.)

- High boiling points
- High melting points
- ✓ Low melting points
- Low vapour pressure

iii) Bucky ball consist of commonly: (C60 has 32 faces — 20 hexagons and 12 pentagons, like a football.)

- C60, having 30 faces (20 hexagonal and 10 pentagonal)
- ✓ C60, having 32 faces (20 hexagonal and 12 pentagonal)
- C60, having 60 faces (40 hexagonal and 20 pentagonal)
- C60, having 62 faces (40 hexagonal and 22 pentagonal)

iv) All the members of homologous series have: (Members share the same functional group and hence similar chemical properties, but their physical properties change gradually with molecular size.)

- Different types of elements
- ✓ Different physical properties
- Different chemical properties
- Different functional groups

v) Which type of functional group is present in the given compound: (The structure shows a C=O group inside a bicyclic ring, bonded to two carbons — a ketone, not a terminal aldehyde or a carboxylic acid.)

- Carboxylic acid
- Aldehyde
- ✓ Ketone
- Acid amide

vi) Homologous compounds differ from each other by: (Successive members differ by one -CH2- methylene group; book Summary, p101.)

- Alkyl group
- ✓ Methylene group
- Methine group
- Methyl group

vii) When hydrogen is removed from alkanes, the product obtained has general formula: (Removing one H from an alkane CnH2n+2 gives the alkyl group CnH2n+1.)

CnH2n+2

CnH2n-2

✓ CnH2n+1

CnH2n-1

viii) Which functional groups are present in the given amino acids: (The structure H2N-CH(CH3)-COOH contains an amino group -NH2 and a carboxylic acid group -COOH; the methyl group is a side chain, not a functional group.)

Amino and hydroxyl group

Amino and alcoholic group

✓ Amino and carboxylic acid group

Amino, carboxylic acid and alkyl group

ix) When nitrogen containing compounds are treated with Lassaigne's solution, the solution obtained is of: (Sodium fusion converts nitrogen to NaCN; with FeSO4 and FeCl3 it gives ferric ferrocyanide, Prussian blue.)

Red colour

✓ Prussian blue colour

Mixture of blue and green colour

Black colour

x) Black colour solution is obtained when sulphur containing compounds are treated with: (FLAGGED: lead acetate gives black lead sulphide (PbS); sodium nitroprusside gives a violet/purple colour, not black, so 'both' is not correct.)

Sodium nitroprusside solution

✓ Lead acetate solution

Tollens reagent

Both a and b

## Exercise - Short Answer Questions

### Q.1. Where natural gas is usually found? Give its two important uses.

Natural gas is usually found in underground reservoirs, often above petroleum deposits, trapped in porous rocks under impermeable layers. Two important uses: (1) As a fuel for domestic heating, cooking and power generation. (2) As a raw material (feedstock) for the manufacture of fertilizers (ammonia/urea), methanol and other petrochemicals.

### Q.2. What are the uses of coal tar?

Uses of coal tar: coal tar is a thick black liquid obtained by the destructive distillation of coal. It is used for road surfacing (tar macadam), as a preservative for wood (creosote), and as a source of aromatic compounds (benzene, toluene, naphthalene, phenol, anthracene) used in dyes, drugs, explosives and plastics.

### Q.3. What is coke? What are its uses?

Coke is the solid, porous, almost pure carbon residue left after the destructive distillation of coal (heating coal out of contact with air). Uses: (1) As a fuel in blast furnaces for the extraction of iron (it is a strong reducing agent and gives intense heat). (2) As a fuel in boilers and for making water gas and producer gas.

Q.4. Who discovered the bucky ball? How many sides are there on a bucky ball?

The bucky ball (buckminsterfullerene, C60) was discovered in 1985 by Harold Kroto, Robert Curl and Richard Smalley (Nobel Prize 1996). A bucky ball has 32 faces (sides): 20 hexagonal and 12 pentagonal faces, arranged like a football, with 60 carbon atoms at the vertices.

Q.5. What is the geometrical shape of a bucky ball?

The geometrical shape of a bucky ball is a hollow sphere (a truncated icosahedron), like a football: 60 carbon atoms arranged at the vertices of 20 hexagons and 12 pentagons, forming a closed cage. (Larger fullerenes can be ellipsoids or tubes.)

Q.6. Can a bucky ball conduct electricity?

Yes, a bucky ball can conduct electricity to a limited extent: its carbon atoms are sp2 hybridized with delocalised pi electrons over the cage, and doped fullerenes (e.g. with potassium, K3C60) become good conductors and even superconductors at low temperature. Pure solid C60 itself is a poor conductor (a semiconductor).

Q.7. Define synthetic compounds and synthetic chemistry.

Synthetic compounds are organic compounds made artificially in the laboratory or industry, rather than obtained from natural sources (e.g. nylon, plastics, aspirin, detergents). Synthetic chemistry is the branch of chemistry concerned with the preparation of such compounds by building molecules from simpler

starting materials, including partial synthesis (modifying a natural product) and total synthesis (building the molecule entirely from simple chemicals).

Q.8. What are the two main functional groups found in amino acids?
The two main functional groups found in amino acids are the amino group (-NH2) and the carboxylic acid group (-COOH), both attached to the same (alpha) carbon atom, together with a hydrogen atom and a side chain (R group).

Q.9. Why sodium metal is stored in kerosene oil?
Sodium metal is stored in kerosene oil because it reacts violently with moisture in the air and with oxygen, forming sodium hydroxide and hydrogen (which may catch fire). Kerosene oil keeps it away from air and water, preventing the reaction. (Same reason as for other alkali metals stored under oil.)

Q.10. Why are organic compounds fused with sodium metal for preparing Lassaigne's solution?
Organic compounds are fused with sodium metal for preparing Lassaigne's solution because sodium converts the covalently bound elements into ionic sodium salts that dissolve in water and can be tested: nitrogen becomes NaCN, sulphur becomes Na2S, and halogens become NaX (NaCl, NaBr, NaI). These ionic forms give the characteristic colour tests (Prussian blue for nitrogen, violet with nitroprusside or black PbS for sulphur, silver halide precipitates for halogens).

### Q.11. Why solution of ferrous sulphate is freshly prepared during detection of nitrogen?

A solution of ferrous sulphate is freshly prepared during the detection of nitrogen because FeSO4 is easily oxidized by air to ferric sulphate (Fe3+). The nitrogen test needs Fe2+ ions to react with cyanide and form sodium ferrocyanide, which then reacts with Fe3+ to give Prussian blue; if the FeSO4 is old, the Fe2+ has already been oxidized and the test fails.

## Exercise - Long Answer Questions

### Q.1. Define organic chemistry. What is the importance of organic chemistry in our daily life?

Organic chemistry is the study of hydrocarbons and their derivatives — the compounds of carbon (with hydrogen, oxygen, nitrogen, halogens, sulphur and other elements). Importance in daily life: almost everything we use is organic — food (carbohydrates, proteins, fats), clothing (cotton, wool, nylon, polyester), fuels (petrol, natural gas, coal), medicines (aspirin, penicillin), soaps and detergents, plastics, rubber, paints, perfumes and cosmetics. Organic chemistry also explains life processes (biochemistry) and provides fertilizers, pesticides and dyes for agriculture and industry.

### Q.2. Define organic compounds. What are the uses of organic compounds?

Organic compounds are the compounds of carbon, chiefly hydrocarbons and their derivatives, in which carbon is bonded covalently to hydrogen and often to oxygen, nitrogen, sulphur or

halogens. Uses of organic compounds: as fuels (petrol, diesel, natural gas); as food and medicines (carbohydrates, proteins, aspirin, antibiotics); as clothing and fibres (cotton, nylon); as solvents (alcohol, ether); as plastics, rubber, paints and detergents; and as fertilizers, pesticides and explosives in agriculture and industry.

Q.3. What are the natural sources of organic compounds? Explain.
Natural sources of organic compounds: (i) Fossil remains — coal, petroleum and natural gas, formed from dead plants and animals over millions of years; they give fuels and thousands of petrochemicals. (ii) Plants — they give carbohydrates (starch, cellulose), proteins, oils, fats, alkaloids (quinine, morphine), dyes and rubber. (iii) Animals — they give proteins, fats, hormones, urea and other products. Coal, petroleum and natural gas are the most important commercial sources because they are available in bulk.

Q.4. What are the common varieties of coal? Which variety of coal has the highest carbon content?
Common varieties of coal: peat, lignite (brown coal), bituminous coal (soft coal) and anthracite (hard coal). They differ in carbon content and age: peat has the lowest carbon (about 60%) and anthracite the highest carbon content (about 90-95%). Anthracite is the hardest, oldest variety, burns with the greatest heat and almost no smoke, and is the most valuable fuel.

Q.5. Define natural product chemistry and describe the use of plants and animals as a source of organic compounds.

Natural product chemistry is the study of organic compounds produced by living organisms (plants, animals, microorganisms) — their isolation, structure and synthesis. Plants as a source: they provide carbohydrates, proteins, fats and oils, alkaloids (quinine from cinchona bark, morphine from poppy), tannins, dyes (indigo), rubber (from latex) and drugs (aspirin originally from willow bark). Animals as a source: they provide proteins (casein from milk, albumin from egg), fats (butter, cod-liver oil), hormones (insulin), urea and musk for perfumes.

Q.6. Describe the synthesis of organic compounds in the lab. What is partial and total synthesis?

Synthesis of organic compounds in the lab: compounds are built step by step from simpler, readily available starting materials by chosen reactions (e.g. ethanol from ethene by hydration; aspirin from salicylic acid and acetic anhydride). Partial synthesis means starting from a natural product and modifying it chemically to get the desired compound (e.g. making a drug from a plant alkaloid). Total synthesis means building the compound entirely from simple, basic chemicals without using a natural product as the starting material (e.g. total synthesis of vitamin B12).

Q.7. What do you know about products of biotechnology?

Products of biotechnology: biotechnology uses living organisms or their products for human benefit. Its products include medicines (insulin, human growth hormone, interferons, vaccines made by genetically engineered bacteria); transgenic crops (herbicide-resistant soybeans, Bt cotton); bio-fertilizers

(Rhizobium); alcoholic drinks and bread made by yeast fermentation; antibiotics (penicillin from fungi); enzymes for detergents and food processing; and biofuels (ethanol, biogas). (See also Unit 26 of Biology for gene technology products.)

Q.8. What do you know about destructive distillation of coal? What are the various products obtained from it? Discuss.

Destructive distillation of coal means heating coal strongly out of contact with air, so that it decomposes instead of burning. Products obtained: (1) Coke — the solid residue (almost pure carbon), used in blast furnaces. (2) Coal gas — a mixture of hydrogen, methane and carbon monoxide, used as a fuel. (3) Ammoniacal liquor — an aqueous solution of ammonia, used for fertilizers. (4) Coal tar — a thick black liquid, a source of aromatic compounds (benzene, toluene, naphthalene, phenol) for dyes, drugs and plastics.

Q.9. How can coal be converted into petroleum? Discuss briefly.

Coal can be converted into petroleum (synthetic petrol) by the Fischer-Tropsch process and by hydrogenation (Bergius process): coal is first gasified with steam to water gas (CO + H2); the water gas is then passed over a catalyst (iron or cobalt) at high temperature and pressure, where CO and H2 combine into liquid hydrocarbons (petrol, diesel). In the Bergius process, powdered coal is heated with hydrogen under high pressure with a catalyst to give liquid fuels directly.

Q.10. What are the important characteristics of organic compounds?

Important characteristics of organic compounds: (1) They are mostly covalent, with weak intermolecular forces, so they have low melting and boiling points and are volatile. (2) They are generally insoluble in water but soluble in organic solvents. (3) They are poor conductors of electricity. (4) They burn in air (combustible), giving CO2 and water. (5) Their reactions are usually slow, complex and give several products (side reactions). (6) They show isomerism and catenation (carbon chains). (7) They are sensitive to heat and decompose easily.

Q.11. Define and explain bucky ball.

Bucky ball (buckminsterfullerene): it is an allotropic form of carbon consisting of molecules composed entirely of carbon atoms arranged in hollow spheres, ellipsoids or tubes. The commonest is C60: 60 carbon atoms at the vertices of 20 hexagonal and 12 pentagonal faces, forming a closed cage like a football (a truncated icosahedron). It was discovered by Kroto, Curl and Smalley (1985). Bucky balls are used in lubricants, drug delivery and superconductors (when doped with metals).

Q.12. Define functional group. How does a functional group affect the properties of an organic compound? Give common examples of functional groups.

A functional group is an atom or group of atoms in a molecule that is mainly responsible for the chemical behaviour of the molecule; it is the active part of organic compounds, and compounds with the same functional group have similar chemical properties. How it affects properties: it decides the type of reactions (e.g. -OH

compounds react as alcohols, -COOH as acids), polarity, solubility and boiling points. Common examples: hydroxyl -OH (alcohols), carboxyl -COOH (acids), aldehyde -CHO, ketone >C=O, amino -NH2, halogen -X, ether -O-, ester -COO-, nitro -NO2, double bond C=C (alkenes) and triple bond C#C (alkynes).

Q.13. What is meant by homologous series of carbon compounds? Give homologous series of alkanes.

A homologous series of carbon compounds is a group of organic compounds with the same functional group, in which each member differs from the next by one methylene group (-CH2-). Homologous series of alkanes: methane CH4, ethane C2H6, propane C3H8, butane C4H10, pentane C5H12 — each differs from the previous by CH2, all have the general formula CnH2n+2 and similar chemical properties.

Q.14. What are the general features of homologous series?

General features of homologous series: (1) All members have the same functional group and general formula. (2) Successive members differ by one -CH2- group (molecular mass differs by 14). (3) Members have similar chemical properties. (4) Physical properties (melting point, boiling point, density, solubility) change gradually with increasing molecular size. (5) Members can be prepared by similar general methods. (6) The first member may differ slightly from the rest in some properties.

# Chapter 16: Hydrocarbons

## Exercise - Multiple Choice Questions

i) The compounds of carbon and hydrogen in which all the four valencies of carbon are fully satisfied are termed as: (Alkanes have only single bonds; every carbon has four single bonds.)

- ✓ Alkanes
- Alkenes
- Alkynes
- Alkanols

ii) Acetylene is the example of: (Acetylene, HC#CH, is the simplest alkyne.)

- Alkane
- Alkene
- ✓ Alkyne
- Enyne

iii) The addition of unsymmetrical reagent to an unsymmetrical alkene follows the rule:

- Pauli's exclusion principle
- Hund's rule
- Auf bau principle
- ✓ Markownikov's rule

iv) The geometry of alkanes is: (sp3 carbon has tetrahedral

geometry, bond angle 109.5 degrees.)

- Linear
- Triangular
- ✓ Tetrahedral
- Octahedral

v) Alkanes are colourless and odourless liquids when they have: (C1-C4 are gases, C5-C17 are liquids, C18 and above are waxy solids.)

- Less than five carbons
- ✓ Five to seventeen carbons
- One to seventeen carbons
- More than seventeen carbons

vi) The isomers of a compound have same: (Isomers share the molecular formula, hence the same molecular mass, but differ in structure and properties.)

- Physical properties
- Chemical properties
- Structural formulas
- ✓ Molecular masses

vii) Which type of substitution reaction occurs in benzene easily? (Benzene's pi cloud attracts electrophiles; substitution preserves aromaticity.)

- ✓ Electrophilic substitution reaction
- Nucleophilic substitution reaction
- Both occurs simultaneously

None of the reaction takes place

viii) The substituent (-CHO) as shown attach to benzene ring is said to be: (The aldehyde group withdraws electrons by resonance and is a meta-directing deactivator.)

Activator to the ring

✓ Deactivator to the ring

It does nothing to the ring

It is just a functional group

ix) The given two compounds are ..... of each other: (Propanone CH3-CO-CH3 and prop-1-en-2-ol H2C=C(OH)-CH3 interconvert by shifting a hydrogen and a double bond — keto-enol tautomers.)

Skeletal isomers

Metameres

✓ Tautomers

Positional isomers

x) Two optical isomers such as (+) lactic acid and (-) lactic acid, have: (Enantiomers have identical physical and chemical properties except that they rotate plane-polarized light in opposite directions and react differently with other chiral substances.)

✓ Same physical and chemical properties

Same physical but different chemical properties

Different physical but same chemical properties

Same behavior towards plane of polarized light

## Exercise - Short Answer Questions

### Q.1. Why alkanes are less reactive than alkenes?

Alkanes are less reactive than alkenes because alkanes contain only strong sigma (C-C and C-H) bonds, which need a great deal of energy to break, and they have no exposed pi electrons for an attacking reagent to attract. Alkenes contain a weaker pi bond whose electrons are exposed above and below the plane, so electrophiles can attack them readily. Hence alkanes undergo only slow free-radical substitution, while alkenes undergo rapid addition reactions.

### Q.2. Why alkenes are more reactive than alkynes?

Alkenes are more reactive than alkynes toward electrophilic addition because the pi electrons of an alkene are held less tightly than those of an alkyne. In alkynes the carbons are sp hybridized (50% s-character), so they are more electronegative and hold the pi electrons closer to the nuclei in a cylindrical cloud, making them less available to an electrophile. The alkyne addition therefore proceeds more slowly and in two steps, while the alkene reacts in one fast step.

### Q.3. Why benzene is less reactive than alkenes?

Benzene is less reactive than alkenes because its six pi electrons are delocalised over the whole ring (resonance), giving a resonance energy of 152 kJ/mol. Addition to benzene would destroy this aromatic delocalisation and cost the resonance energy, so benzene prefers electrophilic substitution, which restores the aromatic ring, instead of addition. Alkenes have localised pi bonds and add readily.

Q.4. Explain why cis-alkenes are generally less stable than their trans-isomers.

cis-Alkenes are generally less stable than their trans-isomers because in the cis isomer the two larger groups lie on the same side of the double bond, where they crowd each other (steric repulsion) and raise the energy. In the trans isomer the groups are on opposite sides, farther apart, so repulsion is less and the molecule is more stable. (For example, trans-but-2-ene is more stable than cis-but-2-ene.)

Q.5. Why a substituent directs an incoming electrophile to a particular position?

A substituent directs an incoming electrophile to a particular position because it changes the electron density at the ortho, meta and para positions of the ring and stabilises (or destabilises) the intermediate carbocation (sigma complex) differently. Electron-donating groups (e.g. -OH, -NH2, -CH3) increase electron density at ortho/para positions and stabilise the intermediate there, so they are ortho/para directors; electron-withdrawing groups (e.g. -NO2, -CHO, -COOH) decrease ortho/para density and stabilise the meta intermediate, so they are meta directors.

Q.6. Define radical, homolytic fission and heterolytic fission.

A radical (free radical) is an atom or group with an unpaired electron, formed by breaking a covalent bond (e.g. Cl. from Cl2). Homolytic fission is the breaking of a bond so that each atom takes one electron of the shared pair, giving two radicals (A-B -> A. + B.).

Heterolytic fission is the breaking of a bond so that one atom takes both electrons of the shared pair, giving a cation and an anion (A-B -> A+ + B-).

Q.7. Give the order of reactivity of alkanes, alkenes, alkynes and benzene.

Order of reactivity (toward electrophilic addition): alkenes > alkynes > benzene > alkanes. Alkenes are the most reactive (exposed pi bond); alkynes are next (pi electrons held more tightly); benzene is less reactive (aromatic delocalisation resists addition); alkanes are the least reactive (only strong sigma bonds).
(FLAGGED: this is the electrophilic-addition order used in the book; for some other reactions, e.g. hydrogenation, the order can differ.)

Q.8. Why alkynes are acidic in nature?

Alkynes are acidic in nature because the terminal alkyne hydrogen is attached to an sp-hybridized carbon (50% s-character), which is highly electronegative and holds the bonding electrons strongly, so the C-H bond is polarized and the hydrogen can be removed as H+. Thus terminal alkynes (RC#CH) react with Na, Ag+ or Cu+ to form acetylides (e.g. HC#CH + AgNO3 -> AgC#C Ag + HNO3), while alkanes and alkenes do not show this acidity.

Q.9. What is electrophilic substitution of benzene?

Electrophilic substitution of benzene is the reaction in which an electrophile (electron-seeking species, e.g. NO2+, Br+, SO3) attacks the benzene ring and replaces a hydrogen atom, while the aromatic ring is restored. Example: nitration — C6H6 + HNO3

(conc. H2SO4) -> C6H5NO2 + H2O. The mechanism passes through a carbocation intermediate (sigma complex) and ends by losing H+ to restore aromaticity.

Q.10. Why benzene does undergo electrophilic substitution reaction?

Benzene undergoes electrophilic substitution because its delocalised pi electron cloud attracts electrophiles, and substitution (unlike addition) restores the aromatic ring and its resonance energy at the end. The reaction proceeds through an arenium (sigma complex) intermediate: the electrophile adds to one carbon, then a proton is lost from that carbon, reforming the aromatic sextet. Hence substitution is energetically favoured over addition.

Q.11. Why benzene does not undergo electrophilic addition?

Benzene does not undergo electrophilic addition because addition would break the delocalised pi system and destroy the aromatic resonance energy (152 kJ/mol), giving a much less stable product. Addition of e.g. Br2 across a double bond would leave a non-aromatic diene. Under ordinary conditions benzene therefore resists addition; only under drastic conditions (e.g. Cl2 with UV light) does it add, giving benzene hexachloride (BHC).

Q.12. Why benzene acts as a nucleophile?

Benzene acts as a nucleophile because its ring carries a cloud of six delocalised pi electrons above and below the plane, which is a region of high electron density. This electron-rich cloud can donate electrons to an electron-deficient species (electrophile), which is

why benzene attacks electrophiles in substitution reactions. In other words, the pi system behaves as a nucleophile toward electrophiles.

### Q.13. What are activating and deactivating groups?

Activating groups are substituents that increase the reactivity of the benzene ring toward electrophilic substitution and direct the new group to ortho and para positions (e.g. -OH, -NH2, -OCH3, -CH3 — they donate electrons into the ring). Deactivating groups decrease the ring's reactivity and direct the new group to the meta position (e.g. -NO2, -CHO, -COOH, -SO3H — they withdraw electrons from the ring). Halogens are a special case: they deactivate the ring but are ortho/para directors.

### Q.14. Is nitro group activating or deactivating?

The nitro group (-NO2) is a deactivating group: it strongly withdraws electrons from the benzene ring by resonance (-M) and inductive (-I) effects, decreasing the ring's electron density and making electrophilic attack slower than in benzene itself. It is a meta director: the incoming electrophile goes to the meta position, where the intermediate carbocation is least destabilised.

## Exercise - Long Answer Questions

### Q.1. What are hydrocarbons? How are they classified? Give an example of each type.

Hydrocarbons are compounds of carbon and hydrogen only. Classification: (1) Aliphatic (open-chain) — alkanes (e.g. methane, CH4), alkenes (e.g. ethene, C2H4), alkynes (e.g. ethyne, C2H2). (2)

Alicyclic (cyclic, non-aromatic) — cycloalkanes (e.g. cyclohexane, C6H12), cycloalkenes (e.g. cyclohexene). (3) Aromatic — compounds containing at least one benzene ring (e.g. benzene, C6H6; toluene, C6H5CH3). Aliphatic and alicyclic are further divided into saturated (single bonds only) and unsaturated (double or triple bonds).

Q.2. Define nomenclature. Give historical background of common and IUPAC systems.

Nomenclature is the system of naming chemical compounds. Historical background: the common (trivial) system named compounds from their source or property (e.g. formic acid from ants, acetic acid from vinegar, methane as marsh gas) with no fixed rules, so the same compound often had several names. The IUPAC system (International Union of Pure and Applied Chemistry) was developed to give every compound one systematic name from its structure: the longest chain is the parent, substituents are named as prefixes with position numbers, and the suffix shows the functional group (-ane, -ene, -yne, -ol, -al, -oic acid).

Q.3. Explain nomenclature of straight chain and branched chain alkanes.

Nomenclature of straight-chain and branched-chain alkanes: the parent chain is the longest continuous chain of carbons; it is named by the number of carbons (meth, eth, prop, but, pent, hex, hept, oct, non, dec) with the suffix -ane. Straight-chain alkanes need no numbers (e.g. CH3CH2CH2CH3 = butane). For branched chains, the side chains (alkyl groups) are named as prefixes (methyl, ethyl)

with the number of the carbon they are attached to, choosing the numbering that gives the lowest locants: e.g. CH3CH(CH3)CH2CH3 = 2-methylbutane; (CH3)3CCH2CH3 = 2,2-dimethylbutane. If several identical substituents occur, di, tri, tetra are used.

Q.4. Describe the nomenclature of cycloalkanes.

Nomenclature of cycloalkanes: the ring is the parent, named by the number of ring carbons with the prefix cyclo- and the suffix -ane (cyclopropane, cyclobutane, cyclopentane, cyclohexane). Substituents are named as prefixes with position numbers, numbering the ring so that the substituents get the lowest possible locants: e.g. methylcyclopentane; 1-ethyl-2-methylcyclopentane; 1,1,2-trimethylcyclopentane. If only one substituent is present, no number is needed.

Q.5. Explain structure, relative stability and reactivity of alkanes.

Structure, relative stability and reactivity of alkanes: alkanes have sp3 hybridized carbons with tetrahedral geometry (109.5 degrees) and only sigma bonds. Stability: they are quite stable and unreactive (paraffins) because C-C and C-H sigma bonds are strong and non-polar; stability increases with branching (tertiary carbons give more stable radicals). Reactivity: they undergo free-radical substitution (halogenation in UV light), combustion (burning to CO2 + H2O), cracking at high temperature, and isomerization; they do not react with acids, bases or oxidizing agents under ordinary conditions.

Q.6. Explain structure, relative stability and reactivity of

cycloalkanes.

Structure, relative stability and reactivity of cycloalkanes: cycloalkanes are rings of sp3 carbons (CnH2n). Stability depends on ring size: cyclopropane (60 degree angles) and cyclobutane are strained and less stable; cyclopentane and cyclohexane are nearly strain-free (cyclohexane adopts the chair conformation with 109.5 degree angles) and are stable. Reactivity: like alkanes they undergo free-radical substitution and combustion, but the small rings also undergo addition reactions that open the ring (e.g. cyclopropane adds Br2/HBr), because relieving the angle strain is energetically favourable.

Q.7. Write down the physical properties of alkanes and cycloalkanes.

Physical properties of alkanes and cycloalkanes: they are colourless and odourless (except the gases, which are odourless; petrol has a characteristic smell from additives). C1-C4 are gases, C5-C17 are liquids, C18 and above are waxy solids. Melting and boiling points increase with molecular size (stronger van der Waals forces) and decrease with branching (less surface contact). They are insoluble in water but soluble in non-polar solvents; they are less dense than water; they are non-conductors of electricity. Cycloalkanes have slightly higher boiling points than the corresponding open-chain alkanes.

Q.8. Give an overview of the radical substitution reactions.

Overview of radical substitution reactions: in radical substitution, a hydrogen atom of an alkane is replaced by another atom or group

through free radicals. The classic example is chlorination of methane in sunlight: Cl2 absorbs light and splits homolytically into two Cl. radicals (initiation); a Cl. radical abstracts H from CH4, giving HCl and CH3. (propagation); CH3. reacts with Cl2, giving CH3Cl and a new Cl. radical (propagation continues); radicals combine to end the chain (termination). Further substitution gives CH2Cl2, CHCl3 and CCl4. Reactivity of hydrogens: tertiary > secondary > primary.

Q.9. Explain the reaction mechanism of free radical.

Reaction mechanism of free radicals: a free-radical chain reaction has three stages. (1) Initiation: a weak bond (e.g. Cl-Cl) breaks homolytically by heat or light, giving two radicals. (2) Propagation: a radical attacks a molecule, forming a new bond and a new radical, so the chain continues (e.g. Cl. + CH4 -> HCl + CH3.; CH3. + Cl2 -> CH3Cl + Cl.). (3) Termination: two radicals combine to form a stable molecule (e.g. Cl. + Cl. -> Cl2; CH3. + CH3. -> C2H6), ending the chain. The product distribution follows radical stability: 3 degree > 2 degree > 1 degree.

Q.10. Describe the oxidation of organic compounds.

Oxidation of organic compounds: oxidation means addition of oxygen or removal of hydrogen (or increase in oxidation number of carbon). Examples: combustion of hydrocarbons (CH4 + 2O2 -> CO2 + 2H2O); controlled oxidation of alkanes gives alcohols, aldehydes and acids; alkenes are oxidized by cold dilute KMnO4 (Baeyer's test) to glycols, and by hot KMnO4 to acids/ketones; primary alcohols oxidize to aldehydes then acids, secondary

alcohols to ketones. Strong oxidizing agents (KMnO4, K2Cr2O7) and combustion are the common methods.

Q.11. What are alkenes? Explain the nomenclature of alkenes.

Alkenes are unsaturated hydrocarbons containing at least one carbon-carbon double bond (general formula CnH2n).

Nomenclature: the parent chain must include the double bond; it is named with the suffix -ene, and the position of the double bond is shown by the number of the first carbon of the double bond (lowest locants): e.g. CH2=CH2 ethene; CH3CH=CH2 propene; CH3CH=CHCH3 but-2-ene. Substituents are named as prefixes with numbers. If two double bonds are present, the suffix is -diene (e.g. CH2=CH-CH=CH2 buta-1,3-diene).

Q.12. Describe the relative stability, structure and reactivity of alkenes.

Relative stability, structure and reactivity of alkenes: alkenes have sp2 carbons (trigonal planar, 120 degrees) with one sigma and one pi bond between the double-bonded carbons. Stability: more substituted alkenes are more stable (hyperconjugation and inductive effects); trans isomers are more stable than cis isomers (less steric crowding). Reactivity: the exposed pi bond makes alkenes reactive toward electrophilic addition (halogens, hydrogen halides, water, hydrogen) and oxidation; they also polymerize (e.g. ethene to polythene).

Q.13. Write down the methods for the preparation of alkenes in detail.

Methods for the preparation of alkenes in detail: (1) Dehydration of alcohols: heating an alcohol with concentrated H2SO4 (or Al2O3) removes water (e.g. ethanol -> ethene at 170 C). (2) Dehydrohalogenation of alkyl halides: heating with alcoholic KOH removes HX (e.g. bromoethane -> ethene). (3) Partial hydrogenation of alkynes: hydrogenation with Lindlar's catalyst gives cis-alkenes (e.g. ethyne -> ethene). (4) Cracking of alkanes: heating higher alkanes gives lower alkenes (e.g. decane -> octane + ethene). (5) Dehalogenation of vicinal dihalides with zinc dust (e.g. 1,2-dibromoethane + Zn -> ethene + ZnBr2).

Q.14. What are the physical and chemical properties of alkenes? Explain.

Physical and chemical properties of alkenes: physical — C2-C4 are gases, C5-C17 liquids, higher members solids; insoluble in water, soluble in organic solvents; boiling points rise with size and are slightly lower than the corresponding alkanes; they burn with a smoky flame. Chemical — electrophilic addition: addition of halogens (Br2 decolourises — the test for unsaturation), hydrogen halides (Markownikov's rule for unsymmetrical alkenes), water (acid-catalysed, giving alcohols), hydrogen (Ni catalyst, giving alkanes); oxidation (cold KMnO4 gives glycols; hot KMnO4 cleaves the double bond); polymerization (ethene -> polythene).

Q.15. Describe the concept of conjugation in alkenes. Your focus should be on dienes.

Concept of conjugation in alkenes (focus on dienes): conjugation means alternating double and single bonds (C=C-C=C), so the pi

electrons are delocalised over four carbons. In buta-1,3-diene the four p-orbitals overlap continuously, giving extra stability (resonance energy about 15 kJ/mol) and special reactions: 1,2-addition (addition across one double bond) and 1,4-addition (addition at the ends, giving the more stable product, favoured at higher temperature). Isolated dienes (double bonds separated by more than one single bond) behave like two independent alkenes; cumulated dienes (C=C=C, allenes) are less stable.

Q.16. What is isomerism? Explain different types of isomerism.

Isomerism is the phenomenon in which two or more compounds have the same molecular formula but different structures or spatial arrangements, and hence different properties. Types: (1) Structural isomerism — chain isomerism (different carbon skeletons, e.g. butane and 2-methylpropane), position isomerism (different position of a functional group or double bond, e.g. but-1-ene and but-2-ene), functional group isomerism (different functional groups, e.g. ethanol and dimethyl ether), metamerism (different alkyl groups on either side of a polyvalent functional group), tautomerism (keto-enol, e.g. propanone and prop-1-en-2-ol). (2) Stereoisomerism — geometrical (cis-trans) isomerism and optical isomerism (enantiomers, non-superimposable mirror images, e.g. (+) and (-) lactic acid).

Q.17. What are alkynes? Explain the nomenclature of alkynes.

Alkynes are unsaturated hydrocarbons containing at least one carbon-carbon triple bond (general formula CnH2n-2). Nomenclature: the parent chain must include the triple bond; it is

named with the suffix -yne, and the position of the triple bond is shown by the number of the first carbon of the triple bond (lowest locants): e.g. HC#CH ethyne; CH3C#CH propyne; CH3C#CCH3 but-2-yne. Substituents are named as prefixes with numbers. If two triple bonds are present, the suffix is -diyne (e.g. hepta-2,4-diyne).

Q.18. Explain relative stability, structure and reactivity of alkynes.
Relative stability, structure and reactivity of alkynes: alkynes have sp carbons (linear, 180 degrees) with one sigma and two pi bonds between the triple-bonded carbons. Stability: alkynes are less stable than alkenes and alkanes (high energy content) but the triple bond itself is strong; terminal alkynes are acidic. Reactivity: they undergo electrophilic addition, but more slowly than alkenes (pi electrons held tightly by sp carbons), adding two molecules of reagent in two steps (e.g. ethyne + 2Br2 -> 1,1,2,2-tetrabromoethane); they also undergo hydration (HgSO4/H2SO4 gives methyl ketones), hydrogenation (Lindlar's catalyst gives cis-alkenes; full hydrogenation gives alkanes) and polymerization.

Q.19. Give the general methods for the preparation of alkynes.
General methods for the preparation of alkynes: (1) From calcium carbide: CaC2 + 2H2O -> C2H2 + Ca(OH)2 (the laboratory and industrial method for ethyne). (2) Dehydrohalogenation of vicinal dihalides: heating with alcoholic KOH and then NaNH2 removes two HX molecules (e.g. 1,2-dibromoethane -> ethyne). (3) Dehalogenation of tetrahalides with zinc dust (e.g. 1,1,2,2-tetrabromoethane + Zn -> ethyne). (4) From haloforms:

chloroform with silver powder gives ethyne. (5) By electrolysis of aqueous sodium/potassium maleate or fumarate (Kolbe's method).

Q.20. Write down the physical properties of alkynes.

Physical properties of alkynes: the first three (ethyne, propyne, but-1-yne) are gases, the next members up to C12 are liquids, higher members are solids. They are colourless and odourless (pure ethyne has a garlic-like smell when impure), insoluble in water but soluble in organic solvents, and less dense than water. Melting and boiling points increase with molecular size and decrease with branching, like other hydrocarbons; they burn with a smoky, luminous flame.

Q.21. What do you know about addition reactions of alkyne? Explain.

Addition reactions of alkyne: alkynes add two molecules of reagent in two steps because the triple bond has two pi bonds. (1) Addition of hydrogen: with Lindlar's catalyst, one molecule adds to give a cis-alkene; with Ni/Pt and excess H2, the alkane forms. (2) Addition of halogens: Br2 adds twice (ethyne -> 1,2-dibromoethene -> 1,1,2,2-tetrabromoethane); bromine water is decolourised. (3) Addition of hydrogen halides: follows Markownikov's rule twice (ethyne + 2HBr -> 1,1-dibromoethane). (4) Addition of water: HgSO4/H2SO4 catalysis gives the enol, which tautomerises to a methyl ketone (ethyne -> ethanal). (5) Addition of HCN, acetic acid and alcohols gives industrial products (acrylonitrile, vinyl acetate).

Q.22. Define benzene and give its classification.

Benzene is an aromatic hydrocarbon, C6H6, consisting of a ring of six carbon atoms with one hydrogen bonded to each carbon and a delocalised pi system of six electrons. Classification: benzene is the parent aromatic hydrocarbon; aromatic hydrocarbons are classified as monocyclic (benzene, toluene, xylene) and polycyclic (naphthalene, anthracene, phenanthrene), and as arenes with one or more benzene rings in their structures. Aromatic compounds follow Huckel's rule (4n+2 pi electrons, planar, cyclic, conjugated).

Q.23. Explain the nomenclature of aromatic hydrocarbons.

Nomenclature of aromatic hydrocarbons: the parent is benzene; substituents are named as prefixes (methyl, ethyl, chloro, nitro) with position numbers. For one substituent no number is needed (chlorobenzene, toluene, phenol, aniline, benzoic acid). For two substituents, positions are ortho (1,2), meta (1,3) and para (1,4), or numbered 1,2 / 1,3 / 1,4 (e.g. 1-bromo-3-chlorobenzene). For three or more substituents, numbers are used, giving the lowest locants (e.g. 1-chloro-3-iodo-5-nitrobenzene; 2,4-dinitrophenol; 3-nitrobenzoic acid). Some retain special names: TNT (2,4,6-trinitrotoluene), picric acid (2,4,6-trinitrophenol), diphenylmethane.

Q.24. Explain the structure of benzene in terms of molecular orbital theory.

Structure of benzene in terms of molecular orbital theory: each carbon in benzene is sp2 hybridized, forming three sigma bonds (to two neighbouring carbons and one hydrogen) in a planar hexagon (120 degree angles). Each carbon has one unhybridized p-orbital

perpendicular to the ring; the six p-orbitals overlap sideways to form a continuous pi molecular orbital system above and below the ring, containing six delocalised pi electrons. These electrons occupy three bonding molecular orbitals, giving benzene its extra stability (resonance energy 152 kJ/mol) and equal C-C bond lengths (139 pm, between single 154 pm and double 134 pm).

Q.25. What is resonance? Explain the important aspects of resonance with reference to benzene.

Resonance is the phenomenon in which a molecule cannot be represented by a single structure but is a hybrid of two or more structures differing only in the arrangement of electrons (shown with a double-headed arrow). Important aspects with reference to benzene: benzene is represented by two Kekule structures (alternating double bonds) whose hybrid is the real molecule; the resonance hybrid is more stable than any single structure by the resonance energy (152 kJ/mol); all C-C bonds are equal (139 pm); the delocalised electrons explain benzene's low reactivity toward addition and its preference for substitution; resonance also explains the equal bond lengths and the planarity of the ring.

Q.26. Define the term resonance energy and discuss the stability of benzene.

Resonance energy is the difference between the energy of the actual molecule (the resonance hybrid) and the energy of the most stable single resonance structure. For benzene the resonance energy is 152 kJ/mol: benzene is 152 kJ/mol more stable than the hypothetical cyclohexatriene with three localised double bonds.

This extra stability (aromaticity) is why benzene resists addition reactions that would destroy the delocalised system, and instead undergoes substitution, which restores the aromatic sextet.

Q.27. Draw the structures for the following hydrocarbons: i) 2-Methylhexane ii) 3-Methylheptane iii) 4-Ethyl-3-methylheptane iv) 2,2,4-trimethylhexane v) 4-Ethyl-2-methylhexane vi) 3-Ethyl-3-methylhexane vii) 2,3-dimethylbutane viii) 2,3,4-trimethylpentane ix) 3-ethyl-2,2-dimethylpentane x) 5-ethyl-2,6-dimethyloctane xi) Methylcyclopentane xii) Ethylcyclohexane xiii) 1-ethyl-2-methylcyclopentane xiv) 1,1,2-trimethylcyclopentane (Structures are given as condensed formulas, since drawings cannot be printed as text.)
Structures (described in condensed form, since drawings cannot be printed as text): i) 2-Methylhexane: CH3-CH(CH3)-CH2-CH2-CH2-CH3. ii) 3-Methylheptane: CH3-CH2-CH(CH3)-CH2-CH2-CH2-CH3. iii) 4-Ethyl-3-methylheptane: CH3-CH2-CH(CH3)-CH(C2H5)-CH2-CH2-CH3. iv) 2,2,4-trimethylhexane: (CH3)3C-CH2-CH(CH3)-CH2-CH3. v) 4-Ethyl-2-methylhexane: (CH3)2CH-CH2-CH(C2H5)-CH2-CH3. vi) 3-Ethyl-3-methylhexane: CH3-CH2-C(CH3)(C2H5)-CH2-CH2-CH3. vii) 2,3-dimethylbutane: (CH3)2CH-CH(CH3)2. viii) 2,3,4-trimethylpentane: (CH3)2CH-CH(CH3)-CH(CH3)2. ix) 3-ethyl-2,2-dimethylpentane: (CH3)3C-CH(C2H5)-CH2-CH3. x) 5-ethyl-2,6-dimethyloctane: (CH3)2CH-CH2-CH2-CH(C2H5)-CH(CH3)-CH2-CH3. xi) Methylcyclopentane: a cyclopentane ring with one methyl group. xii) Ethylcyclohexane: a cyclohexane ring with one ethyl group. xiii)

1-ethyl-2-methylcyclopentane: a cyclopentane ring with ethyl at C1 and methyl at C2. xiv) 1,1,2-trimethylcyclopentane: a cyclopentane ring with two methyls at C1 and one methyl at C2.

Q.28. Write the structural formulas of the following compounds: i) 3-hexene ii) 2,3-heptadiene iii) Hepta-2,4-diyne iv) 1,3-dimethylcyclohexane v) TNT vi) Picric acid

Structural formulas: i) 3-hexene: CH3-CH2-CH=CH-CH2-CH3. ii) 2,3-heptadiene: CH3-CH=C=CH-CH2-CH2-CH3. iii) Hepta-2,4-diyne: CH3-C#C-C#C-CH2-CH3. iv) 1,3-dimethylcyclohexane: a cyclohexane ring with methyl groups at C1 and C3. v) TNT: 2,4,6-trinitrotoluene — a toluene ring with nitro groups at positions 2, 4 and 6. vi) Picric acid: 2,4,6-trinitrophenol — a phenol ring with nitro groups at positions 2, 4 and 6.

Q.29. Give the IUPAC names of the following compounds: i) CH3CH2CH3 ii) CH3(CH2)2CH3 iii) CH3CH(CH3)2 iv) (CH3)3CCH2CH3 v) CH3CH(CH3)CH2CH3 vi) CH3-CH=CH2 vii) CH2=CH-CH=CH2 viii) CH2=CH-C(CH3)=CH2 ix) CH2=CH-CH=C(C2H5)-CH=CH2 x) CH3-C#C-CH3 xi) CH3-C#C-CH2-C#CH xii) CH#C-CH=CH2 xiii) Chlorobenzene xiv) 1-Bromo-3-chlorobenzene xv) 1-Chloro-3-iodo-5-nitrobenzene xvi) 2,4-Dinitrophenol xvii) 3-Nitrobenzoic acid xviii) 3-Chlorobenzenesulphonic acid xix) Diphenylmethane xx) 1,3-Diphenylbutane

IUPAC names: i) CH3CH2CH3 — Propane. ii) CH3(CH2)2CH3 — Butane. iii) CH3CH(CH3)2 — 2-Methylpropane. iv) (CH3)3CCH2CH3 — 2,2-Dimethylbutane. v) CH3CH(CH3)CH2CH3 — 2-Methylbutane.

vi) CH3-CH=CH2 — Propene. vii) CH2=CH-CH=CH2 — Buta-1,3-diene. viii) CH2=CH-C(CH3)=CH2 — 2-Methylbuta-1,3-diene (isoprene). ix) CH2=CH-CH=C(C2H5)-CH=CH2 — 3-Ethylhexa-1,3,5-triene (numbered from the end giving the ethyl group the lowest locant, 3). x) CH3-C#C-CH3 — But-2-yne. xi) CH3-C#C-CH2-C#CH — Hexa-1,4-diyne. xii) CH#C-CH=CH2 — But-1-en-3-yne. xiii) Chlorobenzene. xiv) 1-Bromo-3-chlorobenzene. xv) 1-Chloro-3-iodo-5-nitrobenzene. xvi) 2,4-Dinitrophenol. xvii) 3-Nitrobenzoic acid. xviii) 3-Chlorobenzenesulphonic acid. xix) Diphenylmethane. xx) 1,3-Diphenylbutane.

# Chapter 17: Alkyl Halides and Amines

## Exercise - Multiple Choice Questions

i) RCH2X is the general formula of: (The halogen-bearing carbon is attached to only one other carbon — a primary alkyl halide.)

- ✓ Primary alkyl halide
- Secondary alkyl halide
- Tertiary alkyl halide
- Aryl halide

ii) The formula of chloroform is:

- CCl4
- ✓ CHCl3
- CH2Cl2
- CH3Cl

iii) The boiling point of alkyl halides are in the following order: (Boiling points rise with the size/mass of the halogen: iodides highest, fluorides lowest.)

- RI > RCl > RBr > RF
- RI > RBr > RF > RCl
- ✓ RI > RBr > RCl > RF
- RF > RCl > RBr > RI

iv) The alkyl halide on which a nucleophile attacks is known as: (The substrate is the molecule being attacked; the nucleophile is

the reagent.)

- Reagent
- ✓ Substrate
- Electrophile
- Both reagent and substrate

v) Alkyl halides are considered to be very reactive compounds because they have: (The C-X bond is polarized, making carbon electrophilic, and the halide ion is a good leaving group.)

- an electrophilic carbon
- A nucleophilic carbon
- ✓ an electrophilic carbon and a good leaving group
- a nucleophilic carbon and a good leaving group

vi) Which statement is correct about carbocations? (Carbocations are electron-deficient, with an empty p-orbital, and are highly reactive.)

- ✓ They are reactive species
- They are stable species
- They are electron rich species
- They have filled orbitals

vii) Which statement is true about bases? (Base strength is measured by the base dissociation constant Kb; electron-pair acceptors are acids, not bases.)

- They are electron pair acceptor species.
- They can accept electrons from hydrogen
- Their strength is measured by rate constant

✓ Their strength is measured by base dissociation constant

viii) In SN1 reactions, the rate of reaction depends upon the concentration of: (SN1 is first-order: rate depends only on the substrate, since ionization is the slow step.)

✓ One reactant molecule

Two reactant molecules

Nucleophile

Both nucleophile and electrophile

ix) The SN2 reactions can be best performed with: (SN2 needs backside attack; primary halides are least hindered, tertiary halides are too hindered.)

✓ Primary alkyl halides

Secondary alkyl halides

Tertiary alkyl halides

Aryl halide

x) Primary alcohol is obtained by the reaction of Grignard's reagent with: (RMgX + HCHO, then hydrolysis, gives R-CH2OH, a primary alcohol; with other aldehydes it gives secondary alcohols, with ketones tertiary alcohols.)

Aldehydes other than formaldehyde

Acetone

✓ Formaldehyde

Ethyl formate

## Exercise - Short Answer Questions

Q.1. What is the difference between a base and nucleophile?

A base is a species that accepts a proton (Bronsted-Lowry) or donates an electron pair (Lewis) — the emphasis is on the proton/electron-pair equilibrium (thermodynamic, measured by Kb). A nucleophile is a species that donates an electron pair to an electron-deficient carbon to form a new bond — the emphasis is on the rate of attack at carbon (kinetic). All nucleophiles are Lewis bases, but a good nucleophile need not be a strong base, and a strong base need not be a good nucleophile (e.g. I- is a good nucleophile but a weak base).

Q.2. Can a good nucleophile be a strong base?

No, not necessarily. Nucleophilicity is a kinetic property (how fast a species attacks carbon) while basicity is a thermodynamic property (how strongly it holds a proton). Iodide is a good nucleophile (large, polarizable) but a weak base (its conjugate acid HI is strong). Conversely, a strong base like tert-butoxide is a poor nucleophile because it is bulky and hindered. So a good nucleophile can be a weak base, and vice versa.

Q.3. Is iodine more nucleophilic than chlorine?

Yes, iodine is more nucleophilic than chlorine. Nucleophilicity increases down a group in protic solvents because the larger iodide ion is more polarizable and its electron cloud is more easily distorted to form the new bond, and it is less strongly solvated (less hindered by hydrogen bonding) than the smaller chloride ion. The order is I- > Br- > Cl- > F- in protic solvents.

Q.4. What is a poor leaving group?

A poor leaving group is a group that does not depart readily with the bonding electron pair, usually because it is a strong base that holds its electrons tightly (e.g. OH-, NH2-, OR-, H-). Such groups cannot be displaced easily in substitution reactions unless they are first converted into a good leaving group (e.g. -OH is protonated to -OH2+, which leaves as water).

Q.5. The halogens are good leaving groups why?

Halogens are good leaving groups because the halide ions (Cl-, Br-, I-) are weak bases: they are stable on their own after departure (their conjugate acids HX are strong acids), and the C-X bond is polarized and relatively weak, so it breaks easily. Iodide is the best leaving group among them, fluoride the poorest (F- is a strong base and the C-F bond is very strong).

Q.6. Why alkyl halides undergo nucleophilic substitution reaction?

Alkyl halides undergo nucleophilic substitution because the carbon-halogen bond is polarized (carbon is partially positive, halogen partially negative), so the carbon is electrophilic and attracts nucleophiles, and the halide ion is a good leaving group that departs with the bonding electrons. The nucleophile attacks the electrophilic carbon and replaces the halide, giving a substitution product.

Q.7. What is the carbocation intermediate?

A carbocation intermediate is a positively charged carbon species with only six valence electrons and an empty p-orbital, formed

when the C-X bond breaks heterolytically and the leaving group departs with both bonding electrons. In SN1 reactions the carbocation is formed in the slow, rate-determining step and is then attacked by the nucleophile. Its stability order is tertiary > secondary > primary > methyl, due to inductive effects and hyperconjugation.

Q.8. Do SN2 reactions have carbocation intermediates?

No, SN2 reactions do not have carbocation intermediates. SN2 is a single-step (concerted) reaction: the nucleophile attacks the carbon from the back side at the same time as the leaving group departs, passing through a single transition state in which the carbon is partially bonded to both the nucleophile and the leaving group. The configuration at carbon is inverted (Walden inversion). Carbocations occur only in SN1 reactions.

Q.9. Why is a tertiary carbocation more stable than a secondary?

A tertiary carbocation is more stable than a secondary carbocation because it has three alkyl groups attached to the positive carbon (instead of two), which donate electron density by the inductive effect and by hyperconjugation (overlap of neighbouring C-H sigma bonds with the empty p-orbital). More alkyl groups mean more hyperconjugative structures and greater dispersal of the positive charge, hence greater stability. The order is 3 degree > 2 degree > 1 degree > methyl.

Q.10. Why are organometallic compounds important?

Organometallic compounds are important because they contain a

polar metal-carbon bond in which carbon is nucleophilic (carbanion-like) and can form new carbon-carbon bonds — the most important bond-forming tool in organic synthesis. Grignard reagents (RMgX) are used to prepare alkanes, alkenes, alcohols (primary, secondary, tertiary), aldehydes, ketones and carboxylic acids from simple starting materials, making them among the most versatile reagents in organic chemistry.

Q.11. What is the general formula of Grignard reagent? Give two examples of Grignard's reagents.

The general formula of a Grignard reagent is RMgX, where R is an alkyl or aryl group and X is a halogen (usually Br or I), e.g. CH3MgI (methylmagnesium iodide) and C2H5MgBr (ethylmagnesium bromide). It is prepared by reacting an alkyl (or aryl) halide with magnesium metal in dry ether: RX + Mg -> RMgX.

Q.12. Write the general equation for the preparation of Grignard reagents.

General equation for the preparation of Grignard reagents: RX + Mg -> RMgX (in dry ether), e.g. CH3I + Mg -> CH3MgI; C2H5Br + Mg -> C2H5MgBr. The reaction is carried out in anhydrous ether because the reagent reacts violently with water and any compound containing an acidic hydrogen.

Q.13. What type of solvents can be used in Grignard reactions?

Grignard reactions use dry, aprotic, non-acidic solvents — dry ether (diethyl ether) or tetrahydrofuran (THF). The solvent must be free of water and of any compound with an acidic hydrogen (-OH, -NH,

-COOH, -SH), because the Grignard reagent would react with it at once and be destroyed.

Q.14. Why is ether a good solvent for Grignard reactions?

Ether is a good solvent for Grignard reactions because it is aprotic (has no acidic hydrogen to destroy the reagent), it dissolves both the alkyl halide and the reagent, and its oxygen lone pairs coordinate with (solvate) the magnesium atom, stabilising the reagent. It is also volatile and easily removed after the reaction.

Q.15. Why is a Grignard reagent so reactive?

A Grignard reagent is so reactive because the carbon-magnesium bond is strongly polarized (C delta-, Mg delta+): carbon is much more electronegative than magnesium, so the carbon carries a partial negative charge and behaves like a carbanion. This nucleophilic carbon attacks electrophiles (carbonyl carbons, water, CO2) vigorously, and it also reacts with any acidic hydrogen, which is why the reagent must be kept dry.

Q.16. Is the carbon of Grignard's reagent electrophilic or nucleophilic?

The carbon of a Grignard reagent is nucleophilic, not electrophilic: because magnesium is electropositive, the C-Mg bond is polarized with the negative end on carbon, so the carbon has carbanion character and donates its electron pair to electrophiles (e.g. the carbonyl carbon of aldehydes and ketones, or CO2).

Q.17. Define secondary amine and give two examples.

A secondary amine is an amine in which two hydrogen atoms of ammonia are replaced by alkyl or aryl groups, so the nitrogen is bonded to two carbons and one hydrogen: general formula R2NH. Examples: dimethylamine, (CH3)2NH; diethylamine, (C2H5)2NH; N-methylethanamine, CH3NHC2H5.

Q.18. Why do amines act as bases?

Amines act as bases because the nitrogen atom carries a lone pair of electrons that can accept a proton (Bronsted-Lowry base) or be donated to an electron-deficient species (Lewis base): R-NH2 + H+ -> R-NH3+ (an alkylammonium ion). Amines react with acids to form salts (e.g. CH3NH2 + HCl -> CH3NH3+ Cl-, methylammonium chloride).

Q.19. Why aliphatic amines are more basic while aromatic amines are less basic than ammonia?

Aliphatic amines are more basic than ammonia because the alkyl groups donate electrons by the inductive effect (+I), increasing the electron density on nitrogen and making the lone pair more available for protonation. Aromatic amines (e.g. aniline) are less basic than ammonia because the nitrogen lone pair is delocalised into the benzene ring by resonance, so it is less available to accept a proton; the anilinium ion is also less stabilised than the ammonium ion.

## Exercise - Long Answer Questions

Q.1. What are alkyl halides? Explain both the systems of nomenclature of alkyl halides.

Alkyl halides are compounds in which one or more hydrogen atoms of an alkane are replaced by halogen atoms (F, Cl, Br, I): general formula CnH2n+1X. Two systems of nomenclature: (1) Common system — the alkyl group is named first, followed by the halide (e.g. methyl chloride, ethyl bromide, isopropyl chloride, tert-butyl bromide). (2) IUPAC system — the halogen is named as a prefix (fluoro, chloro, bromo, iodo) with its position number, attached to the parent alkane name (e.g. chloromethane, bromoethane, 2-chloropropane, 2-bromo-2-methylpropane). If several halogens are present, they are listed alphabetically with di, tri, tetra as needed.

Q.2. Draw structures for the following compounds. i) Bromoethane ii) Dichlorodibromomethane iii) 2-bromo-3-chloropentane iv) 1,1,2,2-tetrachloro-3,4-difluorohexane v) 3-bromo-1-pentene (Structures are given as condensed formulas, since drawings cannot be printed as text.)

Structures (given as condensed formulas, since drawings cannot be printed as text): i) Bromoethane: CH3-CH2-Br. ii) Dichlorodibromomethane: CCl2Br2 (carbon with two chlorines and two bromines). iii) 2-bromo-3-chloropentane: CH3-CHBr-CHCl-CH2-CH3. iv) 1,1,2,2-tetrachloro-3,4-difluorohexane: CHCl2-CCl2-CHF-CHF-CH2-CH3. v) 3-bromo-1-pentene: CH2=CH-CHBr-CH2-CH3.

Q.3. Give common and IUPAC names to the following compounds: i) CHI3 ii) CHCl3 iii) CH3I iv) (CH3)2CHCl v) (CH3)3CBr

Common and IUPAC names: i) CHI3 — common: Iodoform; IUPAC: Triiodomethane. ii) CHCl3 — common: Chloroform; IUPAC: Trichloromethane. iii) CH3I — common: Methyl iodide; IUPAC: Iodomethane. iv) (CH3)2CHCl — common: Isopropyl chloride; IUPAC: 2-Chloropropane. v) (CH3)3CBr — common: tert-Butyl bromide; IUPAC: 2-Bromo-2-methylpropane.

Q.4. Explain the structure and reactivity of alkyl halides.

Structure and reactivity of alkyl halides: the C-X bond is polarized (C delta+, X delta-) because halogens are more electronegative than carbon; the carbon is sp3 hybridized and tetrahedral. Reactivity: the polarized bond makes carbon electrophilic, so nucleophiles attack it; the halide ion is a good leaving group, so substitution (SN1, SN2) and elimination (E1, E2) reactions occur readily. Reactivity order: RI > RBr > RCl > RF (weaker C-X bond and better leaving group for larger halides), and tertiary > secondary > primary for SN1, but primary > secondary > tertiary for SN2 (steric hindrance).

Q.5. Write down three different methods for the preparation of alkyl halides. Which one is the best method for the preparation of alkyl halides?

Three different methods for the preparation of alkyl halides: (1) From alcohols: R-OH + HX -> R-X + H2O (e.g. ethanol + HBr -> bromoethane), or with PCl5/SOCl2 (SOCl2 gives the purest product, since the by-products are gases). (2) From alkanes: free-radical halogenation in UV light (e.g. CH4 + Cl2 -> CH3Cl + HCl), though mixtures are formed. (3) From alkenes: electrophilic addition of HX

or X2 (e.g. ethene + HBr -> bromoethane; ethene + Br2 -> 1,2-dibromoethane). The best method is from alcohols with SOCl2 (or HX), because it gives a single, pure product in good yield without rearrangement or mixture of products.

Q.6. Write short note on the following: i) Carbocations and their stability ii) Nucleophile and base iii) Substrate and leaving group
Short notes: i) Carbocations and their stability — a carbocation is a carbon with six valence electrons and a positive charge (empty p-orbital); stability order is tertiary > secondary > primary > methyl, due to inductive donation and hyperconjugation from neighbouring alkyl groups; resonance (e.g. benzyl, allyl) increases stability further. ii) Nucleophile and base — a nucleophile donates an electron pair to carbon (kinetic, rate of attack); a base accepts a proton or donates a pair to H+ (thermodynamic, measured by Kb); all nucleophiles are Lewis bases, but nucleophilicity and basicity do not always run together (I- is a good nucleophile but a weak base). iii) Substrate and leaving group — the substrate is the molecule attacked by the nucleophile (usually the alkyl halide); the leaving group is the group that departs with the bonding electron pair (e.g. X-), and a good leaving group is a weak base.

Q.7. What is the difference between SN1 and SN2 mechanisms?
Difference between SN1 and SN2 mechanisms: SN1 is a two-step, first-order reaction (rate depends only on the substrate) proceeding through a carbocation intermediate; it gives racemization (for chiral substrates) and is favoured by tertiary halides, polar protic solvents and weak nucleophiles. SN2 is a single-step, second-order reaction

(rate depends on substrate and nucleophile) with no intermediate, passing through a single transition state with backside attack; it gives complete inversion of configuration (Walden inversion) and is favoured by primary halides, polar aprotic solvents and strong nucleophiles.

Q.8. What are nucleophilic substitution reactions? Explain in detail the mechanism of nucleophilic substitution (SN) reactions.

Nucleophilic substitution reactions are reactions in which a nucleophile replaces a leaving group (usually a halide) on an electrophilic carbon. Mechanism in detail: SN1 — the C-X bond breaks first (slow, rate-determining ionization) to give a planar carbocation, which is then attacked by the nucleophile from either face, giving a racemic mixture (if chiral). SN2 — the nucleophile attacks from the back side while the leaving group departs, in one concerted step through a pentacoordinate transition state, giving inversion of configuration. The choice depends on the substrate (3 degree favours SN1, 1 degree favours SN2), the nucleophile, the leaving group and the solvent.

Q.9. What are beta-elimination reactions? Explain the types and mechanism of elimination reactions.

Beta-elimination reactions are reactions in which a hydrogen atom is removed from the beta carbon (the carbon next to the carbon bearing the leaving group) together with the leaving group, forming a double bond between the alpha and beta carbons. Types and mechanism: E1 — two-step, through a carbocation intermediate (like SN1), first-order, favoured by tertiary halides and weak bases.

E2 — single-step, concerted removal of H+ by a strong base and departure of the leaving group, second-order, favoured by strong bases and requiring anti-periplanar geometry. Elimination competes with substitution: strong, bulky bases and heat favour elimination; Saytzeff's rule says the more substituted (more stable) alkene is the major product.

Q.10. Discuss the factors that influence nucleophilic substitution and elimination reactions of alkyl halides.

Factors that influence nucleophilic substitution and elimination reactions of alkyl halides: (1) Structure of the substrate — tertiary halides favour SN1/E1 (stable carbocations), primary halides favour SN2, secondary halides give mixtures. (2) Strength and size of the nucleophile/base — strong nucleophiles favour SN2, strong bulky bases favour elimination. (3) Leaving group — better leaving groups (weaker bases, e.g. I-) speed both reactions. (4) Solvent — polar protic solvents favour SN1/E1 (stabilise ions), polar aprotic solvents favour SN2. (5) Temperature — higher temperature favours elimination (entropy increases). (6) Concentration — high nucleophile/base concentration favours second-order SN2/E2.

Q.11. What are organometallic compounds? Discuss preparation and reactivity of Grignard's reagents.

Organometallic compounds are compounds containing a direct metal-carbon bond (e.g. RMgX, RLi, Zn(C2H5)2). Preparation of Grignard reagents: RX + Mg -> RMgX in dry ether (or THF), with the reaction started by a crystal of iodine or gentle warming; the ether solvates and stabilises the reagent. Reactivity: the C-Mg bond is

polarized (C delta-, Mg delta+), so carbon is strongly nucleophilic and basic — it reacts with water and any acidic hydrogen (destroying the reagent), with CO2 to give carboxylic acids, with aldehydes/ketones to give alcohols, and with alkyl halides to give higher alkanes.

Q.12. Explain important reactions of Grignard's reagent.

Important reactions of Grignard's reagent: (1) With water or any acidic hydrogen: RMgX + H2O -> RH + Mg(OH)X (the reagent is destroyed, giving the alkane). (2) With formaldehyde: gives a primary alcohol (RMgX + HCHO -> RCH2OH after hydrolysis). (3) With other aldehydes: gives secondary alcohols. (4) With ketones: gives tertiary alcohols. (5) With carbon dioxide: gives a carboxylic acid (RMgX + CO2 -> RCOOH after hydrolysis). (6) With alkyl halides: gives higher alkanes (coupling). (7) With esters: gives tertiary alcohols (two molecules of reagent add).

Q.13. What are amines? Explain nomenclature of amines.

Amines are derivatives of ammonia in which one or more hydrogen atoms are replaced by alkyl or aryl groups. Nomenclature: (1) Common system — the alkyl groups are named followed by 'amine' (methylamine, ethylamine, dimethylamine, trimethylamine, aniline for phenylamine). (2) IUPAC system — the longest chain containing the -NH2 group is the parent with the suffix '-amine' and the position number of the amino group (methanamine, ethanamine, propan-2-amine); substituents on nitrogen are shown by N- (N-methylmethanamine for dimethylamine; N,N-dimethylmethanamine for trimethylamine;

N-ethyl-3-hexanamine). Primary, secondary and tertiary amines are distinguished by the number of carbons on nitrogen.

Q.14. Draw structures for the following compounds. i) 2-Methyl-2-propanamine ii) 3-Methyl-2-hexanamine iii) N-Methylethanamine iv) N-Ethyl-3-hexanamine v) N-Ethyl-N-methyl-2-butanamine (Structures are given as condensed formulas, since drawings cannot be printed as text.)
Structures (given as condensed formulas, since drawings cannot be printed as text): i) 2-Methyl-2-propanamine: (CH3)3C-NH2. ii) 3-Methyl-2-hexanamine: CH3-CH(NH2)-CH(CH3)-CH2-CH2-CH3. iii) N-Methylethanamine: CH3-NH-CH2-CH3. iv) N-Ethyl-3-hexanamine: CH3-CH2-CH2-CH(NH-C2H5)-CH2-CH3. v) N-Ethyl-N-methyl-2-butanamine: CH3-CH(N(CH3)(C2H5))-CH2-CH3.

Q.15. Give common and IUPAC names to the following compounds: i) CH3NH2 ii) CH3CH2NH2 iii) (CH3)2NH iv) (CH3)3N v) (CH3)2NCH2CH3
Common and IUPAC names: i) CH3NH2 — common: Methylamine; IUPAC: Methanamine. ii) CH3CH2NH2 — common: Ethylamine; IUPAC: Ethanamine. iii) (CH3)2NH — common: Dimethylamine; IUPAC: N-Methylmethanamine. iv) (CH3)3N — common: Trimethylamine; IUPAC: N,N-Dimethylmethanamine. v) (CH3)2NCH2CH3 — common: Dimethylethylamine; IUPAC: N,N-Dimethylethanamine.

Q.16. Discuss the structure, basicity and reactivity of amines.
Structure, basicity and reactivity of amines: structure — nitrogen is

sp3 hybridized, pyramidal, with a lone pair (bond angle about 107 degrees); amines are classified primary (RNH2), secondary (R2NH) and tertiary (R3N). Basicity — amines are basic because of the lone pair; aliphatic amines are stronger bases than ammonia (+I effect of alkyl groups), aromatic amines are weaker (lone pair delocalised into the ring); basicity order in water is usually secondary > primary > tertiary > ammonia (solvation effects). Reactivity — amines act as nucleophiles (alkylation with alkyl halides, acylation with acid chlorides/anhydrides) and as bases (salt formation with acids); primary aromatic amines give diazonium salts with nitrous acid at 0 C.

Q.17. Write down methods for the preparation of amines.

Methods for the preparation of amines: (1) Ammonolysis of alkyl halides: R-X + NH3 -> R-NH2 (with excess ammonia to avoid further alkylation; mixtures of primary, secondary and tertiary amines and quaternary salts are formed). (2) Reduction of nitro compounds: Ar-NO2 + Sn/HCl (or H2/Ni) -> Ar-NH2 (e.g. nitrobenzene -> aniline). (3) Reduction of nitriles: R-CN + LiAlH4 (or H2/Ni) -> R-CH2-NH2. (4) Reduction of amides: R-CONH2 + LiAlH4 -> R-CH2-NH2 (Hoffmann's method is the reverse: amide + Br2/NaOH gives an amine with one carbon less). (5) Gabriel phthalimide synthesis: gives pure primary amines.

Q.18. Describe physical and chemical properties of amines.

Physical properties of amines: lower amines (methylamine, ethylamine) are gases with a fishy smell; higher amines are liquids or solids; they are polar and form hydrogen bonds, so primary and

secondary amines have higher boiling points than comparable alkanes (tertiary amines cannot hydrogen-bond with themselves and boil lower); lower amines are soluble in water (hydrogen bonding with water). Chemical properties: basicity (salt formation with acids); nucleophilic substitution (alkylation with alkyl halides to give secondary/tertiary amines and quaternary ammonium salts); acylation (with acid chlorides/anhydrides to give amides); reaction with nitrous acid (primary aliphatic amines give alcohols with $N_2$ gas; primary aromatic amines give diazonium salts at 0 C); carbylamine reaction (primary amines + $CHCl_3$ + KOH give foul-smelling isocyanides).

# Chapter 18: Alcohols, Phenols and Ethers

## Exercise - Multiple Choice Questions

i) The hydrogen bonding is present in: (Ethanol has an -OH group with H bonded to oxygen, so it forms hydrogen bonds; benzene, bromoethane and dimethyl ether have no O-H or N-H hydrogen.)

- C6H6
- ✓ C2H5OH
- C2H5Br
- CH3-O-CH3

ii) Which compound is more soluble in water? (Ethyl alcohol is completely miscible with water; phenol is only slightly soluble, dimethyl ether slightly soluble, and 1-hexanol poorly soluble because of its long chain.)

- Phenol
- ✓ Ethyl alcohol
- Dimethyl ether
- 1-Hexanol

iii) Oxidation of ethyl alcohol gives: (Controlled oxidation of ethanol first gives ethanal (acetaldehyde); further oxidation gives acetic acid.)

- Formaldehyde
- ✓ Acetaldehyde
- Acetone

Ether

iv) Methanol is also known as: (Methanol was first obtained by destructive distillation of wood, hence wood spirit; absolute alcohol is water-free ethanol; rectified spirit is 95% ethanol.)

Absolute alcohol

Rectified alcohol

✓ Wood spirit

Methylated alcohol

v) Rectified spirit contains about: (Rectified spirit is the constant-boiling mixture, 95.6% ethanol and 4.4% water.)

85% ethyl alcohol

90% ethyl alcohol

✓ 95% ethyl alcohol

100% ethyl alcohol

vi) Alcohols are prepared by the: (Acid-catalysed hydration of alkenes gives alcohols, e.g. ethene + H2O -> ethanol.)

✓ Hydration of alkenes.

Hydrogenation of alkenes.

Halogenation of alkenes.

Oxidation of alkenes.

vii) The protonated ether is called: (R2O + H+ -> R2OH+, an oxonium ion.)

Oxide ion

Carbon ion

Hydronium ion

✓ Oxonium ion

viii) Which statement is true about phenol? (Phenol is more acidic than alcohols because the phenoxide ion is resonance-stabilised, but less acidic than carboxylic acids; it turns blue litmus red and reacts with sodium.)

It is more acidic than acetic acid

✓ It is more acidic than ethyl alcohol.

It turns red litmus blue.

It does not react with sodium.

ix) Which one of the following is the isomer of dimethyl ether? (Dimethyl ether is C2H6O; ethanol CH3CH2OH has the same formula — functional group isomers.)

✓ CH3CH2OH

CH3CHO

CH3COOH

HCOOCH3

x) Which statement is correct about ether? (Ethers are very volatile and highly inflammable — diethyl ether boils at 35 C and catches fire easily; they are less dense than water and are liquids or gases, not crystalline solids.)

Lower ethers are liquids at room temperature.

They are crystalline solids at room temperature.

✓ They are highly inflammable.

They are denser than water.

## Exercise - Short Answer Questions

### Q.1. Why phenols are more acidic than alcohols?

Phenols are more acidic than alcohols because the phenoxide ion left after losing H+ is stabilised by resonance: the negative charge is delocalised into the benzene ring (over the ortho and para positions). In alcohols, the alkoxide ion (RO-) has the charge localised on oxygen with no resonance stabilisation. The resonance-stabilised phenoxide makes phenol lose its proton more readily (phenol Ka about 10^-10, ethanol about 10^-16).

### Q.2. Why the boiling point of alcohol is higher than ether?

The boiling point of an alcohol is higher than that of the isomeric ether because alcohol molecules form intermolecular hydrogen bonds (O-H...O) which need a great deal of energy to break. Ethers have no O-H hydrogen, so their molecules are held only by weak dipole-dipole and van der Waals forces, giving much lower boiling points (e.g. ethanol boils at 78 C, dimethyl ether at -24 C).

### Q.3. Can ether form a hydrogen bond with water?

Yes, an ether can form a hydrogen bond with water: the ether oxygen has lone pairs that accept hydrogen bonds from water's O-H hydrogens (R-O-R...H-OH). But ether molecules cannot hydrogen-bond with each other, since they have no hydrogen bonded to oxygen. This is why lower ethers are slightly soluble in water but have low boiling points.

### Q.4. Why ethers are considered to be good solvents?

Ethers are considered good solvents because they are chemically rather unreactive (they do not react with most reagents), they dissolve a wide range of organic compounds (fats, oils, resins, alkaloids), they are volatile and easily removed by evaporation, and they are miscible with many organic liquids. Diethyl ether is the classic laboratory solvent for Grignard reagents and extractions.

Q.5. Ethers are less dense than liquid water, how?

Ethers are less dense than liquid water because their molecules cannot pack as closely: they have no hydrogen bonding between their own molecules, and the alkyl groups take up space, so the mass per unit volume is lower (diethyl ether density about 0.71 g/mL, water 1.00 g/mL). Hence ether floats on water.

Q.6. Phenols are more reactive than benzene, why?

Phenols are more reactive than benzene because the -OH group donates its oxygen lone pair into the ring by resonance (+M effect), greatly increasing the electron density at the ortho and para positions. Electrophilic substitution therefore occurs much faster and under milder conditions (e.g. phenol gives tribromophenol with bromine water at once, without a catalyst, while benzene needs a Lewis acid catalyst).

Q.7. Explain why the boiling points of alcohols and phenols are much higher than those of alkanes and ethers?

Alcohols and phenols have much higher boiling points than alkanes and ethers of similar molecular mass because their molecules are associated by intermolecular hydrogen bonding (O-H...O), which

needs much energy to overcome. Alkanes have only weak van der Waals forces, and ethers cannot hydrogen-bond among themselves, so both boil much lower (e.g. ethanol 78 C vs propane -42 C vs dimethyl ether -24 C).

Q.8. Why phenols have higher boiling points than alcohols?

Phenols have higher boiling points than alcohols of similar size because, in addition to hydrogen bonding, phenol molecules have larger molecular masses and stronger van der Waals forces (the benzene ring), and the -OH group is attached to the rigid planar ring, allowing closer packing and stronger intermolecular attraction. (Phenol boils at 182 C, while ethanol boils at 78 C.)

Q.9. Which are more soluble in water, alcohols or phenols?

Alcohols are more soluble in water than phenols. Lower alcohols (methanol, ethanol) are completely miscible with water because their small -OH group hydrogen-bonds freely with water. Phenol is only slightly soluble (about 8 g per 100 mL) because its large hydrophobic benzene ring resists water, and phenol molecules prefer to associate with each other. As the alkyl chain grows, alcohol solubility also falls (1-hexanol is poorly soluble).

Q.10. What is the difference between primary and secondary alcohols?

In a primary alcohol the carbon bearing the -OH group is attached to only one other carbon (e.g. ethanol, CH3CH2OH; methanol is also counted primary). In a secondary alcohol the -OH carbon is attached to two other carbons (e.g. propan-2-ol, CH3CH(OH)CH3).

They differ in oxidation: primary alcohols oxidize first to aldehydes then to acids; secondary alcohols oxidize to ketones; and they differ in the Lucas test (secondary alcohols react with Lucas reagent in about 5 minutes, primary alcohols only after long standing/heating).

Q.11. What do you know about rectified and absolute alcohols?

Rectified alcohol (rectified spirit) is 95.6% ethanol and 4.4% water, the constant-boiling (azeotropic) mixture obtained by ordinary distillation of fermented liquors; it cannot be made stronger by distillation alone. Absolute alcohol is water-free (100%) ethanol, obtained by removing the last water chemically (e.g. distilling rectified spirit over quicklime or with benzene) — it is used where water must be absent, as in Grignard reactions and medicine.

## Exercise - Long Answer Questions

Q.1. What are alcohols? How are they classified?

Alcohols are organic compounds containing one or more hydroxyl (-OH) groups attached to a saturated carbon atom: general formula CnH2n+1OH (for monohydric alcohols). Classification: (1) By the number of -OH groups — monohydric (one -OH, e.g. ethanol), dihydric (two -OH, e.g. glycol), trihydric (three -OH, e.g. glycerol), polyhydric. (2) By the type of carbon bearing -OH — primary (carbon attached to one other carbon), secondary (two), tertiary (three). (3) By the group attached — aliphatic alcohols (alkyl) and aromatic alcohols (aryl, e.g. benzyl alcohol; phenols, where -OH is directly on the ring, are a separate class).

Q.2. How will you distinguish between primary, secondary and tertiary alcohols?

Primary, secondary and tertiary alcohols are distinguished by: (1) Oxidation — primary alcohols oxidize to aldehydes and then acids; secondary to ketones; tertiary resist oxidation (they do not oxidize under mild conditions). (2) Lucas test (conc. HCl + $ZnCl_2$) — tertiary alcohols give cloudiness at once, secondary in about 5 minutes, primary only on standing/heating. (3) Victor Meyer's test — primary gives a blood-red colour, secondary a blue colour, tertiary no colour (with nitrolic acid reactions). (4) Dehydrogenation over copper at 300 C — primary gives aldehydes, secondary gives ketones, tertiary gives alkenes.

Q.3. Describe the nomenclature of alcohols.

Nomenclature of alcohols: (1) Common system — the alkyl group is named followed by 'alcohol' (methyl alcohol, ethyl alcohol, isopropyl alcohol, tert-butyl alcohol). (2) IUPAC system — the longest chain containing the -OH group is the parent, the suffix is '-ol', and the position of -OH is shown by the lowest possible number (methanol, ethanol, propan-1-ol, propan-2-ol, butan-2-ol). For two -OH groups the suffix is '-diol' (propane-1,2-diol), for three '-triol' (propane-1,2,3-triol, glycerol). Substituents are named as prefixes with numbers (3-chloro-1,2-propanediol).

Q.4. Draw structures for the following alcohols: i) 1-propanol ii) 2-butanol iii) 1,3-butanediol iv) 5-methyl-2-hexanol v) 4-ethyl-3-heptanol vi) 2,2,4,4-tetramethyl-3-pentanol vii) 3-chloro-1,2-propanediol viii) 2,3,4-pentanetriol (Structures are

given as condensed formulas, since drawings cannot be printed as text.)

Structures (given as condensed formulas, since drawings cannot be printed as text): i) 1-propanol: CH3-CH2-CH2-OH. ii) 2-butanol: CH3-CH(OH)-CH2-CH3. iii) 1,3-butanediol: HO-CH2-CH2-CH(OH)-CH3. iv) 5-methyl-2-hexanol: (CH3)2CH-CH2-CH2-CH(OH)-CH3. v) 4-ethyl-3-heptanol: CH3-CH2-CH(OH)-CH(C2H5)-CH2-CH2-CH3. vi) 2,2,4,4-tetramethyl-3-pentanol: (CH3)3C-CH(OH)-C(CH3)3. vii) 3-chloro-1,2-propanediol: HO-CH2-CH(OH)-CH2-Cl. viii) 2,3,4-pentanetriol: CH3-CH(OH)-CH(OH)-CH(OH)-CH3.

Q.5. Draw structures for the following ethers: i) Methoxymethane ii) Phenoxybenzene iii) 2-methoxypropane iv) 1,2-diphenoxypropane v) 1-ethoxy-4-methylbenzene (Structures are given as condensed formulas, since drawings cannot be printed as text.)

Structures (given as condensed formulas, since drawings cannot be printed as text): i) Methoxymethane: CH3-O-CH3. ii) Phenoxybenzene: C6H5-O-C6H5 (diphenyl ether). iii) 2-methoxypropane: (CH3)2CH-O-CH3. iv) 1,2-diphenoxypropane: C6H5-O-CH2-CH(O-C6H5)-CH3. v) 1-ethoxy-4-methylbenzene: p-CH3-C6H4-O-C2H5.

Q.6. Draw structures and then give IUPAC names to the following compounds. i) Dimethyl ether ii) Ethylmethyl ether iii) Methylphenyl ether iv) Ethyl alcohol v) Sec. propyl alcohol vi) Propylene glycol

IUPAC names: i) Dimethyl ether — Methoxymethane. ii) Ethylmethyl ether — Methoxyethane. iii) Methylphenyl ether — Methoxybenzene

(anisole). iv) Ethyl alcohol — Ethanol. v) Sec. propyl alcohol — Propan-2-ol. vi) Propylene glycol — Propane-1,2-diol.

Q.7. Explain the structure and acidity of alcohols.

Structure and acidity of alcohols: structure — the C-O-H group is bent (C-O-H angle about 105 degrees), oxygen is sp3 hybridized with two lone pairs, and the O-H bond is polarized (O delta-, H delta+), allowing hydrogen bonding. Acidity — alcohols are very weak acids (Ka about 10^-16 to 10^-18): they lose the -OH proton to strong bases (2ROH + 2Na -> 2RONa + H2), but water is a stronger acid than alcohols. Acidity decreases with increasing alkyl size and branching (methyl > primary > secondary > tertiary) because alkyl groups donate electrons (+I) and destabilise the alkoxide ion.

Q.8. Describe the methods for the preparation of alcohols.

Methods for the preparation of alcohols: (1) Hydration of alkenes — acid-catalysed addition of water (ethene + H2O, H3PO4, gives ethanol; Markownikov addition for unsymmetrical alkenes). (2) Hydrolysis of alkyl halides — aqueous NaOH or KOH (R-X + OH- -> R-OH). (3) Reduction of aldehydes and ketones — LiAlH4 or NaBH4 (aldehydes give primary alcohols, ketones give secondary alcohols). (4) From Grignard reagents — RMgX + formaldehyde gives primary alcohols, + other aldehydes gives secondary, + ketones gives tertiary alcohols (after hydrolysis). (5) Fermentation of sugars — glucose -> ethanol + CO2 by yeast (gives rectified spirit).

Q.9. What do you know about reactivity of alcohols?

Reactivity of alcohols: the -OH group is the reactive centre. (1) Acidic behaviour — reaction with active metals (Na, K) gives alkoxides and H2. (2) Nucleophilic substitution — the -OH group is replaced by halogens (with HX, PCl5, SOCl2) giving alkyl halides. (3) Dehydration — heating with concentrated H2SO4 gives alkenes (ethanol at 170 C gives ethene) or ethers (at 140 C gives diethyl ether). (4) Oxidation — primary alcohols give aldehydes then acids, secondary give ketones, tertiary resist oxidation. (5) Esterification — with carboxylic acids (conc. H2SO4) giving esters. Reactivity order for substitution/dehydration: tertiary > secondary > primary.

Q.10. Give physical and chemical properties of alcohols.

Physical properties of alcohols: lower alcohols are colourless liquids with a characteristic smell, completely miscible with water (hydrogen bonding); solubility falls as the chain grows; boiling points are high for their mass (hydrogen bonding) and rise with size, falling with branching; they are less dense than water. Chemical properties: acidic character (reaction with Na); substitution of -OH by halogens (with HX, PCl3, SOCl2); dehydration to alkenes (conc. H2SO4, 170 C) or ethers (140 C); oxidation (primary -> aldehyde -> acid; secondary -> ketone); esterification with acids; combustion (burn with a blue flame to CO2 + H2O).

Q.11. What are thiols? Describe the nomenclature of thiols.

Thiols are sulphur analogues of alcohols, containing the -SH (mercapto or thiol) group attached to carbon: general formula R-SH. Nomenclature: (1) Common system — the alkyl group followed by

'mercaptan' (methyl mercaptan, ethyl mercaptan). (2) IUPAC system — the suffix is '-thiol' with the position number of -SH (methanethiol, ethane-1-thiol, propane-2-thiol). The -SH group takes priority similar to -OH in numbering.

Q.12. Give physical properties and explain structure and acidity of thiols.

Physical properties of thiols: they are colourless liquids (lower members gases) with a strong, unpleasant (garlic-like) smell; they have lower boiling points than the corresponding alcohols (S-H...S hydrogen bonding is very weak, since sulphur is larger and less electronegative) and are less soluble in water; they are slightly denser than the corresponding alcohols. Structure and acidity: the C-S-H group is bent (angle about 96 degrees); thiols are stronger acids than alcohols (Ka about 10^-7 to 10^-11, vs 10^-16 for alcohols) because the S-H bond is weaker and the larger thiolate ion (RS-) disperses the negative charge better; they react with NaOH to give thiolates and with heavy metal ions to give mercaptides.

Q.13. What are phenols? Explain nomenclature of phenols.

Phenols are compounds in which one or more hydroxyl groups are directly attached to a benzene ring (e.g. phenol C6H5OH, cresols, catechol, resorcinol). Nomenclature: the parent is phenol; substituents are named as prefixes with position numbers (2-methylphenol = o-cresol, 3-methylphenol = m-cresol, 4-methylphenol = p-cresol; 2,4,6-trinitrophenol = picric acid; benzene-1,2-diol = catechol; benzene-1,3-diol = resorcinol). If a

higher-priority group (e.g. -COOH) is present, the -OH is named as a prefix 'hydroxy' (e.g. 4-hydroxybenzoic acid).

Q.14. Describe structure and acidity of phenol.

Structure and acidity of phenol: structure — the -OH group is attached directly to an sp2 carbon of the benzene ring; the oxygen lone pair is conjugated with the ring (partial double-bond character in C-O), making the C-O bond shorter and stronger than in alcohols, and the molecule planar. Acidity — phenol (Ka about 10^-10) is about a million times more acidic than ethanol because the phenoxide ion is resonance-stabilised (negative charge delocalised to ortho and para ring positions), whereas the ethoxide ion has no such stabilisation. Electron-withdrawing groups (NO2) increase acidity further; electron-donating groups (CH3) decrease it.

Q.15. Write down four different methods for the preparation of phenols.

Four different methods for the preparation of phenols: (1) From benzene sulphonic acid — fusion with NaOH gives sodium phenoxide, acidified to phenol. (2) From chlorobenzene (Dow's process) — heating with NaOH under pressure gives sodium phenoxide, acidified to phenol. (3) From diazonium salts — hydrolysis of benzene diazonium chloride with warm water gives phenol (with N2). (4) From cumene (industrial) — oxidation of cumene (isopropylbenzene) to cumene hydroperoxide, then acid cleavage gives phenol and acetone. (5) From coal tar — phenol is also obtained by fractional distillation of coal tar (middle oil fraction).

Q.16. Explain reactivity of phenols.

Reactivity of phenols: (1) Acidic reactions — phenol reacts with Na (giving sodium phenoxide + H2) and with NaOH (giving sodium phenoxide), but not with NaHCO3 or Na2CO3 (weaker acid than carbonic acid). (2) Electrophilic substitution — the -OH group strongly activates the ring toward ortho/para substitution: bromination with bromine water gives 2,4,6-tribromophenol at once; nitration gives picric acid; sulphonation gives o/p-phenolsulphonic acids. (3) Reimer-Tiemann reaction — with CHCl3 + NaOH gives salicylaldehyde. (4) Kolbe's reaction — with CO2 + NaOH gives salicylic acid. (5) Reduction and oxidation — phenol is easily oxidized to quinones.

Q.17. Give physical and chemical properties of phenols.

Physical properties of phenols: phenol is a colourless crystalline solid (melting point 42 C) with a characteristic carbolic smell, turning pink on exposure to air (oxidation); it is slightly soluble in water but soluble in alcohol and ether; it has a high boiling point (182 C) due to hydrogen bonding. Chemical properties: acidic character (reaction with Na and NaOH, but not with carbonates); electrophilic substitution (rapid ortho/para bromination, nitration to picric acid, sulphonation); Reimer-Tiemann (salicylaldehyde) and Kolbe (salicylic acid) reactions; oxidation to quinones; condensation with formaldehyde to give bakelite (phenol-formaldehyde resin).

Q.18. What is the difference between alcohols and phenols?

Difference between alcohols and phenols: (1) Structure — in

alcohols -OH is attached to a saturated (sp3) carbon; in phenols it is attached directly to a benzene (sp2) carbon. (2) Acidity — phenols are much more acidic (Ka 10^-10) than alcohols (Ka 10^-16) because the phenoxide ion is resonance-stabilised. (3) Reaction with NaOH — phenols dissolve in NaOH (forming phenoxides); alcohols do not. (4) Reaction with FeCl3 — phenols give a violet/blue colour with neutral ferric chloride; alcohols do not. (5) Electrophilic substitution — phenols undergo rapid ring substitution (bromine water gives tribromophenol); alcohols do not show ring reactions. (6) C-O bond — in phenols it has partial double-bond character and is stronger; in alcohols it is a normal single bond.

### Q.19. What are ethers? What do you know about simple and mixed ethers?

Ethers are compounds in which an oxygen atom is connected by single bonds to two alkyl or aryl groups: general formula R-O-R'. Simple (symmetrical) ethers have the same group on both sides (e.g. dimethyl ether CH3-O-CH3, diethyl ether C2H5-O-C2H5). Mixed (unsymmetrical) ethers have two different groups (e.g. ethyl methyl ether CH3-O-C2H5, methyl phenyl ether C6H5-O-CH3). Ethers are formed by condensation of two molecules of alcohol (with water removed) or by Williamson's synthesis.

### Q.20. Describe nomenclature of ethers.

Nomenclature of ethers: (1) Common system — the two groups are named alphabetically followed by 'ether' (dimethyl ether, ethyl methyl ether, methyl phenyl ether, diphenyl ether). (2) IUPAC system — the smaller group with oxygen is named as an alkoxy

prefix (methoxy, ethoxy, phenoxy) attached to the longer parent chain (methoxymethane for dimethyl ether, methoxyethane for ethyl methyl ether, methoxybenzene for methyl phenyl ether, ethoxybenzene for ethyl phenyl ether). If both groups are equal, one is the prefix and the other the parent.

Q.21. Describe the preparation of ethers by (i) Williamson's Synthesis (ii) the reaction of alkyl halides with dry silver oxide.

Preparation of ethers: (i) Williamson's Synthesis — an alkyl halide reacts with a sodium alkoxide by SN2: R-X + NaOR' -> R-O-R' + NaX (e.g. C2H5Br + NaOCH3 -> C2H5-O-CH3 + NaBr). It works best with primary alkyl halides (tertiary halides give alkenes by elimination). (ii) Reaction of alkyl halides with dry silver oxide — 2R-X + Ag2O -> R-O-R + 2AgX (e.g. 2C2H5I + Ag2O -> C2H5-O-C2H5 + 2AgI); moist silver oxide instead gives alcohols. (iii) Dehydration of alcohols — heating excess alcohol with concentrated H2SO4 at 140 C gives ethers (2C2H5OH -> C2H5-O-C2H5 + H2O).

Q.22. Explain reactivity of ethers.

Reactivity of ethers: ethers are relatively unreactive (the C-O bond is strong and the alkoxy group is a poor leaving group), which is why they are good solvents. Reactions: (1) Protonation — with acids they form oxonium salts (R2O + H+ -> R2OH+). (2) Cleavage by halogen acids — concentrated HI or HBr cleaves ethers (C2H5-O-C2H5 + HI -> C2H5I + C2H5OH, then further to 2C2H5I + H2O). (3) Halogenation — chlorine in the dark gives substitution products. (4) Oxidation — ethers slowly oxidize in air to explosive peroxides (hence old ether bottles are dangerous). (5) Electrophilic

substitution in aryl ethers — the alkoxy group activates the ring toward ortho/para substitution.

Q.23. What are the physical and chemical properties of ethers?
Physical properties of ethers: lower ethers (dimethyl ether) are gases, the rest are colourless, pleasant-smelling, volatile liquids; they are highly inflammable; they are less dense than water and only slightly soluble in water (the oxygen can accept hydrogen bonds from water, but the alkyl groups resist); boiling points are much lower than the isomeric alcohols (no hydrogen bonding among ether molecules) and close to those of alkanes. Chemical properties: they are inert to most reagents (Na, dilute acids, bases, oxidizing and reducing agents under ordinary conditions); they form oxonium salts with acids, are cleaved by concentrated HI/HBr, form peroxides on standing in air, and undergo electrophilic substitution if aryl (anisole gives o/p products).

Q.24. Give IUPAC names to the structures given below: i) CH3-CH(OH)-CH2-CH3 ii) CH3-CH(OH)-CH(CH3)-CH3 iii) CH3-CH(OH)-CH(OH)-CH3 iv) CH3-CH(OH)-C(CH3)2-CH(CH3)-CH3 v) CH3-CH2-O-CH3 vi) CH3-CH2-O-C6H5 vii) CH3-CH(OC6H5)-CH(OCH3)-CH2-OC2H5
IUPAC names: i) CH3-CH(OH)-CH2-CH3 — Butan-2-ol. ii) CH3-CH(OH)-CH(CH3)-CH3 — 3-Methylbutan-2-ol. iii) CH3-CH(OH)-CH(OH)-CH3 — Butane-2,3-diol. iv) CH3-CH(OH)-C(CH3)2-CH(CH3)-CH3 — 3,3,4-Trimethylpentan-2-ol (pentane parent: C1 CH3, C2 CH(OH), C3 C(CH3)2, C4 CH(CH3), C5 CH3). v) CH3-CH2-O-CH3 — Methoxyethane. vi) CH3-CH2-O-C6H5

— Ethoxybenzene. vii) CH3-CH(OC6H5)-CH(OCH3)-CH2-OC2H5 — 1-Ethoxy-2-methoxy-3-phenoxybutane (numbered from the ethoxy end to give the lowest locants 1,2,3).

Q.25. What are locants, prefixes, parent compound and suffixes in 3-methyl-2-pentanol?

In 3-methyl-2-pentanol: locants are 2 and 3 (the numbers showing the positions of the substituent and the functional group). The prefix is 'methyl' (the substituent at position 3). The parent compound is pentane (the five-carbon chain). The suffix is '-ol' (showing the alcohol functional group), with its locant 2 (the carbon bearing the -OH). So the name is built as: prefix (3-methyl) + parent (pent) + suffix (-an-2-ol) = 3-methylpentan-2-ol.

Q.26. Describe isomerism in alcohols and phenols.

Isomerism in alcohols and phenols: (1) Chain isomerism — e.g. butan-1-ol and 2-methylpropan-1-ol (different carbon skeletons). (2) Position isomerism — e.g. butan-1-ol and butan-2-ol (different position of -OH). (3) Functional group isomerism — alcohols are isomeric with ethers of the same formula (e.g. ethanol C2H6O and dimethyl ether C2H6O; C3H8O gives propanols and methoxyethane). (4) Optical isomerism — e.g. butan-2-ol has a chiral centre and exists as (+) and (-) enantiomers. Phenols show position isomerism (o-, m-, p-cresols) and functional group isomerism with aromatic alcohols (phenol vs benzyl alcohol are not isomers — different formulas; cresols vs methyl phenyl ethers are functional group isomers, e.g. p-cresol and methyl phenyl ether derivatives).

Q.27. How many isomers are possible for C4H10O? Draw their structures.

Isomers possible for C4H10O: 7 isomers in total. Alcohols (4): butan-1-ol (CH3CH2CH2CH2OH), butan-2-ol (CH3CH(OH)CH2CH3), 2-methylpropan-1-ol ((CH3)2CHCH2OH), 2-methylpropan-2-ol ((CH3)3COH). Ethers (3): methoxypropane (CH3-O-CH2CH2CH3), ethoxyethane (CH3CH2-O-CH2CH3), 2-methoxypropane (CH3-O-CH(CH3)2). (Structures are given as condensed formulas, since drawings cannot be printed as text.)

# Chapter 19: Aldehydes and Ketones

## Exercise - Multiple Choice Questions

i) The carbon atom of the carbonyl group in aldehydes and ketones is: (Three sigma bonds in a plane, 120 degrees, one pi bond — trigonal planar sp2.)

- sp hybridized
- ✓ $sp^2$ hybridized
- $sp^3$ hybridized
- $dsp^2$ hybridized

ii) The general formula of aldehyde is: (RCHO; RCOR is a ketone, RCOOH a carboxylic acid.)

- ✓ RCHO
- RCOH
- RCOR
- RCOOH

iii) Which one of the following is the example of mild oxidizing agents? (Silver oxide, as in Tollens' reagent, is a mild oxidizer that oxidizes aldehydes but not ketones; KMnO4, K2Cr2O7 and HNO3 are strong oxidizers.)

- Potassium permanganate
- Potassium dichromate
- ✓ Silver oxide
- Nitric acid

iv) Which one of the following has an aldehyde group? (Vanillin has a -CHO group; R-carvone, testosterone and progesterone are ketones.)

✓ Vanillin

R-carvone

Testosterone

Progesterone

v) Which one of the following compounds show acidic properties? (Alcohols react with sodium to give alkoxides, showing acidic character; aldehydes, ketones and alkenes do not.)

Aldehyde

Ketone

✓ Alcohol

Alkene

vi) Aldehydes are reduced to: (RCHO + [H] -> RCH2OH, a primary alcohol.)

✓ Primary alcohols

Secondary alcohols

Tertiary alcohols

Dihydric alcohols

vii) Ketones are reduced to: (RCOR' + [H] -> RCH(OH)R', a secondary alcohol.)

Primary alcohols

✓ Secondary alcohols

Tertiary alcohols

Dihydric alcohols

viii) Aldehydes are obtained by the oxidation of: (Primary alcohols oxidize first to aldehydes, then to acids; secondary alcohols give ketones.)

✓ Primary alcohols

Secondary alcohols

Tertiary alcohols

Dihydric alcohols

ix) Identify that reagent which reacts with both aldehydes and ketones. (Grignard's reagent adds to both aldehydes and ketones; Tollens' reagent reacts only with aldehydes. NOTE — FLAGGED as ambiguous as printed: hydrogen cyanide also adds to both aldehydes and ketones to give cyanohydrins, so option (c) is also chemically correct; the Grignard answer is used as the intended one.)

Water

✓ Grignard's reagent

Hydrogen cyanide

Tollen's reagent

x) Which one of the following reagents can oxidize ketones? (Acidified potassium dichromate is a strong oxidizer and can oxidize ketones (by cleavage); Tollens', Fehling's and Benedict's reagents are mild oxidizers that react only with aldehydes.)

Tollen's reagent

- Potassium dichromate
- Fehling' solution
- Benedict's solution

## Exercise - Short Answer Questions

### Q.1. Why ketones are less reactive toward nucleophilic attack as compared to aldehydes?

Ketones are less reactive toward nucleophilic attack than aldehydes for two reasons: (1) Steric — the carbonyl carbon of a ketone is bonded to two alkyl groups, which hinder the approach of the nucleophile, while an aldehyde has only one alkyl group and a small hydrogen. (2) Electronic — the two alkyl groups donate electrons by the inductive effect, reducing the partial positive charge on the carbonyl carbon; an aldehyde has only one donating group, so its carbonyl carbon is more positive and more attractive to nucleophiles.

### Q.2. What is the similarity between aldehydes and ketones?

The similarity between aldehydes and ketones is that both contain the carbonyl group (C=O) as their functional group. Both are polar compounds, both undergo nucleophilic addition reactions at the carbonyl carbon (with H, C, N and O nucleophiles), both are reduced to alcohols, and both form similar addition products (cyanohydrins, hemiacetals, acetals).

### Q.3. What is the difference between an aldehyde and a ketone?

Difference between an aldehyde and a ketone: an aldehyde has the carbonyl carbon bonded to at least one hydrogen (R-CHO;

formaldehyde has two hydrogens), while a ketone has the carbonyl carbon bonded to two carbon groups (R-CO-R'). Because of this, aldehydes are easily oxidized to carboxylic acids and give positive Tollens' and Fehling's tests, while ketones resist mild oxidation and give negative tests; aldehydes are also more reactive toward nucleophilic addition.

Q.4. Why aldehydes and ketones have higher boiling points than those of ethers and alkanes?

Aldehydes and ketones have higher boiling points than ethers and alkanes of similar molecular mass because their molecules are polar (the C=O bond has a partial positive carbon and partial negative oxygen), so they attract each other by dipole-dipole interactions. Alkanes have only weak van der Waals forces, and ethers are less polar, so both boil lower (e.g. propanal boils at 49 C, butane at -0.5 C).

Q.5. Aldehydes have lower boiling points than ketones, why?

Aldehydes have lower boiling points than ketones of similar molecular mass because ketone molecules are slightly more polar and have a larger surface area for dipole-dipole and van der Waals attractions (two alkyl groups instead of one alkyl group and a hydrogen), so more energy is needed to separate ketone molecules. (For example, propanal boils at 49 C while propanone boils at 56 C.)

Q.6. Which compound has hydrogen bonding, an aldehyde or ketone?

Neither aldehydes nor ketones form hydrogen bonds among themselves, because they have no hydrogen bonded to oxygen — the carbonyl oxygen has no attached hydrogen to donate. However, both can accept hydrogen bonds from water (the carbonyl oxygen's lone pairs attract water's O-H hydrogens), which is why lower aldehydes and ketones are soluble in water. So the question has no 'one of them' answer: neither self-hydrogen-bonds, but both hydrogen-bond with water.

Q.7. What is the product of the following reaction? CH3-CHO + [O] (K2Cr2O7/H2SO4) -> ?

The product is acetic acid (ethanoic acid): CH3-CHO + [O] (K2Cr2O7/H2SO4) -> CH3-COOH. Aldehydes are easily oxidized to carboxylic acids with the same number of carbon atoms by acidified potassium dichromate (orange to green) or potassium permanganate.

Q.8. What is the major product of the reaction given below? What type of reaction is this? Acetaldehyde + Zn-Hg / Conc. HCl -> ? + H2O

The major product is ethane (CH3-CH3), and the reaction is a reduction — specifically Clemmensen reduction. Acetaldehyde is reduced by zinc amalgam and concentrated hydrochloric acid: CH3-CHO + 4[H] -> CH3-CH3 + H2O. The carbonyl oxygen is removed entirely and the carbonyl carbon becomes a -CH2- group.

## Exercise - Long Answer Questions

Q.1. What are aldehydes and ketones? Explain nomenclature of

aldehydes and ketones.

Aldehydes are organic compounds containing the aldehyde group -CHO (carbonyl carbon bonded to H and to R or H): general formula RCHO; ketones contain the keto group >C=O (carbonyl carbon bonded to two carbon groups): general formula RCOR'. Nomenclature: (1) Common system — aldehydes are named from the corresponding acid (formaldehyde, acetaldehyde, benzaldehyde); ketones as dialkyl ketones (dimethyl ketone = acetone, methyl ethyl ketone). (2) IUPAC system — aldehydes take the suffix '-al' with the -CHO carbon as C1 (methanal, ethanal, propanal, 2-methylpropanal); ketones take the suffix '-one' with the carbonyl position numbered to give the lowest locant (propanone, butan-2-one, pentan-3-one). If a higher-priority group is present, -CHO is named as 'formyl' and >C=O as 'oxo'.

Q.2. Discuss the structure and reactivity of aldehydes and ketones.

Structure: the carbonyl carbon is sp2 hybridized, trigonal planar (bond angles about 120 degrees), with a sigma bond plus a pi bond to oxygen; the C=O bond is polar (carbon delta+, oxygen delta-) because oxygen is more electronegative, and the pi bond is exposed, making the carbon electrophilic. Reactivity: the characteristic reaction is nucleophilic addition — a nucleophile attacks the electrophilic carbonyl carbon, the pi bond breaks, and the oxygen becomes O- (then protonated to -OH). Aldehydes are more reactive than ketones (less steric hindrance, more positive carbonyl carbon). Other reactions: oxidation (aldehydes to acids), reduction (to alcohols, or to alkanes by Clemmensen/Wolff-Kishner), condensation with ammonia/amines (imines), aldol condensation

(alpha-hydrogen reactions), and addition of HCN, alcohols, Grignard reagents, NaHSO3.

Q.3. What are the physical properties of aldehydes and ketones?

Physical properties of aldehydes and ketones: (1) Lower members (formaldehyde gas, acetaldehyde liquid) are colourless with a pungent smell; higher members have a pleasant fruity smell (used in perfumes). (2) They are polar, so their boiling points are higher than alkanes/ethers but lower than alcohols (no hydrogen bonding among themselves); boiling points rise with molecular mass. (3) Lower members are soluble in water (hydrogen bonding with water through the carbonyl oxygen); solubility falls as the chain grows. (4) They are less dense than water and are good solvents for organic compounds.

Q.4. Write down three methods for the preparation of aldehydes and ketones?

Three methods for the preparation of aldehydes and ketones: (1) Oxidation of alcohols — primary alcohols (with PCC or controlled K2Cr2O7) give aldehydes; secondary alcohols give ketones. (2) Ozonolysis of alkenes — cleavage of the double bond by ozone followed by Zn/H2O gives aldehydes and/or ketones depending on substitution. (3) From acid chlorides — Rosenmund reduction (H2/Pd-BaSO4) of RCOCl gives aldehydes; reaction of acid chlorides with dialkylcadmium or Grignard reagents (controlled) gives ketones. (4) Hydration of alkynes — HgSO4/H2SO4 hydration of ethyne gives ethanal; other terminal alkynes give methyl ketones. (5) From calcium salts of carboxylic acids — dry distillation of

calcium formate with another calcium carboxylate gives aldehydes; of two carboxylates gives ketones.

### Q.5. Discuss base and acid catalysed nucleophilic addition reactions.

Base-catalysed nucleophilic addition: the base generates the nucleophile (e.g. OH- + HCN -> CN- + H2O), the nucleophile attacks the carbonyl carbon, forming an alkoxide ion, which is then protonated by water or HCN to give the addition product (e.g. cyanohydrin) and regenerate the catalyst. Acid-catalysed nucleophilic addition: the acid first protonates the carbonyl oxygen, making the carbonyl carbon even more electrophilic, so a weak nucleophile (e.g. an alcohol) can attack; the intermediate is then deprotonated to give the product (e.g. hemiacetal/acetal formation). In short: base catalysis works by making the nucleophile stronger; acid catalysis works by making the carbonyl carbon more positive.

### Q.6. How can aldehyde reacts with the following reagents? (i) Hydrogen cyanide (ii) Grignard's reagent (iii) Ammonia (iv) Primary amines

How an aldehyde reacts with the following reagents: (i) Hydrogen cyanide — nucleophilic addition gives a cyanohydrin: RCHO + HCN -> R-CH(OH)-CN (base-catalysed). (ii) Grignard's reagent — nucleophilic addition followed by hydrolysis gives a secondary alcohol: RCHO + R'MgX -> R-CH(R')-OMgX, then H2O/H+ -> R-CH(OH)-R'. (iii) Ammonia — condensation gives an imine (aldimine): RCHO + NH3 -> R-CH=NH + H2O (acid-catalysed; the

imine from ammonia itself is unstable and tends to polymerize). (iv) Primary amines — condensation gives an imine (Schiff base): RCHO + R'NH2 -> R-CH=N-R' + H2O (acid-catalysed, water removed).

Q.7. What are the IUPAC names of the following compounds? i) H-CHO ii) CH3-CO-CH3 iii) CH3-CH(CH3)-CHO iv) CH3-CH(CH3)-CO-CH3 v) CH3-CH(CH3)-CO-CH(C2H5)-CH3 (Structures are read from the book's diagrams.)
IUPAC names: i) H-CHO — Methanal (formaldehyde). ii) CH3-CO-CH3 — Propanone (acetone). iii) CH3-CH(CH3)-CHO — 2-Methylpropanal. iv) CH3-CH(CH3)-CO-CH3 — 3-Methylbutan-2-one. v) CH3-CH(CH3)-CO-CH(C2H5)-CH3 — 2-Methylhexan-3-one (longest chain containing the carbonyl group is six carbons: CH3-CH(CH3)-CO-CH-CH2-CH3).

Q.8. Write structural formulas for the following compounds: i) Butanal ii) 2-Methyl-2-hexanal iii) 2-Chlorobutanal iv) 2-Bromo-3-chloro-2-methylheptanal v) Butanone vi) 2,3-Pentadione vii) 2-Phenyl-3-pentanone viii) 4-nitroacetophenone ix) 3-Hydroxybutanal x) 3-methyl-3-phenylpentanal
Structural formulas (given as condensed formulas, since drawings cannot be printed as text): i) Butanal — CH3-CH2-CH2-CHO. ii) 2-Methyl-2-hexanal — OHC-C(CH3)2-CH2-CH2-CH2-CH3. iii) 2-Chlorobutanal — CH3-CH2-CH(Cl)-CHO. iv) 2-Bromo-3-chloro-2-methylheptanal — OHC-C(Br)(CH3)-CH(Cl)-CH2-CH2-CH2-CH3. v) Butanone — CH3-CO-CH2-CH3. vi) 2,3-Pentadione — CH3-CO-CO-CH2-CH3. vii) 2-Phenyl-3-pentanone — CH3-CH(C6H5)-CO-CH2-CH3. viii)

4-nitroacetophenone — p-O2N-C6H4-CO-CH3. ix) 3-Hydroxybutanal — CH3-CH(OH)-CH2-CHO. x) 3-methyl-3-phenylpentanal — CH3-CH2-C(CH3)(C6H5)-CH2-CHO.

# Chapter 20: Carboxylic Acids and their Functional Derivatives

## Exercise - Long Answer Questions

SOURCE GAP — FLAGGED, NOT INVENTED: the source scan (Chemistry-12.pdf) jumps from printed page 281 straight to printed page 284, so printed pages 282-283 are missing from the scan. The exercise's Multiple Choice Questions, Short Answer Questions, Long Answer Questions Q.1-Q.3, and Long Q.4 parts (i)-(ii) therefore cannot be verified from this source and are NOT covered here. They must be added from a complete copy of the book when one is obtained; nothing has been guessed to fill the gap.

Q.4 (parts iii-v). Give IUPAC names to the following structures: iii) CH3-C(OH)(CH3)-COOH iv) CH3-CH(C2H5)-COOH v) CH3-CH(Cl)-CH(NH2)-COOH (Parts (i)-(ii) are on the missing printed pages 282-283 and are not covered — see the source-gap note above.)

IUPAC names: iii) CH3-C(OH)(CH3)-COOH — 2-Hydroxy-2-methylpropanoic acid (the central carbon carries CH3, OH, CH3 and COOH). iv) CH3-CH(C2H5)-COOH — 2-Methylbutanoic acid (the parent chain including the carboxyl carbon is four carbons, with a methyl on C2). v) CH3-CH(Cl)-CH(NH2)-COOH — 2-Amino-3-chlorobutanoic acid (COOH carbon is C1, the amino group is on C2, chlorine on C3).

Q.5. Write structural formulas for the following compounds: i) Propionic acid ii) Valeric acid iii) Lactic acid iv) Phenylacetic acid v)

2-Chloro-3-phenylpentanoic acid

Structural formulas (given as condensed formulas, since drawings cannot be printed as text): i) Propionic acid — CH3-CH2-COOH. ii) Valeric acid — CH3-CH2-CH2-CH2-COOH (pentanoic acid). iii) Lactic acid — CH3-CH(OH)-COOH (2-hydroxypropanoic acid). iv) Phenylacetic acid — C6H5-CH2-COOH. v) 2-Chloro-3-phenylpentanoic acid — CH3-CH2-CH(C6H5)-CH(Cl)-COOH.

Q.6. Describe structure, acidity, and reactivity of carboxylic acids.

Structure: the carboxyl group -COOH combines a carbonyl (C=O) and a hydroxyl (-OH) on the same carbon; the carbon is sp2 hybridized, planar, and the group is strongly polar, forming hydrogen-bonded dimers. Acidity: carboxylic acids are weak acids (Ka about 10^-5 for acetic acid) but much stronger than alcohols and phenols, because the carboxylate ion (RCOO-) is resonance-stabilised with the negative charge shared equally by both oxygens; electron-withdrawing groups (Cl) increase acidity, electron-donating groups (CH3) decrease it. Reactivity: the -OH is replaced by halogens (with PCl5/SOCl2 giving acid chlorides), by alkoxy groups (esterification with alcohols), by -NH2 (amide formation); the carbonyl carbon undergoes nucleophilic acyl substitution; decarboxylation on heating with soda lime gives alkanes; alpha-hydrogen reactions (halogenation, Hell-Volhard-Zelinsky) occur at the alpha carbon.

Q.7. Write down five physical properties of carboxylic acids.

Five physical properties of carboxylic acids: (1) Lower members

(formic, acetic) are colourless liquids with a pungent vinegar-like smell; higher members are waxy solids, odourless. (2) They have higher boiling points than the corresponding alcohols (e.g. acetic acid 118 C vs ethanol 78 C) because they form hydrogen-bonded dimers. (3) Lower members are completely miscible with water (hydrogen bonding); solubility decreases rapidly with chain length (higher members insoluble). (4) They are denser than the corresponding alcohols and their density increases with molecular mass. (5) They are weak electrolytes — they ionize only slightly in water and turn blue litmus red.

Q.8. Write down five different methods for preparing carboxylic acids.

Five different methods for preparing carboxylic acids: (1) Oxidation of primary alcohols or aldehydes (e.g. ethanol -> ethanal -> acetic acid with K2Cr2O7/H2SO4 or KMnO4). (2) Hydrolysis of nitriles (R-CN + 2H2O, acid or base, heat -> R-COOH + NH3, via the amide). (3) From Grignard reagents (RMgX + CO2 -> RCOOMgX, then hydrolysis -> R-COOH). (4) Hydrolysis of esters, acid chlorides, acid anhydrides or amides (e.g. CH3COOC2H5 + H2O -> CH3COOH + C2H5OH). (5) Oxidation of alkylbenzenes (e.g. toluene with KMnO4 -> benzoic acid; the side chain oxidizes to -COOH regardless of its length).

Q.9. Write short note on the conversion of carboxylic acids to carboxylic acid derivatives (acyl halides, acid anhydrides, esters and amides).

Short note on the conversion of carboxylic acids to carboxylic acid

derivatives: (1) Acyl halides — R-COOH + PCl5 (or PCl3, SOCl2) -> R-COCl + POCl3 + HCl (SOCl2 is preferred since its by-products SO2 and HCl are gases). (2) Acid anhydrides — 2R-COOH, heated with P2O5 or by dehydration -> (RCO)2O + H2O; acetic anhydride is made industrially from ketene and acetic acid. (3) Esters — Fischer esterification: R-COOH + R'OH (excess alcohol, conc. H2SO4 catalyst, heat) -> R-COOR' + H2O. (4) Amides — R-COOH + NH3 -> ammonium salt R-COO-NH4+, heated -> R-CONH2 + H2O. All four conversions proceed by nucleophilic acyl substitution: the nucleophile attacks the carbonyl carbon and -OH (as water, after protonation) leaves. Reactivity order of derivatives toward nucleophiles: acid chloride > acid anhydride > ester > amide.

# Chapter 21: Biochemistry

## Exercise - Multiple Choice Questions

i) A large molecule that is made up of small repeating units is called: (A macromolecule is built from many small repeating monomer units.)

- Isomer
- Monomer
- Micromolecule
- ✓ Macromolecule

ii) Which one of the following is a macromolecule? (Proteins are polymers of amino acids; amino acids and sugars are small units, and lipids are not polymers.)

- Amino acids
- Sugars
- Lipids
- ✓ Proteins

iii) $C_{12}H_{22}O_{11}$ is the chemical formula of: (Sucrose, cane sugar; glucose and fructose are $C_6H_{12}O_6$.)

- Blood sugar
- Glucose
- Fructose
- ✓ Cane sugar

iv) The most abundant biopolymer on the earth is: (Carbohydrates — cellulose in plant cell walls is the most abundant organic compound on earth.)

- Proteins
- Lipids
- ✓ Carbohydrates
- Nucleic acids

v) The reaction between fat and NaOH is known as: (Alkaline hydrolysis of fats gives soap and glycerol — saponification.)

- Hydration
- ✓ Saponification
- Esterification
- Fermentation

vi) Which one of the following is the example of conjugated proteins? (Phosphoproteins carry a phosphate prosthetic group; albumins and histones are simple proteins.)

- Albumins
- Histones
- ✓ Phosphoproteins
- Poly-peptides

vii) Proteins are essential for: (Proteins form protoplasm, act as digestive enzymes, and casein protein was used to manufacture buttons — all of these.)

- Formation of protoplasm
- Digestion of food

Manufacture of buttons

✓ All of the these

viii) Which one of the following transports oxygen from lungs to the cellular tissues? (Haemoglobin in red blood cells carries oxygen.)

✓ Haemoglobin

Nucleoprotein

Albumins

Globulins

ix) Which one of the following enzymes is present in the pancreas? (Trypsin is secreted by the pancreas; pepsin works in the stomach, zymase in yeast.)

Pepsin

✓ Trypsin

Zymase

Isomerase

x) The mineral, which is essential for immune system: (Zinc is essential for immune function; iron carries oxygen, calcium builds bones, magnesium supports enzymes.)

Iron

✓ Zinc

Magnesium

Calcium

## Exercise - Short Answer Questions

Q.1. Why boiling points of saturated fatty acid is higher than

unsaturated fatty acid?

Saturated fatty acids have higher boiling points than unsaturated fatty acids because their straight hydrocarbon chains pack closely together, giving strong van der Waals forces over a large contact area. Unsaturated fatty acids have cis double bonds that put kinks (bends) in the chains, so the molecules cannot pack closely, the intermolecular forces are weaker, and less heat is needed to separate them.

Q.2. What do you mean by carbohydrate?

A carbohydrate is a polyhydroxy aldehyde or polyhydroxy ketone, or a compound that gives such compounds on hydrolysis. The name means 'hydrate of carbon' from the old general formula Cx(H2O)y. Examples: glucose (a monosaccharide), sucrose (a disaccharide), starch and cellulose (polysaccharides).

Q.3. What happens when you eat too much protein?

Eating too much protein overworks the liver and kidneys, which must remove the extra nitrogen as urea; the excess amino acids are deaminated and the carbon skeletons converted to fat or glucose, so surplus protein is stored as fat and can cause weight gain. It can also cause dehydration (water is needed to excrete urea) and, over time, strain the kidneys.

Q.4. Why do weightlifters take amino acids?

Weightlifters take amino acids (protein supplements) because amino acids are the building blocks of muscle protein: after heavy training, muscles need extra amino acids to repair damaged fibres

and build new muscle tissue (muscle growth). The essential amino acids, especially leucine, stimulate muscle protein synthesis, and a ready supply speeds recovery.

Q.5. Where enzymes are found?

Enzymes are found in every living cell and in the digestive fluids of the body: in the mouth (salivary amylase), stomach (pepsin), small intestine and pancreas (trypsin, lipase, amylase), and inside cells (e.g. mitochondria, cytoplasm) where they run metabolism. They are also found in plants, yeasts and bacteria.

Q.6. Where are enzymes produced in the body?

Enzymes are produced in the body by the cells of glands and organs: digestive enzymes are produced by the salivary glands (amylase), the stomach lining (pepsin), the pancreas (trypsin, lipase, pancreatic amylase) and the intestinal wall; metabolic enzymes are produced inside the cells themselves (ribosomes make the enzyme proteins).

Q.7. What are the three main functions of lipids?

Three main functions of lipids: (1) Energy storage — fats are the most concentrated energy store in the body (about 9 kcal per gram, more than double carbohydrates or proteins). (2) Structural — phospholipids form cell membranes, and cholesterol maintains membrane fluidity. (3) Protection and insulation — fat under the skin insulates against heat loss and cushions vital organs; waxes protect plant surfaces and skin.

Q.8. Why are lipids a better source of energy?

Lipids are a better source of energy than carbohydrates or proteins because they contain a much higher proportion of carbon and hydrogen and very little oxygen, so complete oxidation releases more energy per gram (about 9 kcal/g vs 4 kcal/g). They are also stored without water (anhydrous), so a given mass stores far more usable energy.

Q.9. Can lipids dissolve in polar solvents like water or not?

No, lipids do not dissolve in polar solvents like water, because lipid molecules are non-polar (long hydrocarbon chains) while water is polar — like dissolves like. Lipids dissolve in non-polar organic solvents such as ether, chloroform, benzene and carbon tetrachloride.

Q.10. What is the best solvent for lipids?

The best solvent for lipids is a non-polar organic solvent: ether, chloroform, benzene or carbon tetrachloride dissolve fats and oils readily. (In the laboratory, a chloroform-methanol mixture is commonly used to extract lipids from tissues.)

Q.11. What are the two main types of nucleic acids?

The two main types of nucleic acids are: (1) DNA (deoxyribonucleic acid) — double-stranded, stores hereditary information in the nucleus. (2) RNA (ribonucleic acid) — usually single-stranded, transfers genetic information and builds proteins (mRNA, tRNA, rRNA).

## Exercise - Long Answer Questions

### Q.1. What is biochemistry? Discuss in four to five lines.

Biochemistry is the study of the chemical processes associated with living organisms: the structures, functions and interactions of biological molecules (biomolecules) such as carbohydrates, proteins, enzymes, lipids and nucleic acids, the mechanisms of enzyme action, the chemistry of vitamins, and the energy transformations in the cell. It explains life in chemical terms — how food is broken down, how energy (ATP) is made and used, and how genetic information is stored and expressed.

### Q.2. What are carbohydrates? Explain briefly.

Carbohydrates are the most abundant naturally occurring compounds, widely distributed in plants and animals (about three-fourths of the dry weight of plants). They are polyhydroxy aldehydes or ketones, or compounds that give these on hydrolysis, with the general formula Cx(H2O)y. Examples: glucose, fructose, sucrose, starch, glycogen and cellulose. They are the main energy source for the brain and body (glucose, glycogen, starch) and provide structural support (cellulose in plants).

### Q.3. Explain various classes of carbohydrates.

Various classes of carbohydrates: (1) Monosaccharides — simple sugars that cannot be hydrolyzed into simpler carbohydrates (glucose, fructose, galactose); divided into aldoses (aldehyde group, e.g. glucose) and ketoses (keto group, e.g. fructose), and by carbon number (trioses, tetroses, pentoses, hexoses). (2) Oligosaccharides — yield 2-10 monosaccharides on hydrolysis;

disaccharides are the most important (sucrose = glucose + fructose; lactose = glucose + galactose; maltose = glucose + glucose). (3) Polysaccharides — yield many monosaccharides on hydrolysis; storage polysaccharides (starch in plants, glycogen in animals) and structural polysaccharides (cellulose in plant cell walls, chitin in insect shells).

Q.4. Write down main functions of carbohydrates.

Main functions of carbohydrates: (1) Energy source — glucose is the immediate fuel of cells; oxidation of glucose gives ATP for all life processes. (2) Energy storage — starch in plants and glycogen in animals store glucose for later use. (3) Structural — cellulose forms plant cell walls; chitin forms insect exoskeletons. (4) Building blocks — ribose and deoxyribose form the backbones of RNA and DNA. (5) Other roles — carbohydrates on cell surfaces act in recognition (glycoproteins), and lactose is the sugar of milk.

Q.5. What is the Nutritional Importance of Carbohydrates?

Nutritional importance of carbohydrates: they are the cheapest and most important energy food (about 4 kcal per gram), providing the glucose the brain and muscles run on; fibre (indigestible cellulose) keeps the digestive system healthy and prevents constipation; they spare proteins from being burned for energy, so proteins can be used for growth and repair; and they give foods sweetness and bulk (sugars, starch in bread, rice, potatoes). A diet lacking carbohydrates causes weakness, ketosis and the breakdown of body protein for energy.

Q.6. What are proteins? Describe different classes of proteins.

Proteins are polymers of amino acids linked by peptide bonds — the building blocks of all living organisms and essential components of protoplasm. Classes of proteins: (1) Simple proteins — made only of amino acids (albumins in egg white and blood, globulins, histones in nuclei, glutelins in wheat). (2) Conjugated proteins — a simple protein plus a non-protein prosthetic group (phosphoproteins with phosphate, e.g. casein in milk; glycoproteins with carbohydrate; lipoproteins with lipid; nucleoproteins with nucleic acid; chromoproteins with a coloured group, e.g. haemoglobin with haem). (3) Derived proteins — products of partial hydrolysis (proteoses, peptones, peptides).

Q.7. Write a short note on the properties of proteins. Discuss the importance of protein.

Short note on the properties of proteins: proteins are high molecular mass polymers of amino acids; they are amphoteric (both acidic -COOH and basic -NH2 groups), so they act as buffers; they are denatured by heat, acids, alkalis and heavy metals (their shape unfolds, e.g. egg white on heating); many are soluble in water or salt solutions (albumins, globulins) while fibrous proteins (keratin, collagen) are insoluble; they give colour tests (biuret test gives violet with CuSO4 + NaOH; xanthoproteic test gives yellow with nitric acid). Importance of protein: proteins build and repair body tissues (muscles, skin, hair), form enzymes and hormones, carry oxygen (haemoglobin), fight disease (antibodies), and maintain protoplasm — life cannot exist without them.

Q.8. What are enzymes? Explain the role of enzymes as a biocatalyst.

Enzymes are proteins that catalyse chemical reactions in living organisms — biological catalysts. They speed up reactions enormously (millions of times) without being used up, by lowering the activation energy: the substrate fits into the enzyme's active site (lock-and-key or induced-fit), the enzyme-substrate complex forms, the reaction occurs, and the products are released while the enzyme is regenerated. Enzymes are highly specific (one enzyme usually catalyses one reaction), work best at an optimum temperature (about 37 C in man) and pH, and are destroyed by high temperature (denaturation). Examples: amylase digests starch, pepsin digests proteins, lipase digests fats.

Q.9. What are the factors that affect enzyme activity? Discuss.

Factors that affect enzyme activity: (1) Temperature — activity rises with temperature up to an optimum (about 37 C for human enzymes), then falls sharply as the enzyme denatures. (2) pH — each enzyme has an optimum pH (pepsin about pH 2 in the stomach; trypsin about pH 8 in the intestine); extremes denature it. (3) Substrate concentration — activity rises with substrate concentration until all active sites are saturated, then levels off. (4) Enzyme concentration — more enzyme gives a faster rate (with excess substrate). (5) Inhibitors — competitive inhibitors block the active site, non-competitive inhibitors change the enzyme's shape; both slow or stop the reaction. (6) Activators/cofactors — metal ions (Zn2+, Mg2+) and coenzymes (vitamins) are needed by many enzymes.

Q.10. What are the commercial uses of enzymes.

Commercial uses of enzymes: (1) Food industry — amylase in bread-making and brewing, rennin (chymosin) in cheese-making, pectinase in fruit-juice clarification, invertase in sweets. (2) Detergents — proteases and lipases in biological washing powders remove protein and fat stains. (3) Leather industry — proteases in dehairing and softening hides. (4) Medicine — streptokinase dissolves blood clots, digestive enzyme preparations treat deficiencies. (5) Paper and textiles — cellulases soften fabrics and process paper pulp. (6) Biotechnology — restriction enzymes cut DNA in genetic engineering.

Q.11. What are lipids? Describe the classification of lipids.

Lipids are high molecular mass organic compounds of animal and plant origin that are soluble in organic solvents (ether, chloroform, benzene) but insoluble in water. Classification of lipids: (1) Simple lipids — esters of fatty acids with alcohols: fats and oils (triglycerides, glycerol + three fatty acids) and waxes (long-chain alcohol + fatty acid). (2) Compound (conjugated) lipids — simple lipids plus another group: phospholipids (glycerol + fatty acids + phosphate, cell membranes), glycolipids (with carbohydrate), lipoproteins (with protein). (3) Derived lipids — obtained by hydrolysis of the above: fatty acids (saturated, e.g. palmitic, stearic; unsaturated, e.g. oleic, linoleic), glycerol, and steroids (cholesterol, hormones like testosterone, vitamin D).

Q.12. Explain the structure of lipids.

Structure of lipids: fats and oils are triglycerides — one glycerol molecule esterified with three fatty acid molecules (R1COO-CH2, R2COO-CH, R3COO-CH2). The fatty acids may be saturated (no double bonds, straight chains, e.g. palmitic acid C16, stearic acid C18 — solid fats) or unsaturated (one or more cis double bonds, kinked chains, e.g. oleic acid — liquid oils). Phospholipids replace one fatty acid by a phosphate group, giving a polar head and non-polar tails (the basis of cell membranes). Steroids have a characteristic four-ring (tetracyclic) structure of three six-membered rings and one five-membered ring (e.g. cholesterol).

Q.13. Explain briefly the physical and chemical properties of lipids.

Physical and chemical properties of lipids (briefly): Physical — fats are solids and oils liquids at room temperature; they are insoluble in water but soluble in organic solvents; they are lighter than water, greasy to the touch, and poor conductors of heat. Chemical — (1) Hydrolysis: fats + water (acid, base or lipase) give glycerol + fatty acids; alkaline hydrolysis is saponification (soap-making). (2) Hydrogenation: unsaturated oils + H2 (Ni catalyst) give solid fats (vanaspati ghee). (3) Halogenation: halogens add across double bonds (iodine number measures unsaturation). (4) Rancidity: exposure to air, moisture and light oxidizes/hydrolyzes fats, giving an unpleasant smell. (5) Saponification value and acid value measure soap-making quality and free fatty acids.

Q.14. What is the nutritional and biological importance of lipids?

Nutritional and biological importance of lipids: nutritionally, fats are the most concentrated energy food (9 kcal/g), carry the fat-soluble

vitamins A, D, E and K, provide essential fatty acids (linoleic, linolenic) the body cannot make, and make food tasty. Biologically, phospholipids and cholesterol form all cell membranes; fats insulate the body and cushion organs; steroids act as hormones (sex hormones, cortisone); waxes protect plant leaves and animal skin; and stored fat is the body's long-term energy reserve.

Q.15. What are nucleic acids? What are the structural components of DNA and RNA?

Nucleic acids are high molecular mass biopolymers of nucleotides, present in the nucleus of all living cells and in viruses; they are the carriers of hereditary information. Structural components of DNA and RNA: both are built from nucleotides, each made of (1) a pentose sugar — deoxyribose in DNA, ribose in RNA; (2) a phosphate group; (3) a nitrogenous base — adenine (A), guanine (G), cytosine (C) plus thymine (T) in DNA, uracil (U) in RNA instead of thymine. DNA is a double helix of two strands (A pairs with T, G with C) joined by hydrogen bonds; RNA is usually single-stranded (mRNA carries the message, tRNA brings amino acids, rRNA builds ribosomes).

Q.16. What are sources and biological significance of minerals? Describe briefly.

Sources and biological significance of minerals (briefly): Iron — sources: meat, liver, spinach, pulses; significance: forms haemoglobin (oxygen transport), deficiency causes anaemia. Calcium — sources: milk, cheese, green vegetables; significance: builds bones and teeth, needed for blood clotting, muscle

contraction and nerve impulses. Phosphorus — sources: milk, meat, fish, cereals; significance: builds bones/teeth, forms ATP, DNA and RNA. Zinc — sources: meat, shellfish, nuts, pulses; significance: essential for the immune system, wound healing, growth and many enzymes. Iodine — sources: seafood, iodized salt; significance: makes thyroxine (thyroid hormone), deficiency causes goitre. Magnesium — sources: green vegetables, nuts; significance: activates enzymes, builds bones, needed for chlorophyll in plants.

# Chapter 22: Industrial Chemistry

## Exercise - Multiple Choice Questions

i) Which one of the following statements is true regarding chemical industries? (Chemical industries produce medicines and drugs that save lives; the other statements are not true of chemical industries in general.)

- ✓ Chemical industries provide medicines and drugs for saving life of peoples
- Chemical industries are located near automobile industries
- Chemical industries provide fish and plants to the farmers
- Chemical industries provide infra structure for industries

ii) Which of the following statements is/are correct? (Herbicides kill unwanted plants (weeds). NOTE — FLAGGED as defective as printed: option (b) is also true — nylon was indeed introduced as a silk substitute — so two options are correct; (a) is used as the intended answer since it matches the chapter's pesticide section. Options (c) and (d) are false: excess fertilizer harms soil, and cotton is a natural fibre.)

- ✓ Herbicides are used for killing unwanted plants
- Nylon was first introduced as an early substitute for silk
- Excessive use of fertilizers is good for the fertility of soil
- Cotton is an artificial fibre of almost pure cellulose

iii) Adhesive that become soft on heating are: (Thermoplastic

adhesives soften when heated and harden on cooling; thermosetting adhesives set permanently.)

- Thermosetting
- ✓ Thermoplastic
- Elastomeric
- Hybrid

iv) Natural adhesives are also known as: (Natural adhesives from plants/animals are bioadhesives.)

- ✓ Bioadhesives
- Synthetic adhesives
- Spray adhesives
- Fabric adhesives

v) Which one of the following is a synthetic fibre? (Polyester is man-made; wool, cotton and silk are natural fibres.)

- Wool
- Cotton
- Silk
- ✓ Polyester

vi) What is the fuel that is mostly used to run the buses, trucks and trains? (Diesel engines power most buses, trucks and trains.)

- Petrol
- ✓ Diesel
- Kerosene
- Bitumen

vii) Ethylene is used in the manufacture of: (Ethylene is polymerized to polyethylene and is the basic raw material of the plastics industry. NOTE — FLAGGED: PVC is also made from ethylene (via vinyl chloride), so option (d) is also defensible; (b) Plastics is the general intended answer.)

- Fertilizers
- ✓ Plastics
- Silks
- PVC

viii) Nail polish remover contains: (Acetone dissolves nail varnish quickly.)

- Benzene
- ✓ Acetone
- Acetic acid
- Benzoic acid

ix) Henna usually gives you a shade of: (Henna (lawsone) stains skin and hair reddish brown.)

- Golden yellow
- Bluish green
- ✓ Reddish brown
- Light pink

x) The term petrochemical refers to the chemicals derived from: (Petrochemicals are made from natural gas liquids and the streams of oil refineries, not from any single fuel alone.)

- Gasoline oil only

Natural gas only

Crude petroleum only

✓ Natural gas liquids and oil refinery streams

## Exercise - Short Answer Questions

### Q.1. What do you know about chemical industry?

The chemical industry is the industry that converts raw materials (from the lithosphere, hydrosphere, atmosphere and biosphere) into useful chemical products by chemical processes: medicines and drugs, fertilizers, pesticides, dyes, paints, plastics, synthetic fibres, soaps and detergents, cosmetics, cement, glass and petrochemicals. It is one of the largest industries and the basis of the prosperity of a nation, since almost every other industry depends on its products.

### Q.2. Where do colour dyes come from?

Colour dyes come from two sources: natural dyes (from plants, animals and minerals — e.g. indigo from the indigo plant, henna from the henna plant, cochineal from insects, madder from roots) and synthetic dyes (made in laboratories and factories from coal tar and petroleum chemicals — e.g. azo dyes, vat dyes, reactive dyes). Today most dyes are synthetic because they are cheaper, brighter and more fast.

### Q.3. How do pesticides affect human health?

Pesticides affect human health when their residues reach food, water or air: acute poisoning causes nausea, headache, dizziness and skin/eye irritation; long-term exposure can damage the liver,

kidneys and nervous system, disturb hormones, and increase cancer risk. Farm workers spraying without protection are most at risk, and residues in food and drinking water expose the general population.

Q.4. What are the most commonly used pesticides?

The most commonly used pesticides are: insecticides (kill insects, e.g. DDT — now banned in many countries, malathion, pyrethroids), herbicides/weedicides (kill weeds, e.g. 2,4-D, glyphosate), fungicides (kill fungi, e.g. copper sulphate, mancozeb), and rodenticides (kill rats and mice, e.g. zinc phosphide, warfarin).

Q.5. What are the four major ingredients in nail varnish?

The four major ingredients in nail varnish (nail polish) are: (1) Film former — nitrocellulose, which forms the hard shiny film. (2) Resin — e.g. tosylamide-formaldehyde resin, which improves adhesion, gloss and hardness. (3) Plasticizer — e.g. dibutyl phthalate or camphor, which keeps the film flexible and prevents cracking. (4) Solvent — e.g. ethyl acetate, butyl acetate or acetone, which dissolves the ingredients and evaporates on application, plus pigments for colour.

Q.6. What are primary and secondary raw materials?

Primary raw materials are materials obtained directly from nature for chemical industries: from the lithosphere (minerals, ores, coal, rock salt, limestone), the hydrosphere (water, sea salt), the atmosphere (nitrogen, oxygen) and the biosphere (plants, animals, wood, agricultural products). Secondary raw materials are materials

already processed by one industry and used by another: e.g. coke from coal, sulphuric acid, ammonia, ethylene and other petrochemicals, which serve as feedstocks for further chemical manufacture.

Q.7. Write down the name of three main fossil fuels?

The three main fossil fuels are: (1) Coal — solid fuel from ancient plants. (2) Petroleum (crude oil) — liquid fuel from ancient marine organisms. (3) Natural gas — gaseous fuel, mainly methane, found with petroleum deposits.

Q.8. Give some of the important examples of synthetic and natural polymers.

Important examples of synthetic polymers: polyethylene (polythene), polyvinyl chloride (PVC), nylon, polyester (terylene), polystyrene, bakelite. Important examples of natural polymers: cellulose (cotton, wood), starch, natural rubber, proteins (silk, wool), nucleic acids (DNA, RNA).

Q.9. Name the four most commonly used cosmetic products.

The four most commonly used cosmetic products are: (1) Lipsticks. (2) Nail varnish (nail polish) and nail varnish remover. (3) Hair dyes. (4) Face powders/creams (skin-care creams and lotions). (Perfumes and synthetic adhesives are also covered in this chapter.)

Q.10. What type of nail polish remover works quickly and effectively?

Acetone-based nail polish removers work quickly and effectively,

because acetone is a strong solvent that rapidly dissolves the nitrocellulose film of nail varnish. (Less harsh alternatives — ethyl acetate or isopropyl alcohol based removers — work more slowly.)

Q.11. Define resins. What are the different types of resins used in nail varnish?

Resins are solid or semi-solid organic substances (natural or synthetic) that are insoluble in water but soluble in organic solvents and form hard, glossy films on drying. Different types of resins used in nail varnish: (1) Tosylamide-formaldehyde resin — the most common, gives adhesion, gloss and hardness. (2) Alkyd resins. (3) Acrylic resins. (4) Natural resins such as shellac and benzoin (used in some formulations).

Q.12. Why is wax used in lipstick?

Wax is used in lipstick because it gives the lipstick its solid shape and structure at room temperature, raises its melting point so it does not melt in the hand or on the lips, provides a smooth, firm texture for application, and holds the oils, pigments and emollients together in a stable stick. Common waxes: beeswax, carnauba wax and candelilla wax.

Q.13. Is there any other name for nail polish?

Yes — nail polish is also called nail varnish, nail enamel, or nail lacquer. (In this book the term 'nail varnish' is used.)

Q.14. What chemicals make up nail polish?

Nail polish (nail varnish) is made up of: nitrocellulose (film former),

a resin such as tosylamide-formaldehyde resin (adhesion and gloss), a plasticizer such as dibutyl phthalate or camphor (flexibility), solvents such as ethyl acetate and butyl acetate (which evaporate after application), and pigments/dyes for colour (plus pearl essence for frosted shades).

Q.15. Give names of less harsh and toxic nail polish removers.
Less harsh and toxic nail polish removers: ethyl acetate-based removers, isopropyl alcohol-based removers, and acetone-free removers containing propylene carbonate or soy-based (methyl soyate) solvents. These are gentler on nails and skin but remove polish more slowly than acetone.

Q.16. Name the four different types of synthetic adhesives.
The four different types of synthetic adhesives are: (1) Thermoplastic adhesives (soften on heating, e.g. polyvinyl acetate — white glue). (2) Thermosetting adhesives (set permanently on heating, e.g. epoxy resins, urea-formaldehyde). (3) Elastomeric adhesives (rubber-based, flexible, e.g. contact adhesives). (4) Hybrid adhesives (combinations, e.g. epoxy-polyurethane hybrids, silicone-modified polymers).

Q.17. What is the difference between glue and adhesive?
Difference between glue and adhesive: 'glue' traditionally means an adhesive made from natural materials (animal glue from bones and hides, fish glue, starch paste) used mainly for wood and paper, while 'adhesive' is the general term for any substance (natural or synthetic) that bonds two surfaces together. In modern use the

words are often interchangeable, but 'adhesive' covers the whole class including synthetic products (epoxies, cyanoacrylates, silicones), whereas 'glue' usually refers to the simpler, natural or household types.

## Exercise - Long Answer Questions

### Q.1. What is industrial chemistry? Discuss the importance of industrial chemistry.

Industrial chemistry is the branch of chemistry that deals with the large-scale manufacture of chemical products from raw materials by chemical processes. Importance of industrial chemistry: it provides the materials of modern life — fertilizers that feed the world, medicines that save lives, pesticides that protect crops, dyes and paints, plastics, synthetic fibres, soaps and detergents, cement, glass and paper; it creates employment and earns foreign exchange; it supports agriculture, health, construction, transport and defence; and the prosperity of a nation is largely determined by the strength of its chemical industry and its petroleum products.

### Q.2. What is the role of chemical industries in the economy of a country?

Role of chemical industries in the economy of a country: they provide employment to millions directly and indirectly; they supply essential goods (fertilizers, medicines, pesticides, plastics, textiles, cement) that raise living standards; they add value to local raw materials (minerals, agricultural products, petroleum) instead of exporting them raw; they earn foreign exchange through exports and save it by import substitution; they support other industries

(agriculture, construction, transport, textiles) and contribute a large share of national income and tax revenue.

Q.3. Discuss various raw materials that are used in chemical industries.

Various raw materials used in chemical industries: (1) From the lithosphere — minerals and ores (iron ore, bauxite, rock salt, limestone, sulphur, phosphate rock), coal and fossil fuels. (2) From the hydrosphere — water (solvent, coolant, reactant) and sea water (common salt, magnesium, bromine). (3) From the atmosphere — nitrogen (ammonia, fertilizers), oxygen (steel-making, combustion), argon. (4) From the biosphere — agricultural products (sugar cane, cotton, wood, vegetable oils), animal products (hides, bones, milk). (5) Secondary raw materials — coke, ammonia, sulphuric acid, petrochemicals (ethylene, propylene, benzene) produced by other industries.

Q.4. Explain how the lithosphere is an important source of natural raw materials for the chemical industry?

The lithosphere (the earth's crust) is an important source of natural raw materials for the chemical industry because it supplies: minerals and ores (iron, copper, aluminium, zinc ores for metals and their compounds); fossil fuels (coal, petroleum, natural gas — the source of petrochemicals, fuels and organic chemicals); industrial minerals (rock salt for chlorine and caustic soda; limestone for cement, lime and soda ash; sulphur for sulphuric acid; phosphate rock for fertilizers; clay for ceramics; silica sand for glass). Without lithosphere materials there would be no metals,

cement, glass, fertilizers or petrochemicals.

Q.5. How chemical process safety can be managed in industries?

Chemical process safety can be managed in industries by: identifying hazards (HAZOP studies, risk assessment); designing plants with safety margins (pressure relief valves, containment, automatic shut-down systems); following standard operating procedures and permits for hazardous work; training workers and providing personal protective equipment; storing chemicals properly (segregation of incompatibles, ventilation, labelling); controlling ignition sources near flammable materials; installing fire-fighting and gas-detection systems; planning emergency response and evacuation; and regularly inspecting and maintaining equipment to prevent leaks and failures.

Q.6. What are dyes? What is the chemical composition of dyes? How are they classified on the basis of origin?

Dyes are coloured organic compounds that impart colour to materials (textiles, leather, paper, hair) by being absorbed or chemically bonded to them. Chemical composition of dyes: most dyes are aromatic organic compounds containing a chromophore (a colour-producing group such as -N=N- azo, -NO2, C=O, C=C) and an auxochrome (a group such as -OH, -NH2, -SO3H that deepens colour and helps the dye attach to the fibre). Classification on the basis of origin: (1) Natural dyes — from plants (indigo, madder, henna), animals (cochineal, Tyrian purple from shellfish) and minerals (ochre). (2) Synthetic dyes — made from coal tar and petroleum chemicals (azo dyes, vat dyes, reactive dyes, disperse

dyes); they are cheaper, brighter and more wash-fast, and dominate modern use.

Q.7. What are pesticides? Describe its types.

Pesticides are chemical substances used to kill or control pests (insects, weeds, fungi, rodents, mites) that damage crops, stored food and human health. Types of pesticides: (1) By target pest — insecticides (kill insects), herbicides/weedicides (kill weeds), fungicides (kill fungi), rodenticides (kill rodents), nematicides (kill nematodes), acaricides (kill mites). (2) By chemical nature — organochlorines (DDT, BHC), organophosphates (malathion, parathion), carbamates (carbaryl), pyrethroids (synthetic pyrethrins), and inorganic pesticides (copper sulphate, sulphur). (3) By mode of action — contact, stomach, systemic and fumigant pesticides.

Q.8. What is meant by petrochemicals? Describe different types of petrochemicals in detail.

Petrochemicals are chemicals derived from petroleum and natural gas (natural gas liquids and oil refinery streams). Different types of petrochemicals in detail: (1) Olefins — ethylene (polyethylene, PVC, ethanol, ethylene glycol), propylene (polypropylene, propylene oxide, cumene), butadiene (synthetic rubber). (2) Aromatics — benzene (styrene, phenol, nylon, detergents), toluene (TNT, benzoic acid, solvents), xylenes (polyester, terephthalic acid). (3) Synthesis gas — carbon monoxide + hydrogen (methanol, ammonia, urea). (4) Other products — acetylene, carbon black, waxes, lubricating oils, bitumen, and a huge range of solvents,

plastics, fibres, rubbers, detergents, fertilizers and pharmaceuticals.

Q.9. What are polymers? What are natural and synthetic polymers?

Polymers are large molecules (macromolecules) formed by joining many small repeating units (monomers) by covalent bonds. Natural polymers occur in nature: cellulose and starch (from glucose), natural rubber (from isoprene), proteins (from amino acids), nucleic acids (from nucleotides), silk and wool. Synthetic polymers are man-made: polyethylene, polypropylene, PVC, polystyrene, nylon, polyester (terylene), bakelite, teflon. Polymers are formed by addition polymerization (monomers add without loss, e.g. ethene -> polythene) or condensation polymerization (monomers join with loss of a small molecule like water, e.g. nylon-6,6 from hexamethylenediamine and adipic acid).

Q.10. Write down the formation and uses of PVC and Nylon?

Formation and uses of PVC and Nylon: PVC (polyvinyl chloride) is formed by addition polymerization of vinyl chloride (CH2=CHCl) under heat and pressure with a catalyst/initiator: nCH2=CHCl -> -(CH2-CHCl)n-. Uses: pipes, cables insulation, flooring, window frames, raincoats, credit cards, bottles. Nylon (nylon-6,6) is formed by condensation polymerization of hexamethylenediamine and adipic acid with loss of water: nH2N-(CH2)6-NH2 + nHOOC-(CH2)4-COOH -> nylon-6,6 + 2nH2O. Uses: fibres for clothes, carpets, ropes, parachutes, fishing nets, toothbrush bristles, machine gears and bearings.

Q.11. What are cosmetics? Explain the role and uses of cosmetics.

Cosmetics are substances applied to the human body (skin, hair, nails, lips) to clean, beautify, protect or change appearance without affecting the body's structure or functions. Role and uses of cosmetics: they clean and protect the skin (creams, lotions, sunscreens), beautify (lipsticks, face powders, eye make-up, nail varnish), colour hair (hair dyes, henna), care for nails (nail varnish and removers), give fragrance (perfumes), and protect against sun, wind and infection. They play a large social and economic role, but must be safe — harmful ingredients (lead, mercury, harsh solvents) are restricted.

Q.12. Describe preparation and applications of lipsticks.

Preparation and applications of lipsticks: Preparation — lipsticks are made by melting together waxes (beeswax, carnauba wax, candelilla wax for shape and hardness), oils (castor oil, lanolin for smooth spreading and gloss), and pigments/dyes (for colour) with perfumes; the molten mixture is poured into moulds, cooled, and the sticks are removed and finished. Applications — lipsticks colour and beautify the lips, protect them from drying and cracking (moisturizing oils and waxes), and provide a long-lasting, even film; medicated lipsticks also protect against sun and cold.

Q.13. Describe preparation and applications of nail varnish and nail varnish remover.

Preparation and applications of nail varnish and nail varnish remover: Nail varnish preparation — nitrocellulose (film former) is dissolved in solvents (ethyl acetate, butyl acetate) with a resin

(tosylamide-formaldehyde resin for adhesion and gloss), a plasticizer (dibutyl phthalate, camphor for flexibility) and pigments; the mixture is blended, filtered and bottled. Applications — nail varnish beautifies and protects nails, giving a hard, glossy, coloured film. Nail varnish remover preparation — usually acetone (fast-acting) or an acetone-free solvent (ethyl acetate, isopropyl alcohol, propylene carbonate) with oils and moisturisers to reduce drying. Applications — it dissolves and removes old varnish quickly without damaging the nail when used properly.

Q.14. What are hair dyes? What ingredients are most commonly used in hair dyes?

Hair dyes are cosmetic products used to change the colour of hair. Ingredients most commonly used in hair dyes: (1) Permanent (oxidation) dyes — p-phenylenediamine (PPD) or p-aminophenol as the primary intermediate, coupled with resorcinol or m-aminophenol, with hydrogen peroxide as the developer/oxidizer and ammonia to open the hair cuticle. (2) Temporary/semi-permanent dyes — direct dyes (azo dyes, nitro dyes) that coat the hair shaft. (3) Natural dyes — henna (lawsone, reddish brown), indigo, chamomile. (4) Metallic dyes — lead acetate (progressive dyes). (5) Bleaches — hydrogen peroxide with ammonia to lighten hair before dyeing.

Q.15. Discuss types and applications of hair dyes.

Types and applications of hair dyes: (1) Temporary dyes — coat the hair surface, wash out in one shampoo; used for parties and trial colours. (2) Semi-permanent dyes — penetrate slightly into the

cuticle, last 4-6 washes; used to refresh colour and cover a little grey. (3) Permanent (oxidation) dyes — penetrate the cortex and form large colour molecules inside the hair, lasting until the hair grows out; used to cover grey completely and for major colour changes. (4) Natural dyes — henna and indigo, used traditionally for reddish-brown and black shades and for conditioning. (5) Bleaches/lighteners — hydrogen peroxide based, used to lighten hair before dyeing or for blonde shades.

Q.16. What are adhesives? Explain different types of adhesives.

Adhesives are substances that bond two surfaces together by surface attachment (adhesion) and internal strength (cohesion). Different types of adhesives: (1) Natural adhesives (bioadhesives) — from plants (starch, dextrin, natural rubber) and animals (casein, gelatin, fish glue). (2) Synthetic adhesives — thermoplastic (polyvinyl acetate, hot melts — soften on heating), thermosetting (epoxy, urea-formaldehyde — set permanently), elastomeric (rubber-based contact adhesives), and hybrid (silicone-modified, epoxy-polyurethane). (3) By form — liquid, paste, powder, tape and hot-melt adhesives. (4) By use — wood adhesives, paper adhesives, metal adhesives, medical and dental adhesives.

Q.17. Explain types and applications of synthetic adhesives.

Types and applications of synthetic adhesives: (1) Thermoplastic adhesives (e.g. polyvinyl acetate, hot-melt EVA) — soften on heating, used for woodwork, paper, packaging and bookbinding. (2) Thermosetting adhesives (e.g. epoxy resins, phenol-formaldehyde, urea-formaldehyde) — form strong permanent bonds, used for

aircraft, automobiles, plywood and construction. (3) Elastomeric adhesives (e.g. neoprene, nitrile rubber contact adhesives) — flexible, used for shoes, leather, laminates and flooring. (4) Hybrid adhesives (e.g. silyl-modified polymers, epoxy-polyurethane) — combine strength and flexibility, used for construction joints, vehicle assembly and sealing.

# Chapter 23: Environmental Chemistry

## Exercise - Multiple Choice Questions

i) The segments of environments are: (Four segments: atmosphere, hydrosphere, lithosphere and biosphere.)

- One
- Two
- Three
- ✓ Four

ii) Polar ice caps, glaciers, rivers, rocks, mountains, soils are the parts of: (NOTE — FLAGGED as ambiguous as printed: the list mixes hydrosphere items (ice caps, glaciers, rivers) with lithosphere items (rocks, mountains, soils), so no single segment fits exactly; (d) Geosphere is used as the intended umbrella answer.)

- Lithosphere
- Hydrosphere
- biosphere
- ✓ Geosphere

iii) Most of mass of atmosphere is present in: (The troposphere holds about 70-80% of the atmosphere's mass.)

- ✓ Troposphere
- Stratosphere
- Mesosphere
- Thermosphere

iv) Rust is: (Rust is hydrated iron(III) oxide, $Fe_2O_3.xH_2O$.)

- Iron (III) oxide
- Iron (II) oxide
- ✓ Hydrated Iron (III) oxide
- Hydrated Iron (II) oxide

v) Acid rain is mainly caused in the atmosphere by the emission of: (Sulphur oxides and nitrogen oxides oxidize to sulphuric and nitric acids in rain.)

- NOx and $KNO_3$
- SOx and charcoal
- COx and sulphur
- ✓ SOx and NOx

vi) Smog is derived from: (Smog = smoke + fog.)

- Smoke and water droplet
- Fog and water droplet
- ✓ Fog and smoke
- Mist and water droplet

vii) London smog or gray smog is caused by: (Classical London smog comes from coal smoke — $SO_2$, soot particulates and fog moisture.)

- ✓ $SO_2$ + particulates + moisture of fog
- $NO_2$ + particulates + moisture of fog
- $CO_2$ + particulates + moisture of fog
- PAN + particulates + moisture of fog

viii) Brown smog results from the mixing of: (Photochemical (brown) smog forms when NO2 and volatile organic compounds react in sunlight.)

- NO2 + particulates + sun light
- ✓ NO2 + VOCS + sun light
- CO2 + VOCS + sun light
- PAN + particulates + sun light

ix) Which two gases of the following are the major contributors to global warming: (Carbon dioxide is the largest contributor and methane the second; nitrous oxide contributes less.)

- Carbon dioxide and nitrous oxide
- Methane and nitrous oxide
- ✓ Carbon dioxide and methane
- Methane and water vapours

x) The ozone layer lies in which of the following layers of the atmosphere? (The ozone layer is in the stratosphere, about 15-30 km up.)

- Troposphere
- Mesosphere
- Exosphere
- ✓ Stratosphere

## Exercise - Short Answer Questions

Q.1. Why Environmental Chemistry is important?

Environmental chemistry is important because it studies the

sources, reactions, effects, transport and fate of chemical substances in air, water and soil, and their effects on human health and the natural environment. It explains pollution, acid rain, smog, global warming and ozone depletion, and it provides the scientific basis for controlling pollution, treating water and protecting ecosystems.

Q.2. Why it is important to protect the environment?

It is important to protect the environment because we depend on it for air, water, food and shelter; pollution damages human health (respiratory diseases, cancer), destroys ecosystems and wildlife, reduces crop yields, and changes the climate. Protecting the environment preserves resources for future generations and keeps the natural systems that sustain life working.

Q.3. Why is the environment important to us?

The environment is important to us because it provides everything we need to live: the air we breathe (oxygen), the water we drink, the food we eat (soil, plants, animals), the climate that suits us, and the raw materials for industry. It also supports all other living things and the natural cycles (water cycle, carbon cycle, nitrogen cycle) that keep the earth habitable.

Q.4. How do human activities affect the environment?

Human activities affect the environment by burning fossil fuels (releasing CO2, SO2, NOx and particulates that cause air pollution, acid rain and global warming), cutting forests (deforestation reduces oxygen and wildlife habitat), dumping industrial and

domestic waste into rivers and seas (water pollution), overusing fertilizers and pesticides (soil and water contamination), and producing plastic and chemical waste that persists in ecosystems.

Q.5. What are the effects of pollution in our environment?

Effects of pollution in our environment: air pollution causes respiratory diseases, smog, acid rain and global warming; water pollution spreads water-borne diseases, kills aquatic life and makes water unfit to drink; soil pollution reduces fertility and contaminates food crops; noise pollution causes stress and hearing loss; and the build-up of greenhouse gases changes the climate, melts polar ice and raises sea levels.

Q.6. Earth is called blue planet, why?

Earth is called the blue planet because about 71% of its surface is covered by water (oceans, seas, lakes, rivers and ice), and seen from space the oceans make the earth look blue. Water is also essential for all life, and the hydrosphere dominates the earth's surface.

Q.7. How does the hydrosphere affect us?

The hydrosphere affects us because it supplies the water we drink, cook and wash with, irrigates our crops, provides food (fish), carries transport (ships), generates hydroelectric power, and controls the climate (oceans store heat and drive rainfall). Polluted or scarce water directly harms health, agriculture and industry.

Q.8. What is the importance of the hydrosphere?

Importance of the hydrosphere: it provides all the water for life (drinking, agriculture, industry); it regulates the earth's temperature and climate through ocean currents and evaporation; it is the home of aquatic life; it drives the water cycle (evaporation, condensation, precipitation); and it supplies hydroelectric energy and transport routes.

### Q.9. How atmosphere is important for us?

The atmosphere is important for us because it provides the oxygen we breathe and the carbon dioxide plants need; it protects us from harmful ultraviolet radiation (ozone layer) and from meteorites (which burn up in it); it keeps the earth warm (greenhouse effect); it carries weather and rainfall; and it provides nitrogen for fertilizers and industry.

### Q.10. What is the Earth's atmosphere made up of?

The Earth's atmosphere is made up of: nitrogen about 78%, oxygen about 21%, argon about 0.9%, carbon dioxide about 0.04%, plus water vapour (variable), and trace gases (neon, helium, methane, ozone) and dust particles. The troposphere (up to 11 km) holds most of its mass.

### Q.11. What are the main pollutants in the air?

The main pollutants in the air are: carbon oxides (CO, CO2), sulphur oxides (SO2, SO3), nitrogen oxides (NO, NO2), volatile organic compounds (VOCs, e.g. hydrocarbons from fuels and solvents), particulate matter (dust, smoke, soot), ozone (in the troposphere) and PAN (peroxyacetyl nitrate).

Q.12. How much do automobiles contribute to air pollution?

Automobiles contribute a large share of air pollution: in cities, vehicles produce roughly 50-60% of the carbon monoxide, a large share of nitrogen oxides and hydrocarbons (which form photochemical smog and ozone), plus particulates from exhaust and lead compounds (from leaded petrol, now reduced). The exact share varies by city, but transport is one of the biggest single sources of urban air pollution.

Q.13. What is the role of the catalytic converter?

A catalytic converter is a device fitted in the exhaust system of vehicles: it contains catalysts (platinum, palladium, rhodium) on a ceramic or metal support, which convert harmful exhaust gases into harmless ones before they leave the vehicle: carbon monoxide is oxidized to carbon dioxide, unburnt hydrocarbons are oxidized to carbon dioxide and water, and nitrogen oxides are reduced to nitrogen gas.

Q.14. What is the importance of catalytic converter?

Importance of the catalytic converter: it greatly reduces the harmful pollutants from vehicle exhaust (CO, hydrocarbons and NOx), so it reduces photochemical smog, acid rain and respiratory diseases in cities; it is one of the most effective and widely used pollution-control devices, and it is now required by law in most countries.

Q.15. How is smog caused?

Smog is caused when smoke (from coal, factories, vehicles) mixes with fog (water droplets) — classical smog — or when nitrogen oxides and volatile organic compounds from vehicles react in sunlight to form ozone and PAN — photochemical smog. Calm, humid weather and temperature inversions trap the pollutants near the ground and make smog worse.

Q.16. How is smog harmful to the environment?

Smog is harmful to the environment because it damages human health (irritates eyes, nose and lungs, causes asthma and bronchitis, and worsens heart disease), damages plants (ozone and PAN injure leaves and reduce crop yields), corrodes buildings and monuments, reduces visibility (traffic accidents), and harms animals and aquatic life.

Q.17. What is the main cause of greenhouse gases?

The main cause of greenhouse gases is the burning of fossil fuels (coal, oil, natural gas) for energy, transport and industry, which releases carbon dioxide; other causes are deforestation (fewer trees to absorb CO2), agriculture (methane from livestock and rice fields, nitrous oxide from fertilizers), and industrial processes (CFCs, methane leaks).

Q.18. How are greenhouse gases produced by humans?

Greenhouse gases are produced by humans through: burning fossil fuels in power stations, vehicles and factories (CO2); agriculture — livestock digestion and manure, rice paddies (methane), and nitrogen fertilizers (nitrous oxide); deforestation and land-use

change ($CO_2$ released, less absorption); waste disposal in landfills (methane); and industrial use of fluorinated gases (CFCs, HFCs) in refrigeration and aerosols.

Q.19. How does acid rain affect humans?

Acid rain affects humans by damaging the respiratory system (sulphur dioxide and fine sulphate/nitrate particles irritate lungs and worsen asthma and bronchitis), contaminating drinking water (it leaches toxic metals like aluminium and lead from soil and pipes), damaging skin and eyes on contact, and harming health indirectly by reducing crop yields and damaging buildings and monuments.

Q.20. Why is chlorine added to remaining treated waste water?

Chlorine is added to remaining (treated) waste water to disinfect it: chlorine kills bacteria, viruses and other disease-causing microorganisms left after treatment, so the discharged water does not spread water-borne diseases (cholera, typhoid, dysentery). It also oxidizes remaining organic matter and removes bad odours, making the effluent safer to release into rivers or reuse.

## Exercise - Long Answer Questions

Q.1. What is environmental chemistry? Briefly describe the segments of the environment.

Environmental chemistry is the study of the chemical phenomena that occur in the natural environment — the sources, reactions, effects, transport and fate of chemical substances in air, water and soil, and their effects on human health and the natural

environment. Segments of the environment: (1) Atmosphere — the layer of gases around the earth (troposphere, stratosphere, mesosphere, thermosphere, exosphere). (2) Hydrosphere — all water on earth (oceans, rivers, lakes, glaciers, groundwater, polar ice caps). (3) Lithosphere — the upper rigid rocky layer of the earth's crust (soil, rocks, mountains). (4) Biosphere — the region where all living things exist, above and below sea level, overlapping the other three segments.

Q.2. Explain the chemical reactions of nitrogen oxides and sulphur oxides in the atmosphere.

Chemical reactions of nitrogen oxides and sulphur oxides in the atmosphere: Nitrogen oxides (NO from combustion and lightning; NO2) react with oxygen and ozone: 2NO + O2 -> 2NO2; NO2 absorbs sunlight and splits: NO2 + sunlight -> NO + O, and the oxygen atom forms ozone: O + O2 -> O3. NO2 also reacts with water to give nitric acid: 3NO2 + H2O -> 2HNO3 + NO (acid rain). Sulphur dioxide (SO2 from fossil fuels and volcanoes) is oxidized by oxygen (catalysed by particulates) to SO3: 2SO2 + O2 -> 2SO3; SO3 reacts with water to give sulphuric acid: SO3 + H2O -> H2SO4 (acid rain). Both acids fall as wet deposition (rain, snow) or dry deposition (gases, particulates).

Q.3. Discuss the chemical reactions of oxygen in atmosphere.

Chemical reactions of oxygen in the atmosphere: Oxygen (O2, 21% of air) is produced by photosynthesis (6CO2 + 6H2O -> C6H12O6 + 6O2) and consumed by respiration and combustion. In the stratosphere, ultraviolet light splits oxygen: O2 + UV -> 2O; the

oxygen atoms combine with O2 to form ozone: O + O2 -> O3 (the ozone layer, which absorbs harmful UV). Ozone itself is broken down by UV and by pollutants (CFCs release chlorine, which destroys ozone catalytically: Cl + O3 -> ClO + O2; ClO + O -> Cl + O2). Oxygen also oxidizes pollutants (SO2 -> SO3, NO -> NO2) and metals (rusting: 4Fe + 3O2 + 2xH2O -> 2Fe2O3.xH2O).

Q.4. How do the chemical reactions of organic compounds carried out in the atmosphere?

Chemical reactions of organic compounds in the atmosphere: Volatile organic compounds (VOCs — hydrocarbons from fuels, solvents, vegetation) are oxidized in the atmosphere by hydroxyl radicals (OH.), ozone and sunlight. In the presence of nitrogen oxides and sunlight, VOCs take part in photochemical reactions that produce ozone and peroxyacetyl nitrate (PAN): e.g. hydrocarbons + NO2 + sunlight -> O3 + PAN + aldehydes. These secondary pollutants form photochemical (brown) smog. Organic compounds are also oxidized to carbon dioxide and water over time, and some (e.g. methane) act as greenhouse gases; methane reacts with OH. radicals: CH4 + OH. -> CH3. + H2O, beginning a chain that eventually gives CO2.

Q.5. Define air pollutants. Describe the sources and harmful effects of oxides of carbon, nitrogen and Sulphur.

Air pollutants are substances in the air that have adverse effects on human health, quality of life and the natural functioning of ecosystems. Sources and harmful effects of oxides of carbon, nitrogen and sulphur: Carbon oxides — CO (incomplete combustion

in vehicles) is poisonous (binds haemoglobin, causing suffocation); CO2 (combustion, respiration) is a greenhouse gas causing global warming. Nitrogen oxides — NO and NO2 (vehicles, power stations, lightning) cause respiratory diseases, form acid rain (nitric acid) and photochemical smog. Sulphur oxides — SO2 and SO3 (fossil fuels, volcanoes, smelters) cause respiratory diseases, form acid rain (sulphuric acid), damage plants, buildings and monuments, and reduce visibility.

Q.6. What are the sources of volatile organic compounds in the atmosphere? What are the adverse effects indicated by them?

Sources of volatile organic compounds (VOCs) in the atmosphere: evaporation of petrol and solvents, vehicle exhaust, industrial processes (paints, chemicals, plastics), natural sources (trees release terpenes, e.g. pine scent), and burning of wood and waste. Adverse effects: VOCs irritate eyes, nose and throat, cause headaches and dizziness, and some are carcinogenic (e.g. benzene); in sunlight with nitrogen oxides they form ozone and PAN (photochemical smog), which damage lungs, crops and materials; and some VOCs (e.g. methane) are greenhouse gases.

Q.7. Discuss the characteristic and undesirable effects of ozone and peroxyacetyl nitrate (PAN).

Ozone and peroxyacetyl nitrate (PAN) — characteristics and undesirable effects: Ozone (O3) in the stratosphere is beneficial (absorbs harmful UV), but in the troposphere it is a pollutant: it is a strong oxidizing gas with a sharp smell, formed photochemically from NO2 and VOCs in sunlight. PAN (CH3COOONO2) is a secondary

pollutant formed in photochemical smog from VOCs and NO2; it is a powerful eye irritant (lachrymator) even at low concentrations. Undesirable effects: both damage the respiratory system (coughing, asthma, reduced lung function), injure plant leaves and reduce crop yields, damage rubber, textiles and paints, and reduce visibility.

Q.8. What do you know about automobile pollutants? What are the importance and role of catalytic converters?

Automobile pollutants: vehicles emit carbon monoxide (incomplete combustion), nitrogen oxides (high-temperature combustion), unburnt hydrocarbons/VOCs, particulates (soot), sulphur dioxide (from fuel sulphur) and lead compounds (from leaded petrol). Importance and role of catalytic converters: fitted in the exhaust, they use platinum, palladium and rhodium catalysts to convert CO to CO2, hydrocarbons to CO2 + H2O, and NOx to N2, greatly reducing smog-forming and poisonous emissions; without them, city air would be far more polluted.

Q.9. Define smog and explain industrial and photochemical smog.

Smog is a mixture of smoke and fog (classical smog) or a photochemical haze (photochemical smog) that hangs over cities. Industrial (classical/London/gray) smog: forms in cool, humid weather from coal smoke — SO2, soot particulates and fog moisture; it is grey, acidic and reducing in nature. Photochemical (Los Angeles/brown) smog: forms in warm, sunny weather when NO2 and VOCs from vehicles react in sunlight to give ozone, PAN and aldehydes; it is brownish, oxidizing in nature, and irritates eyes

and lungs. Both reduce visibility and harm health, but their chemistry, colour and weather conditions differ.

Q.10. What is the effect of smog on the health of peoples? How to prevent smog?

Effect of smog on the health of peoples: smog irritates eyes, nose and throat, causes coughing, asthma attacks, bronchitis and reduced lung function, worsens heart disease, and long exposure increases the risk of lung cancer and early death; children and the elderly are most vulnerable. How to prevent smog: reduce vehicle emissions (catalytic converters, public transport, cleaner fuels), control industrial emissions (filters, scrubbers), use renewable energy instead of coal, avoid open burning, plant trees, and enforce air-quality laws with monitoring and public warnings on high-smog days.

Q.11. How does car exhaust produce the smog that hangs over our cities?

Car exhaust produces the smog that hangs over cities because exhaust gases contain nitrogen oxides and unburnt hydrocarbons (VOCs); in sunlight, $NO_2$ splits to give oxygen atoms that form ozone, and VOCs oxidize to PAN, aldehydes and other secondary pollutants. These mix with particulates and moisture to form a brownish haze (photochemical smog) that hangs over the city, especially on calm, sunny days when temperature inversions trap the pollutants near the ground.

Q.12. What is greenhouse effect? What are greenhouse gases?

Discuss their contribution to global warming? What are the consequences of global warming?

Greenhouse effect: the warming of the earth because greenhouse gases (CO2, CH4, N2O, CFCs, water vapour) in the atmosphere absorb infrared radiation emitted by the earth and re-radiate part of it back, trapping heat — like the glass of a greenhouse. Greenhouse gases: carbon dioxide (fossil fuels, deforestation), methane (livestock, rice fields, landfills), nitrous oxide (fertilizers), CFCs (refrigeration), ozone and water vapour. Contribution to global warming: rising concentrations trap more heat, raising the average global temperature. Consequences: melting of polar ice and glaciers, sea-level rise, changed rainfall patterns (droughts and floods), heat waves, loss of habitats and species, reduced crop yields, and spread of tropical diseases.

Q.13. What is the acid rain? Discuss briefly. What are bad effects of acid rain on the environment and public health? How can we minimize its bad effects?

Acid rain: rain (or snow, sleet, hail) with a pH below 5.6 (the book's Summary uses 5.3), formed when sulphur dioxide and nitrogen oxides from fossil fuels, vehicles and industry oxidize in the atmosphere to sulphuric acid and nitric acid and mix with rainwater. Bad effects on the environment and public health: it acidifies lakes and rivers (killing fish), damages forests and crops (leaches nutrients, injures leaves), corrodes buildings, bridges and monuments (stone cancer on marble), leaches toxic metals into drinking water, and worsens respiratory diseases. How to minimize its bad effects: burn low-sulphur fuels, fit scrubbers and catalytic

converters, use renewable energy, treat acidic lakes with lime, and enforce emission limits on industry and vehicles.

Q.14. Define water pollution and water pollutants? What are the different types of water pollutants and how they are produced? What are their effects?

Water pollution is the addition of any harmful substance to water that makes it unfit for its intended use (drinking, bathing, irrigation, aquatic life). Water pollutants: (1) Biological — bacteria, viruses, parasites from sewage (cause cholera, typhoid, dysentery). (2) Chemical — industrial chemicals, heavy metals (lead, mercury, cadmium), fertilizers, pesticides, acids, oils. (3) Physical — sediments, silt, heat (thermal pollution from power stations), radioactive substances. They are produced by sewage discharge, industrial effluents, agricultural run-off, oil spills and dumping. Effects: spread of water-borne diseases, death of fish and aquatic life (oxygen depletion by organic waste), contamination of drinking water, damage to crops, and foul smell and appearance.

Q.15. Explain the various parameters of water analysis.

Various parameters of water analysis: (1) Physical — temperature, colour, odour, taste, turbidity (cloudiness), total dissolved solids (TDS), electrical conductivity. (2) Chemical — pH, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD), hardness ($Ca^{2+}$, $Mg^{2+}$), chlorides, nitrates, phosphates, heavy metals, residual chlorine. (3) Biological — total coliform count and E. coli (indicators of sewage contamination), algal count. Safe drinking water must have acceptable values for all

of these (e.g. pH 6.5-8.5, low BOD, no E. coli).

Q.16. Define waste water and write an informative note on waste water treatment. What are the steps involved in waste water treatment?

Waste water (sewage) is the mixture of wastes from the human body and used water from homes, institutions and industries (including toilet waste, kitchen and washing water, and industrial effluents). Informative note on waste water treatment: treatment removes contaminants so the water can be safely discharged or reused. Steps involved: (1) Preliminary treatment — screening (removing large solids) and grit removal. (2) Primary treatment — sedimentation in tanks (settling of suspended solids, removal of floating grease/oil). (3) Secondary treatment — biological treatment: aerobic microbes in trickling filters or activated sludge digest dissolved organic matter. (4) Tertiary/advanced treatment — filtration, nutrient removal (nitrogen, phosphorus), and disinfection (chlorination, UV, ozonation) to kill pathogens. The treated water is then discharged into rivers or reused for irrigation, and the sludge is digested and disposed of safely.

# Chapter 24: Analytical Chemistry

## Exercise - Multiple Choice Questions

i) Volumetric analysis method is aimed for: (Volumetric (titrimetric) analysis determines the concentration of a solution by titration.)

- ✓ Analyzing solutions
- Analyzing liquids only
- Analyzing solids only
- Analyzing volume only

ii) The combustion analysis methods are: (Combustion analysis burns an organic compound to $CO_2$ and $H_2O$ to find its carbon and hydrogen content.)

- ✓ Only limited to organic compounds
- Only limited to Inorganic compounds
- Cover both organic and inorganic compounds
- Covered only physical characters of a chemical compound

iii) The salt analysis covers: (Salt analysis is the qualitative analysis of inorganic salts — their cations and anions.)

- Analysis of organic compounds
- ✓ Analysis of Inorganic compounds
- Both organic and inorganic compounds
- Analysis the sodium chloride only

iv) Mass spectrometers separates ions on which of the following:

(Ions are separated by their mass-to-charge ratio, m/z.)

- Mass
- Charge
- ✓ Mass-to-charge ratio
- Molecular mass

v) The Ultraviolet light has energy: (UV photons are more energetic than visible red light, but less energetic than X-rays and gamma rays.)

- ✓ More than red light
- Less than red light
- More than X-rays
- More than gamma rays

vi) The red light has wave length from: (Red light is the longest visible wavelength, about 620-750 nm; 620-800 nm is the listed range.)

- 400-450nm
- 450-550nm
- 550-620nm
- ✓ 620-800nm

vii) The portion of the fire that feels us warm is: (The heat felt from a fire is infrared radiation.)

- Ultraviolet rays
- ✓ Infrared rays
- Visible radiations
- Yellow light of fire

viii) The globar rod in IR spectrometer is made of: (A globar is a silicon carbide rod; the oxides of zirconium/cerium describe the Nernst glower, a different source.)

- Oxides of zirconium
- Silver carbide
- Oxides of cerium
- ✓ Silicon carbide

ix) The colour of a chemical compound results from: (Colour comes from absorption of visible light, which promotes electrons to higher levels — electronic transitions; the absorbed colour's complement is seen.)

- Stretching of molecules
- Bending of molecules
- Release of energy
- ✓ Electronic transitions

x) Mass spectroscopy is used to determine: (Mass spectrometry gives molecular masses, isotopic masses and isotopic abundances.)

- ✓ Molecular masses
- Functional group
- Type of atoms
- Unsaturation

## Exercise - Short Answer Questions

Q.1. Define chemical analysis. What are the main types of chemical analysis?

Chemical analysis is the branch of analytical chemistry that deals with the separation, identification and quantification of the components of a sample of matter. Main types of chemical analysis: (1) Qualitative analysis — identifies what is present (which elements, ions or functional groups). (2) Quantitative analysis — determines how much of each component is present (by mass, volume or moles).

Q.2. What is the importance of qualitative analysis and quantitative analyses?

Importance of qualitative and quantitative analyses: qualitative analysis tells us what substances are present (e.g. which cation or anion is in a salt, which functional group is in an organic compound), which is essential for identification and for choosing the right quantitative method; quantitative analysis tells us how much is present (e.g. the percentage of iron in an ore, the concentration of a drug, the purity of a sample), which is essential for quality control, medicine, industry and research. Together they fully characterize a sample.

Q.3. What are the different types of spectroscopy?

Different types of spectroscopy: (1) UV-Visible spectroscopy (electronic transitions, 200-800 nm). (2) Infrared (IR) spectroscopy (molecular vibrations, 2.5-25 micrometres). (3) Nuclear magnetic resonance (NMR) spectroscopy (nuclear spin transitions in a magnetic field). (4) Mass spectrometry (mass-to-charge ratio of ions). (5) Atomic absorption and atomic emission spectroscopy (elemental analysis). (6) Raman spectroscopy (scattered light,

vibrations). (7) X-ray spectroscopy and microwave spectroscopy.

### Q.4. What is the main use of IR spectroscopy?

The main use of IR spectroscopy is the identification of functional groups in organic compounds: each functional group absorbs infrared radiation at a characteristic frequency (e.g. O-H about 3300 cm-1, C=O about 1710 cm-1, C-H about 2900 cm-1), so the IR spectrum acts as a fingerprint that identifies the groups present and helps determine the structure of an unknown compound.

### Q.5. Why most chemist use wave number unit rather than wavelength in IR spectroscopy?

Most chemists use wave number units (cm-1) rather than wavelength in IR spectroscopy because wave number is directly proportional to energy (E = hc x wave number) and to frequency, so spectra are linear in energy; also, wave numbers are convenient whole numbers (400-4000 cm-1) whereas wavelengths are small decimals in micrometres, and characteristic group frequencies are easier to remember and compare in cm-1.

### Q.6. How UV-Vis spectroscopy differs from IR spectroscopy?

How UV-Vis spectroscopy differs from IR spectroscopy: UV-Vis uses ultraviolet and visible light (200-800 nm) and causes electronic transitions (electrons move to higher energy levels), giving information about conjugation, unsaturation and colour; IR uses infrared radiation (2.5-25 micrometres) and causes molecular vibrations (stretching and bending of bonds), giving information about functional groups. UV-Vis spectra show broad bands; IR

spectra show sharper peaks at characteristic wave numbers.

Q.7. How does the detector in NMR spectrometer work?

How the detector in an NMR spectrometer works: the detector (a receiver coil wound around the sample) picks up the radio-frequency signal emitted by nuclei when they fall back from the higher-energy spin state to the lower-energy state (resonance). As the magnetic field or the radio frequency is swept, each chemically different set of protons comes into resonance in turn and emits a signal; the detector converts these tiny radio signals into electrical signals, which are amplified and recorded as peaks in the NMR spectrum, whose positions (chemical shifts) and areas reveal the chemical environments and numbers of protons.

Q.8. What are the principles of mass spectrometer?

Principles of the mass spectrometer: (1) Vaporization — the sample is vaporized. (2) Ionization — the vapour is bombarded with high-energy electrons (or other ionization methods), producing positive ions (molecular ions and fragment ions). (3) Acceleration — the ions are accelerated by an electric field. (4) Deflection — a magnetic field deflects the ions; lighter ions (smaller m/z) are deflected more than heavier ones. (5) Detection — the separated ions strike a detector in order of increasing m/z, giving a mass spectrum of relative abundance vs m/z, from which molecular mass, formula and structure are deduced.

Q.9. Name the factors that affect vibrational frequency.

Factors that affect vibrational frequency: (1) Masses of the bonded

atoms — heavier atoms vibrate more slowly (lower frequency); frequency is proportional to 1/sqrt(reduced mass). (2) Bond strength (force constant) — stronger bonds (triple > double > single) vibrate at higher frequency. (3) Bond type and hybridization — e.g. C-H, O-H, C=O each have characteristic ranges. (4) Chemical environment — hydrogen bonding lowers the frequency (e.g. O-H stretching shifts and broadens), and neighbouring electronegative atoms shift frequencies.

Q.10. Give three applications of mass spectroscopy.

Three applications of mass spectroscopy: (1) Determination of molecular masses and molecular formulas of compounds. (2) Determination of relative isotopic masses and isotopic abundances, and hence average atomic masses (e.g. neon, chlorine). (3) Structure determination — the fragmentation pattern identifies functional groups and the carbon skeleton of organic compounds. (Other applications: detecting impurities, drug and forensic analysis, and dating of rocks by isotope ratios.)

## Exercise - Long Answer Questions

Q.1. What is chemical analysis? Explain qualitative and quantitative analysis with at least one example.

Chemical analysis is the study of the separation, identification and quantification of the components of a sample of matter. Qualitative analysis identifies the components: example — a salt solution gives a white precipitate with silver nitrate solution, showing chloride ions are present (Ag+ + Cl- -> AgCl, white ppt). Quantitative analysis measures how much is present: example — titration of a

hydrochloric acid solution with standard sodium hydroxide gives its exact concentration (HCl + NaOH -> NaCl + H2O), or gravimetric analysis weighs the silver chloride precipitate to find the mass of chloride in the sample.

Q.2. What do you know about classical and modern methods? Compare classical methods of analysis with modern methods.

Classical and modern methods: classical methods are the traditional wet-chemical methods — qualitative analysis (flame tests, precipitation reactions, borax bead test) and quantitative analysis by volumetric methods (titrations) and gravimetric methods (weighing a precipitate). Modern methods are instrumental methods — spectroscopy (UV-Vis, IR, NMR, atomic absorption/emission), mass spectrometry, chromatography (separation + identification), electroanalytical methods (pH meter, conductometry). Comparison: classical methods are simple, cheap and need little equipment but are slow, less sensitive and need larger samples; modern methods are fast, very sensitive (trace amounts), accurate and automated, but need expensive instruments and skilled operators. Modern methods have largely replaced classical methods for routine analysis, though classical methods are still used for teaching and as reference methods.

Q.3. What is the difference between volumetric and gravimetric methods of analysis?

Difference between volumetric and gravimetric methods of analysis: Volumetric analysis measures the volume of a standard solution (titrant) that reacts completely with the analyte (e.g.

acid-base titration with an indicator to find the end point); it is fast and convenient for solutions, and the result comes from a volume measurement. Gravimetric analysis converts the analyte into a pure, stable, insoluble compound (a precipitate), which is filtered, washed, dried and weighed; the result comes from a mass measurement. Volumetric methods are quicker and easier; gravimetric methods are more accurate and absolute (no standard solution needed) but slower and more laborious.

Q.4. What is combustion analysis? Discuss the procedure of combustion analysis.

Combustion analysis is the method of determining the carbon, hydrogen (and sometimes nitrogen and sulphur) content of an organic compound by burning a known mass of it completely in oxygen and weighing the products ($CO_2$ and $H_2O$). Procedure: (1) A weighed sample is placed in a combustion tube and heated in a stream of pure oxygen (with a catalyst such as copper(II) oxide to ensure complete combustion). (2) The carbon of the compound is converted to $CO_2$ and the hydrogen to $H_2O$. (3) The gases pass through weighed absorption tubes: calcium chloride (or magnesium perchlorate) absorbs $H_2O$, and potassium hydroxide (or soda lime) absorbs $CO_2$. (4) The increase in mass of each tube gives the masses of $H_2O$ and $CO_2$ formed. (5) From these, the masses (and percentages) of C and H in the sample are calculated: mass of C = (12/44) x mass of $CO_2$; mass of H = (2/18) x mass of $H_2O$; oxygen is found by difference. (Nitrogen is measured separately by Dumas' method, sulphur by Carius' method.)

Q.5. What are electromagnetic radiations? Describe the regions in electromagnetic spectrum in detail?

Electromagnetic radiations are waves of electric and magnetic fields travelling at the speed of light (3 x 10^8 m/s) in vacuum, differing in wavelength and frequency (energy E = hf). Regions of the electromagnetic spectrum in detail (from longest wavelength/lowest energy to shortest/highest): (1) Radio waves (wavelength > 1 m) — broadcasting, NMR. (2) Microwaves (1 mm - 1 m) — radar, cooking, rotational spectroscopy. (3) Infrared (700 nm - 1 mm) — heat radiation, IR spectroscopy (vibrations). (4) Visible light (400-700 nm) — the colours violet to red, UV-Vis spectroscopy. (5) Ultraviolet (10-400 nm) — electronic transitions, sterilization. (6) X-rays (0.01-10 nm) — medical imaging, crystal structure. (7) Gamma rays (< 0.01 nm) — nuclear transitions, cancer treatment. As wavelength decreases, frequency and energy increase.

Q.6. Define and explain atomic absorption and emission spectra.

Atomic absorption and emission spectra: Atomic absorption spectrum — when white light passes through a vapour of an element, the atoms absorb light of certain wavelengths (those that promote their electrons to higher levels), producing dark lines on a continuous bright background; each element has its own set of absorbed wavelengths. Atomic emission spectrum — when atoms are excited (by heat or electricity), their electrons jump to higher levels and then fall back, emitting light of definite wavelengths, producing bright lines on a dark background. Explanation: both arise from electronic transitions between fixed energy levels (Bohr's

theory); the difference between two levels equals the photon energy (delta E = hf). Each element's line pattern is unique (its fingerprint), so both spectra are used to identify elements and determine their concentration.

Q.7. Define spectroscopy. What are the applications of spectroscopy?

Spectroscopy is the study of the interaction of electromagnetic radiation with matter — how matter absorbs, emits or scatters radiation — used to identify substances and determine their structure. Applications of spectroscopy: (1) Identification of unknown compounds (IR functional groups, NMR structure, mass spectra). (2) Quantitative analysis (UV-Vis and atomic absorption measure concentrations by Beer-Lambert's law). (3) Elemental analysis (atomic emission/absorption identify elements). (4) Medical diagnosis (MRI uses NMR; X-rays image bones). (5) Environmental monitoring (pollutants in air and water). (6) Forensic and pharmaceutical analysis (drugs, poisons). (7) Astronomy (composition of stars from their spectra).

Q.8. What is UV-Vis spectroscopy? Explain the principle and applications of UV-Vis spectroscopy.

UV-Vis spectroscopy is the study of the absorption of ultraviolet (200-400 nm) and visible (400-800 nm) radiation by a substance. Principle: UV-Vis radiation has enough energy to promote electrons from bonding or non-bonding orbitals to anti-bonding orbitals (electronic transitions: sigma->sigma*, n->sigma*, pi->pi*, n->pi*); a molecule absorbs at the wavelength corresponding to the energy

gap, and the absorbance follows Beer-Lambert's law (A = epsilon x c x l), so concentration can be measured. Applications: determining concentrations of coloured and UV-absorbing substances (e.g. transition metal ions, organic compounds with conjugation); identifying compounds and checking purity; studying reaction kinetics; determining the colour of transition metal complexes from their spectra (the absorbed colour's complement is seen); and detecting unsaturation and conjugation in organic molecules.

Q.9. Explain the various electronic transitions possible on the absorption of UV-Visible radiation.

Various electronic transitions possible on the absorption of UV-Visible radiation: (1) sigma -> sigma* — electrons in a sigma bond promoted to the anti-bonding sigma orbital; needs high energy (vacuum UV, < 200 nm), e.g. saturated hydrocarbons. (2) n -> sigma* — a non-bonding (lone pair) electron promoted to sigma*; needs less energy, e.g. alcohols, amines, halides (about 200 nm). (3) pi -> pi* — an electron in a pi bond promoted to pi*; common in alkenes and conjugated systems (200-400 nm), intensity high. (4) n -> pi* — a lone-pair electron promoted to pi* of a carbonyl or C=C; lowest energy (about 270-300 nm), low intensity, e.g. aldehydes and ketones. Energy order: sigma->sigma* > n->sigma* > pi->pi* > n->pi*. Conjugation lowers the pi->pi* gap, shifting absorption to longer wavelengths (bathochromic shift).

Q.10. Write a note on infrared spectroscopy? Describe the principle and applications of IR spectroscopy.

Note on infrared spectroscopy: IR spectroscopy studies the absorption of infrared radiation (wave numbers 400-4000 cm-1) by molecules, which causes their bonds to vibrate (stretching and bending). Principle: a bond absorbs IR radiation only if its vibration changes the molecule's dipole moment, and it absorbs at its natural vibrational frequency, which depends on the bond strength and the masses of the atoms. The spectrum (percentage transmittance vs wave number) shows peaks for each vibrating group: the functional group region (4000-1500 cm-1) identifies groups (O-H, C=O, C-H, C=C), and the fingerprint region (1500-400 cm-1) is unique to each compound. Applications: identification of functional groups and unknown organic compounds; checking purity; distinguishing isomers; studying hydrogen bonding; and quantitative analysis of mixtures.

Q.11. Explain the various vibrations possible when a molecule absorbs infrared radiation.

Various vibrations possible when a molecule absorbs infrared radiation: (1) Stretching vibrations — the bond length changes rhythmically: (a) Symmetric stretching (both bonds stretch together, e.g. CH2 both C-H bonds lengthen together); (b) Asymmetric stretching (one bond lengthens while the other shortens). (2) Bending vibrations — the bond angle changes: (a) Scissoring (two bonds move toward/away from each other in the plane, like scissors); (b) Rocking (the group swings back and forth in the plane); (c) Wagging (the group swings back and forth out of the plane); (d) Twisting (one bond twists out of the plane relative to the other). Stretching vibrations need more energy (higher wave

number) than bending vibrations. A non-linear molecule with n atoms has 3n-6 fundamental vibrations; a linear molecule has 3n-5.

## Exercise - Problems

Q.1. When 2.203g of sample of an organic compound was burnt in a combustion tube, 4.401g of CO2 and 1.802g of H2O were produced. Determine the % age of carbon, hydrogen and oxygen atoms.

Step 1 — Mass of carbon in the CO2 formed: mass of C = (12/44) x mass of CO2 = (12/44) x 4.401 g = 1.200 g.

Step 2 — Percentage of carbon: %C = (1.200 / 2.203) x 100 = 54.48%.

Step 3 — Mass of hydrogen in the H2O formed: mass of H = (2/18) x mass of H2O = (2/18) x 1.802 g = 0.200 g.

Step 4 — Percentage of hydrogen: %H = (0.200 / 2.203) x 100 = 9.09%.

Step 5 — Percentage of oxygen by difference: %O = 100 - 54.48 - 9.09 = 36.43%.

%C = 54.48%, %H = 9.09%, %O = 36.43%.

Q.2. An unknown compound contains 14.31% sodium, 9.97% sulphur, 6.22% hydrogen and 69.5% oxygen. Calculate the empirical formula of the compound.

Step 1 — Take 100 g of the compound: it contains 14.31 g Na, 9.97 g S, 6.22 g H and 69.5 g O.

Step 2 — Convert to moles: Na = 14.31/23 = 0.622 mol; S = 9.97/32 = 0.312 mol; H = 6.22/1 = 6.22 mol; O = 69.5/16 = 4.344 mol.

Step 3 — Divide by the smallest (0.312): Na = 0.622/0.312 = 1.99 = 2; S = 0.312/0.312 = 1; H = 6.22/0.312 = 19.9 = 20; O = 4.344/0.312 = 13.9 = 14.

Step 4 — The simplest whole-number ratio is Na : S : H : O = 2 : 1 : 20 : 14.

Empirical formula = Na2SH20O14 (sodium sulphate decahydrate, Na2SO4.10H2O).

Q.3. The empirical formula of glucose is CH2O. It molecular mass is 180 amu. Determine it molecular formula.

Step 1 — Empirical formula mass of CH2O = 12 + 2(1) + 16 = 30 amu.

Step 2 — Find n: n = molecular mass / empirical formula mass = 180 / 30 = 6.

Step 3 — Molecular formula = n x empirical formula = 6 x (CH2O) = C6H12O6.

Molecular formula = C6H12O6 (glucose).

Q.4. Calculate the average atomic mass of neon with the help of mass spectrum given below: 20Ne (90.92%), 21Ne (0.26%), 22Ne (8.82%).

Step 1 — Read the spectrum: the isotopes and their relative abundances are 20Ne = 90.92%, 21Ne = 0.26%, 22Ne = 8.82%.

Step 2 — Multiply each mass number by its abundance: 20 x 90.92 = 1818.40; 21 x 0.26 = 5.46; 22 x 8.82 = 194.04.

Step 3 — Add the products: 1818.40 + 5.46 + 194.04 = 2017.90.

Step 4 — Divide by 100 (total abundance): 2017.90 / 100 = 20.179 = 20.18 amu.

Average atomic mass of neon = 20.18 amu.